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(54) **COMPUTER IMPLEMENTED INTERACTIVE
SURGICAL SYSTEMS**

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- (63) Continuation of application No. 17/203,884, filed on Mar. 17, 2021, now Pat. No. 12,318,152, which is a continuation of application No. 15/940,629, filed on Mar. 29, 2018, now abandoned.
- (60) Provisional application No. 62/611,341, filed on Dec. 28, 2017, provisional application No. 62/611,340, filed on Dec. 28, 2017, provisional application No. 62/611,339, filed on Dec. 28, 2017, provisional application No. 62/649,310, filed on Mar. 28, 2018.

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A61B 50/13 (2016.01)

A61B 90/00 (2016.01)

G16H 40/60 (2018.01)

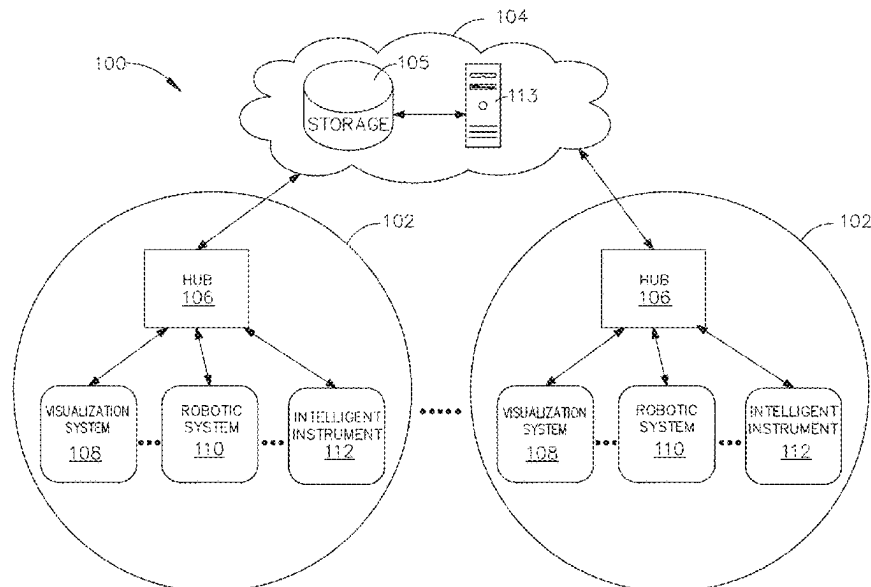
(52) **U.S. Cl.**

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(2013.01); **A61B 18/12** (2013.01); **A61B**
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A61B 34/76 (2016.02); **A61B 50/13** (2016.02);
G16H 40/60 (2018.01); **A61B 2017/00199**
(2013.01); **A61B 2017/00221** (2013.01); **A61B**
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2034/302 (2016.02); **A61B 2034/304**
(2016.02); **A61B 2034/305** (2016.02); **A61B**
34/37 (2016.02); **A61B 34/77** (2016.02); **A61B**
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(2016.02); **A61B 2090/066** (2016.02); **A61B**
90/361 (2016.02); **A61B 90/37** (2016.02);
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(2013.01); **A61B 2560/0443** (2013.01)

(57)

ABSTRACT

A surgical hub is for use with a surgical instrument configured to deliver therapeutic energy to tissue at a surgical site of a surgical procedure. The surgical hub comprises: a hub enclosure, comprising a docking station including a docking port comprising data and power contacts; and a combo generator module removably retainable in the docking station. The combo generator module comprises: an ultrasonic energy generator component; a radio frequency (RF) energy generator component; a smoke evacuation component; and a connection port. At least one of the ultrasonic energy generator component and the radio frequency (RF) generator component are couplable to the surgical instrument through the connection port. The combo generator module further comprises at least one smoke evacuation component, configured to evacuate smoke generated by an application of therapeutic energy to the tissue by the surgical instrument.



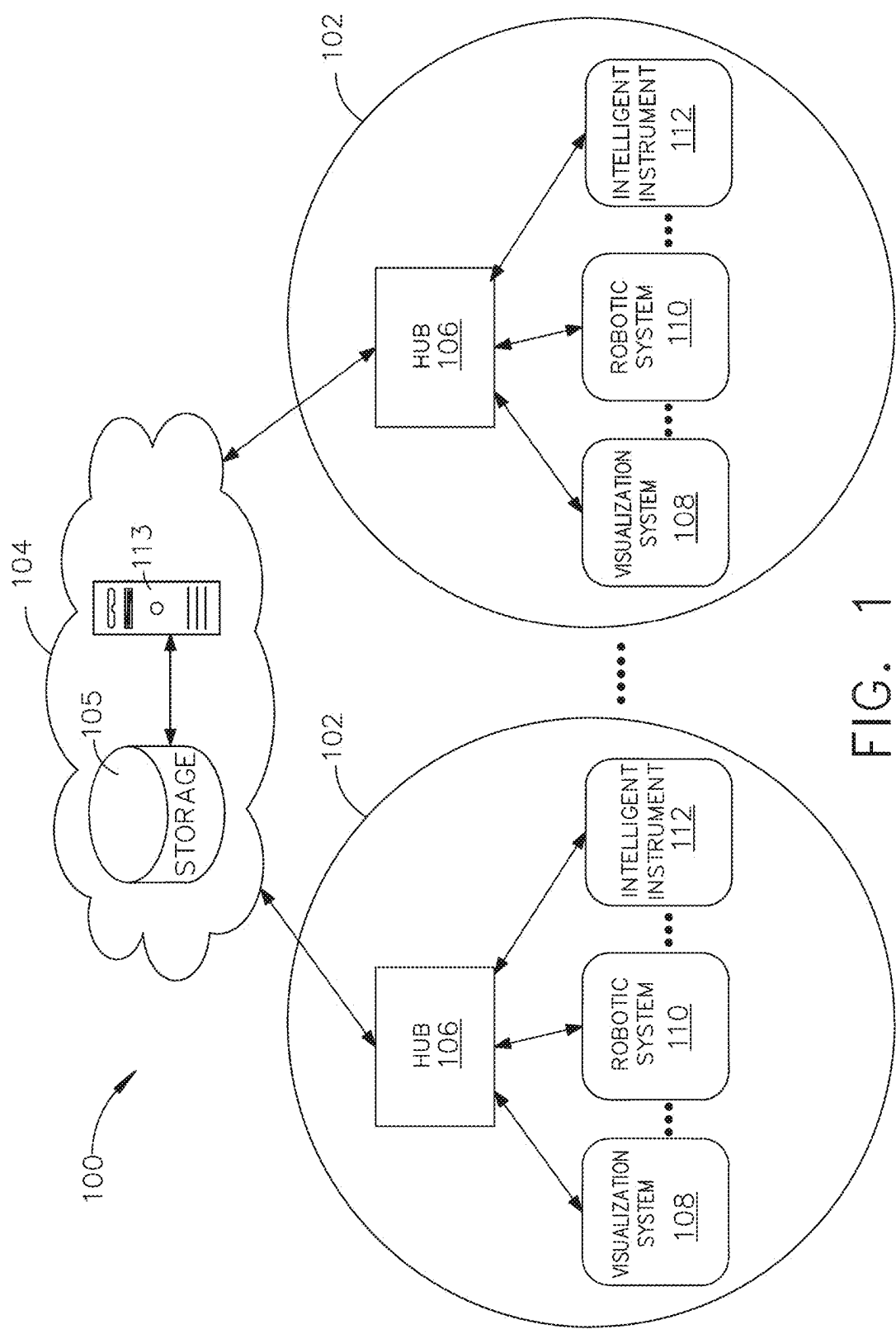


FIG. 1

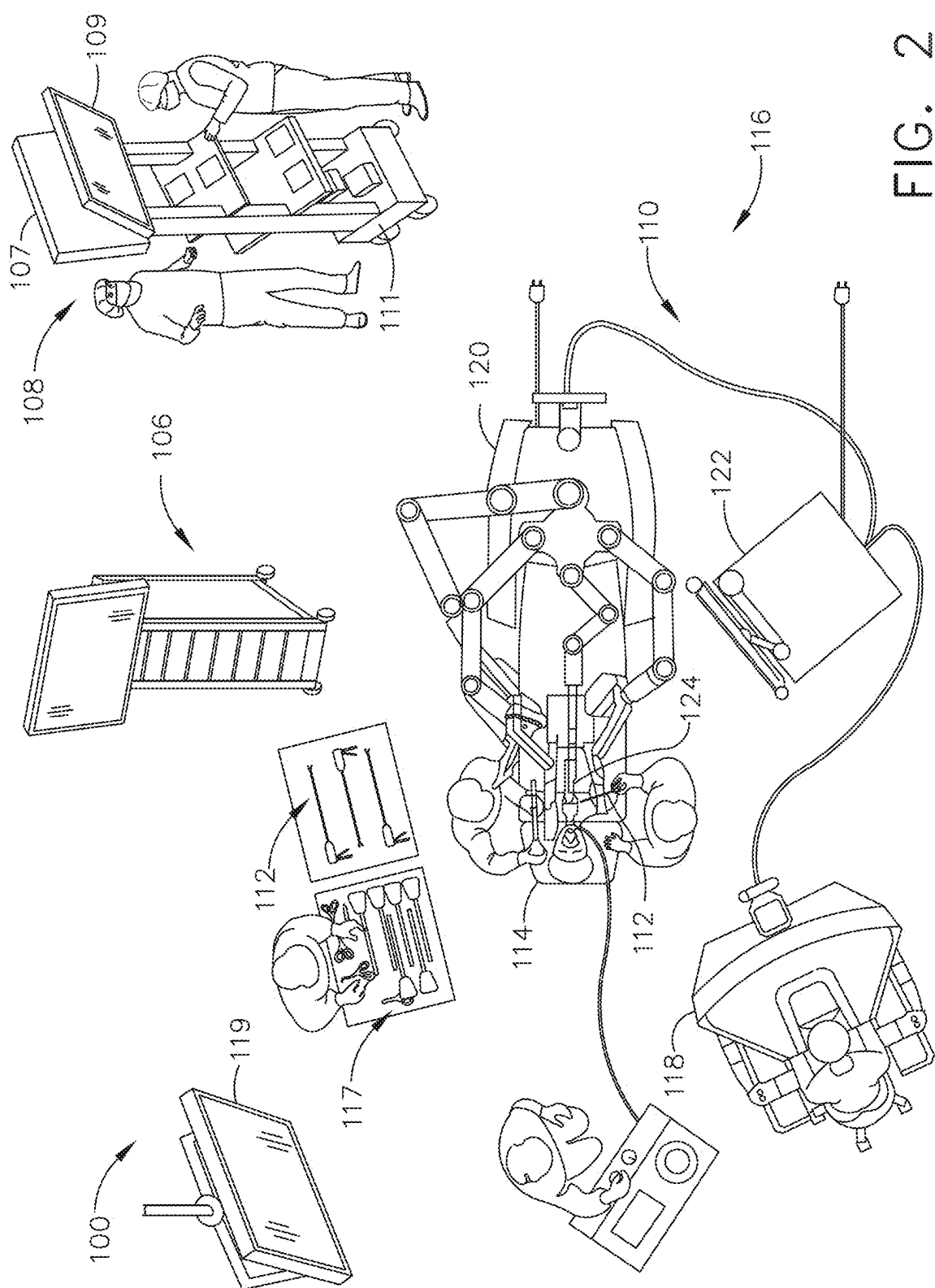


FIG. 2

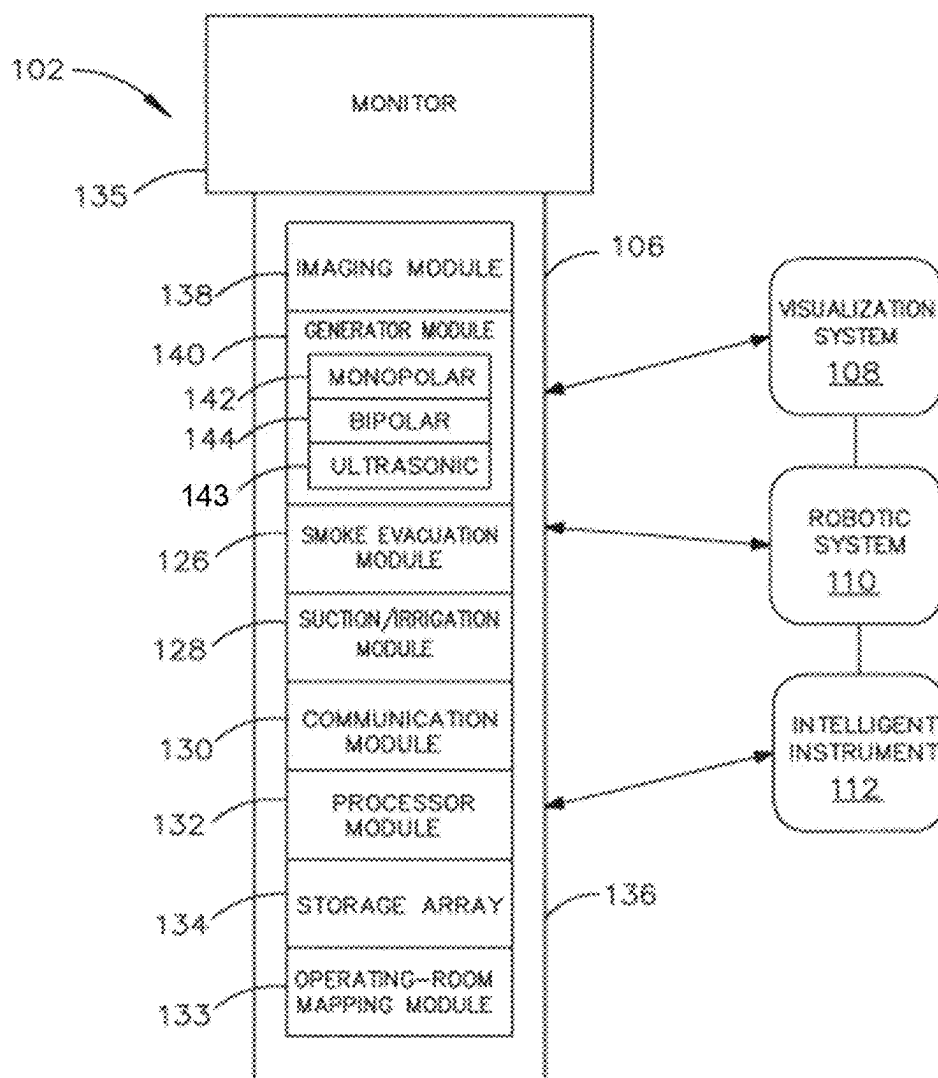


FIG. 3

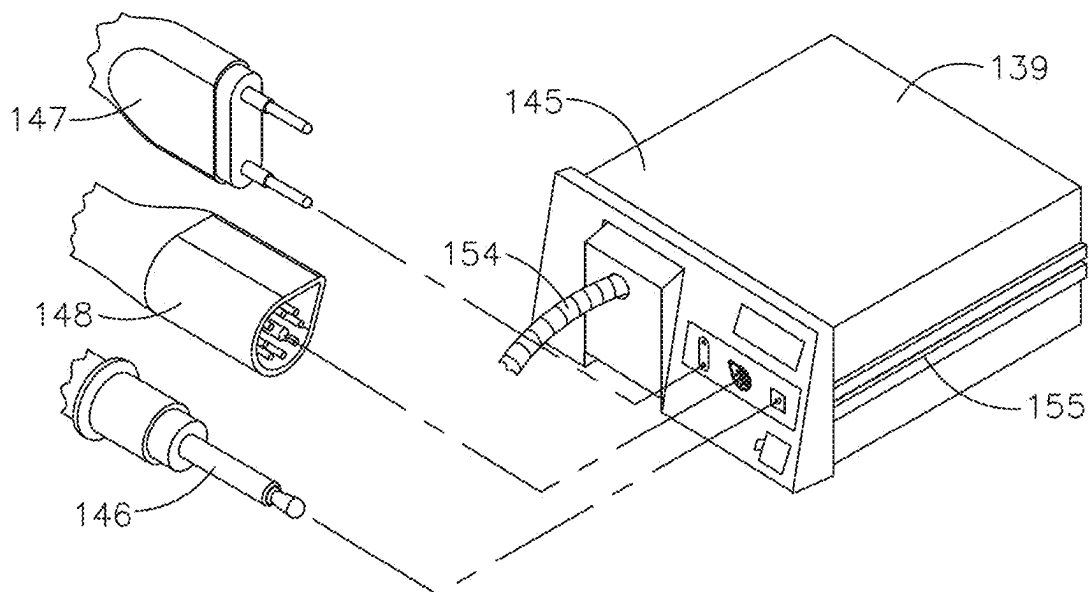


FIG. 5

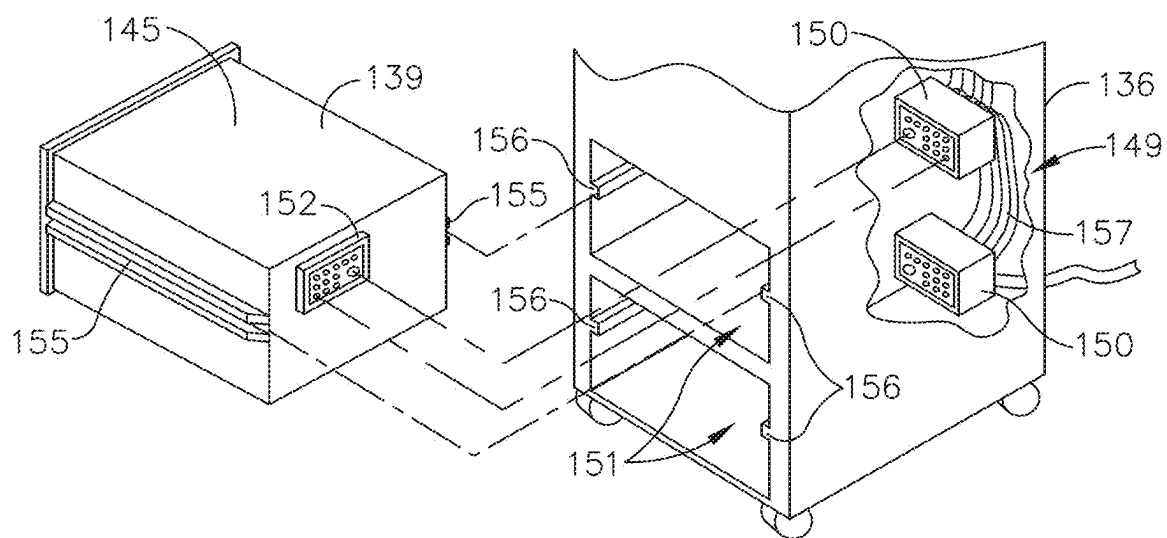


FIG. 4

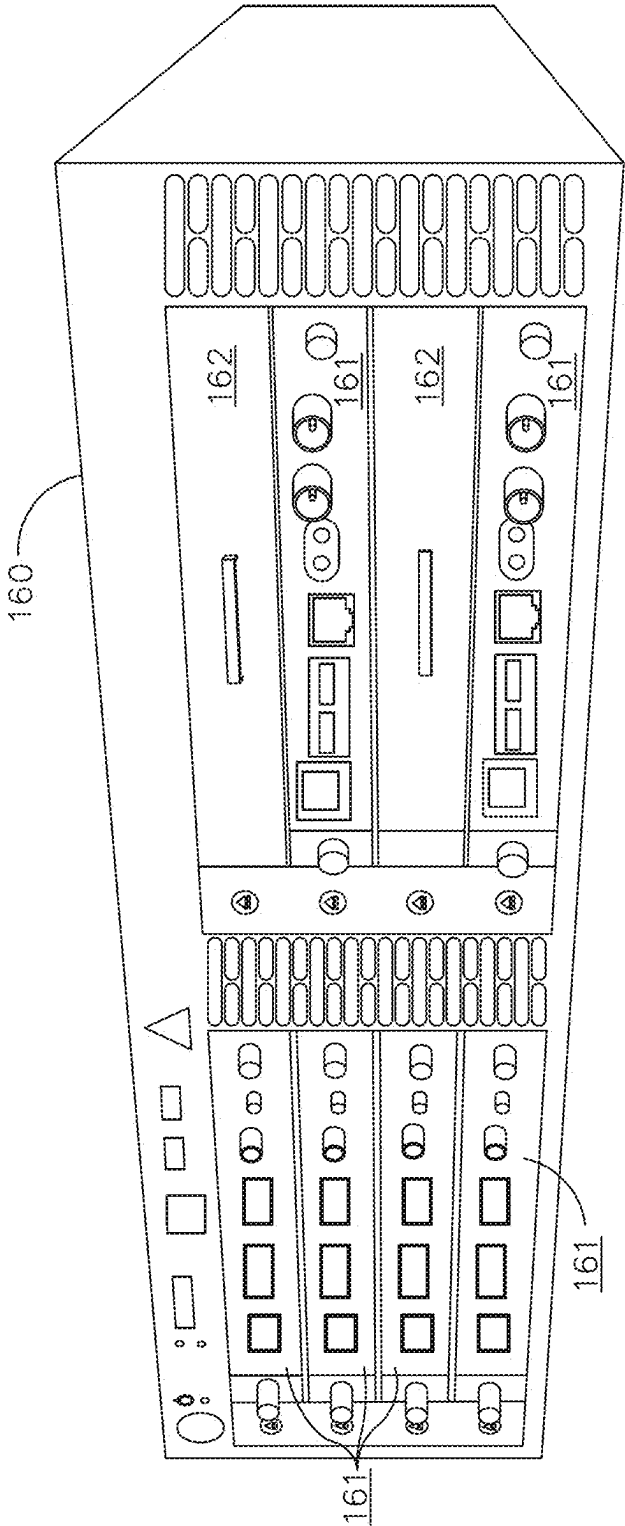


FIG. 6

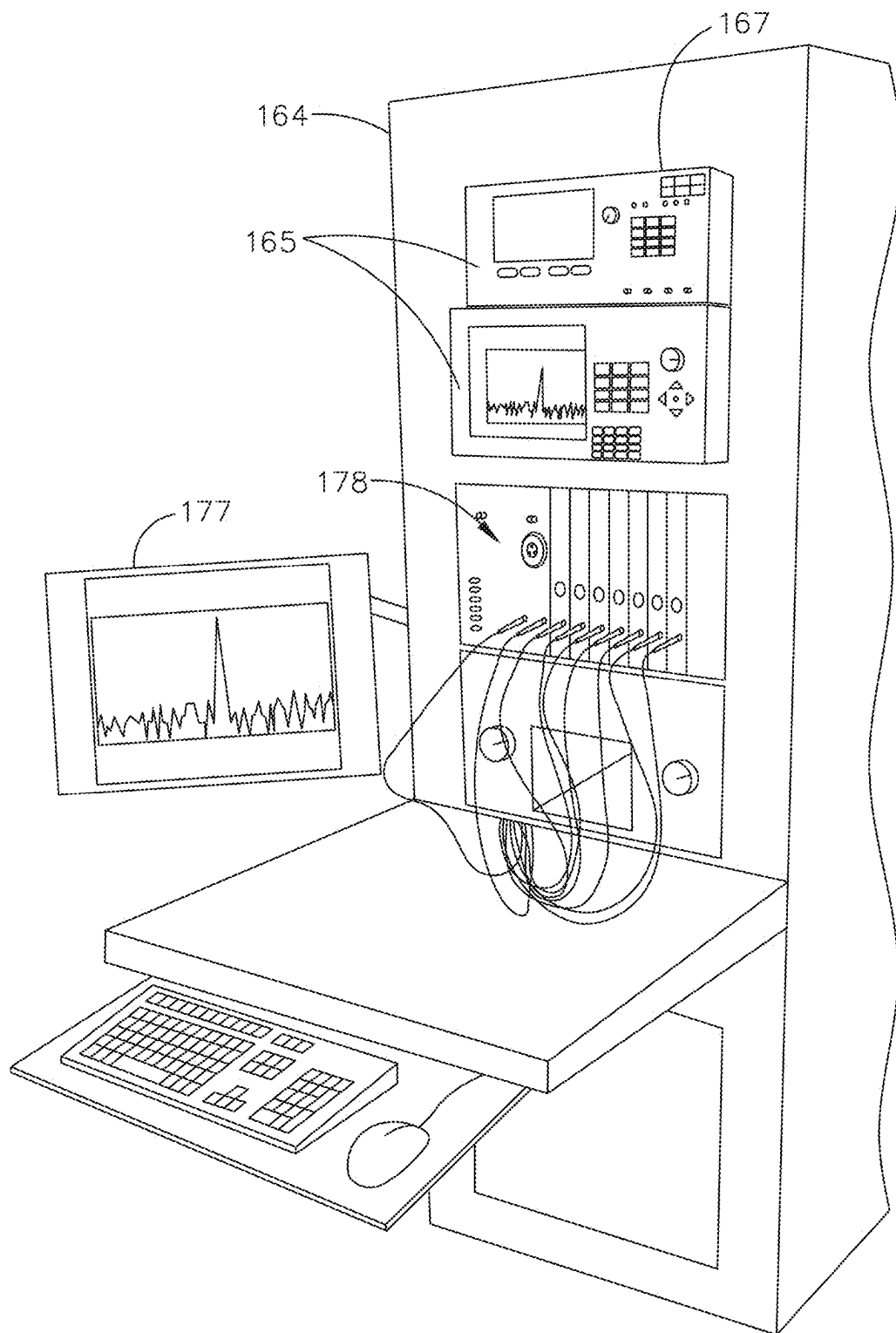


FIG. 7

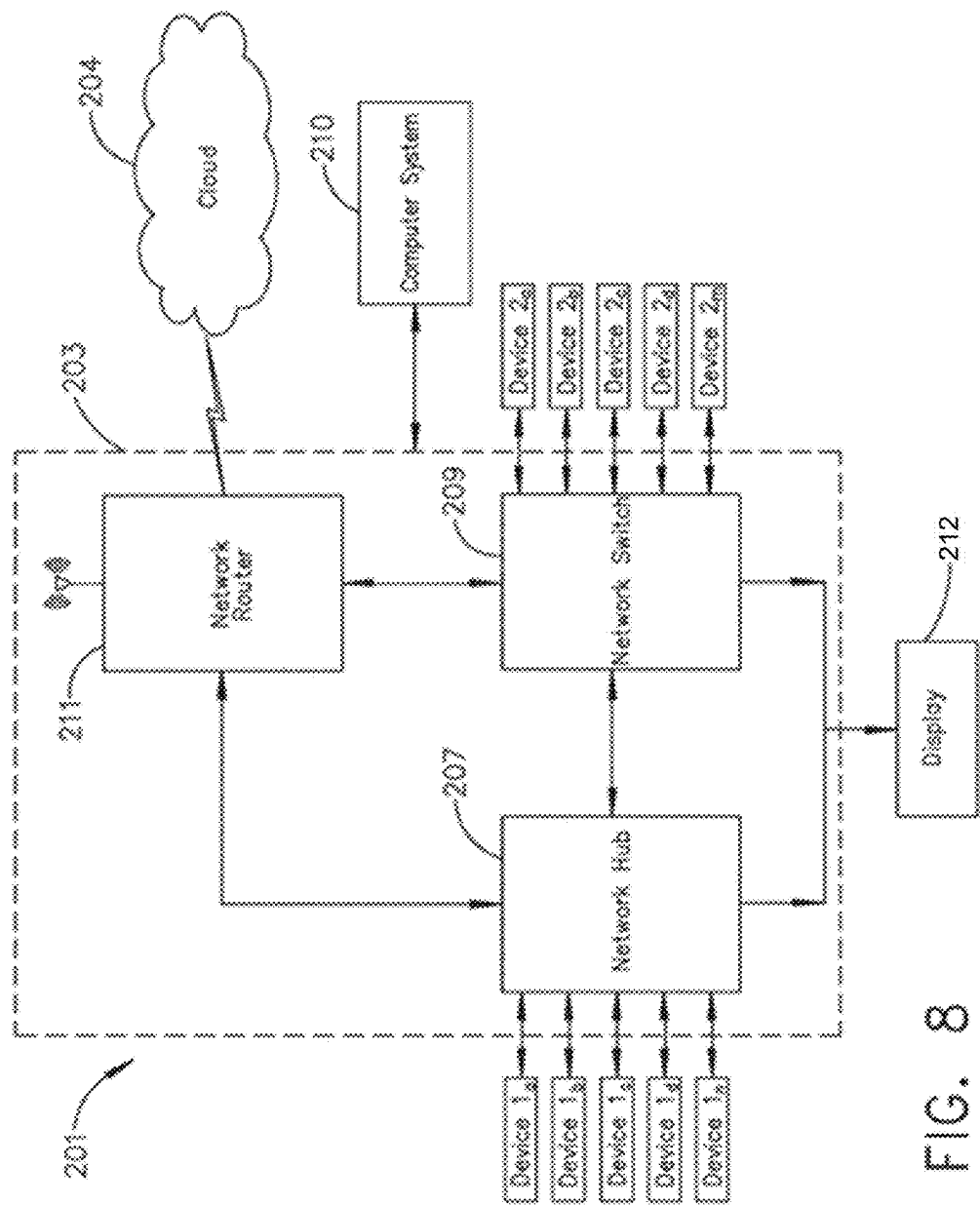


FIG. 8

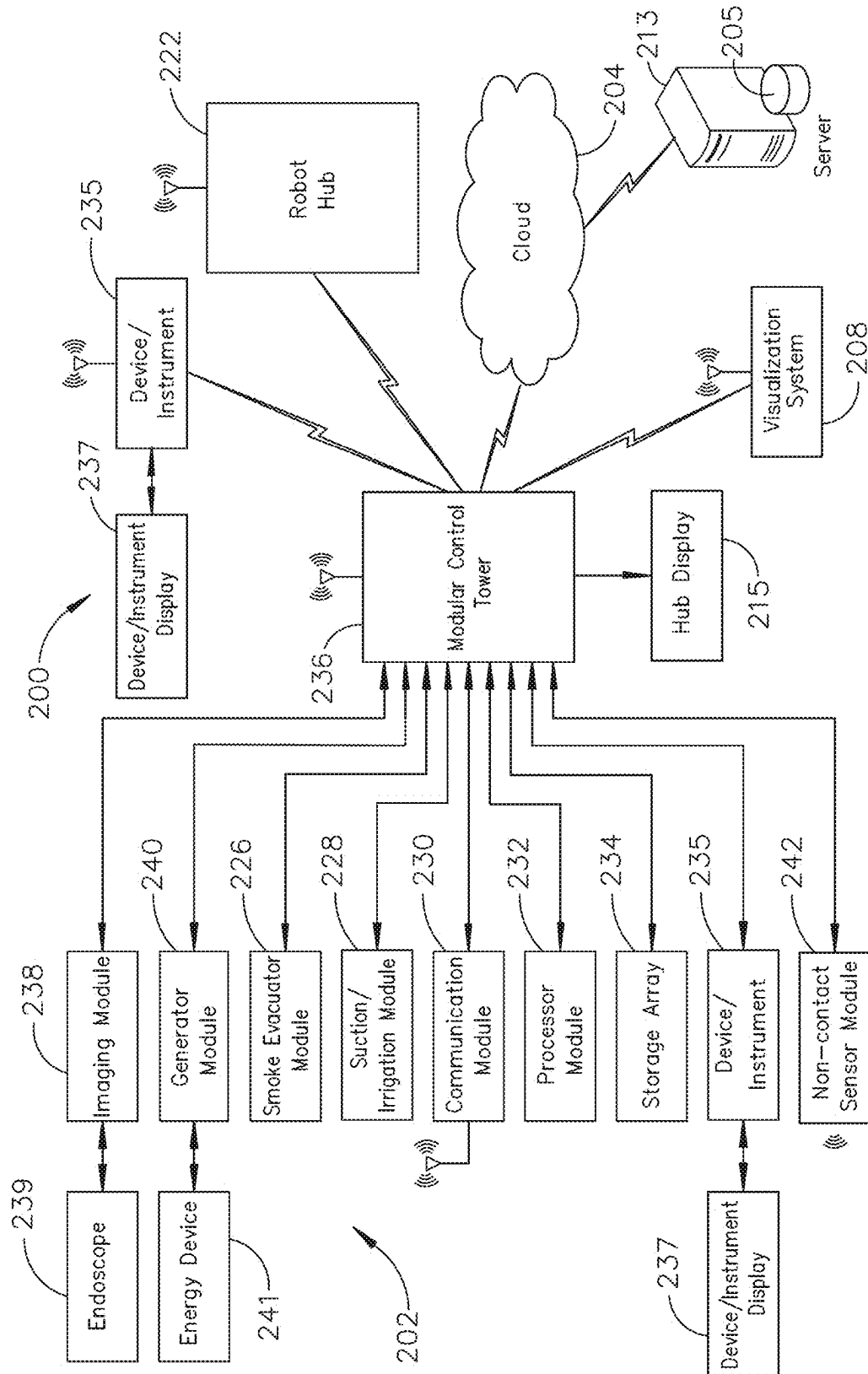


FIG. 9

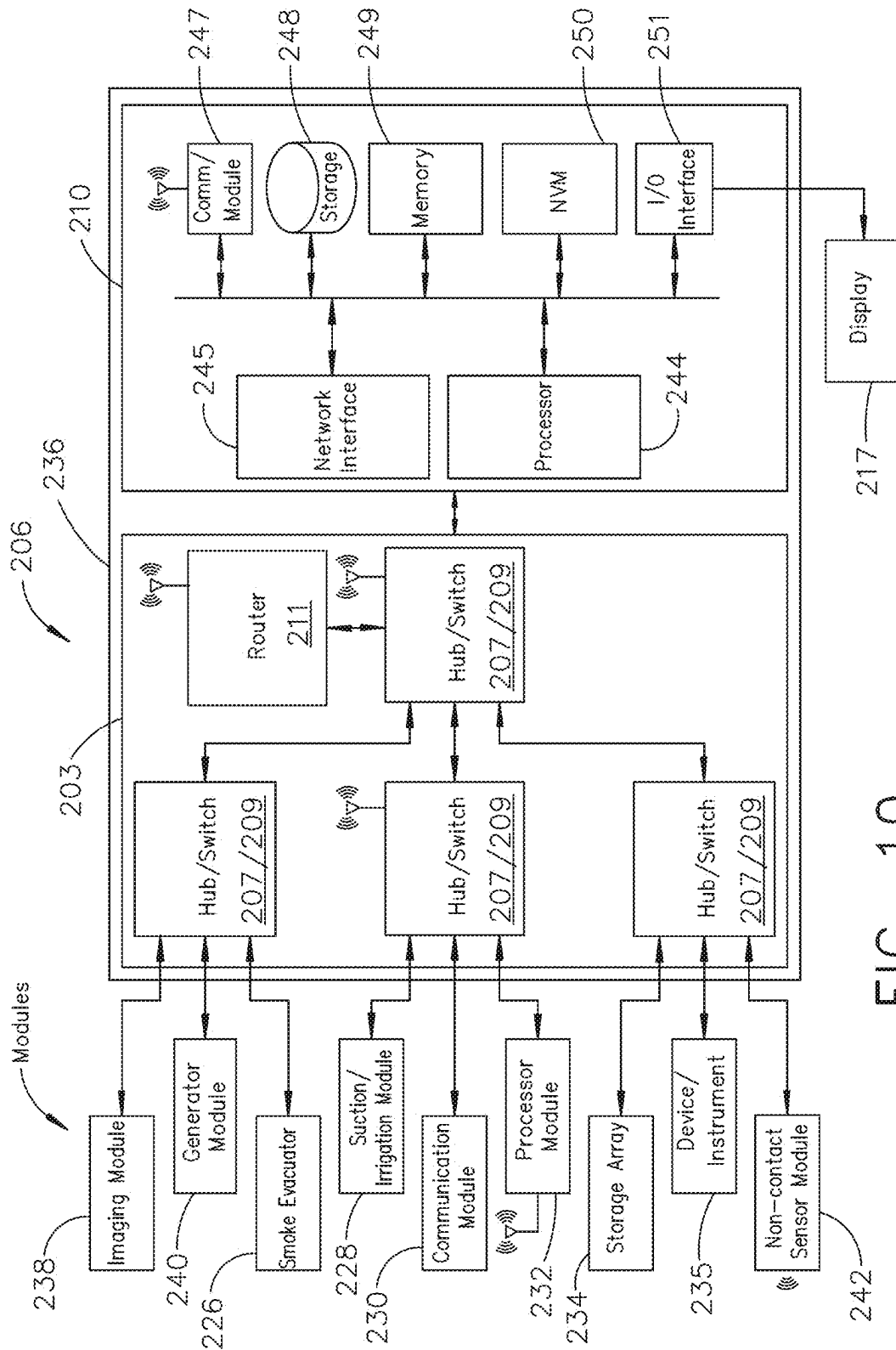


FIG. 10

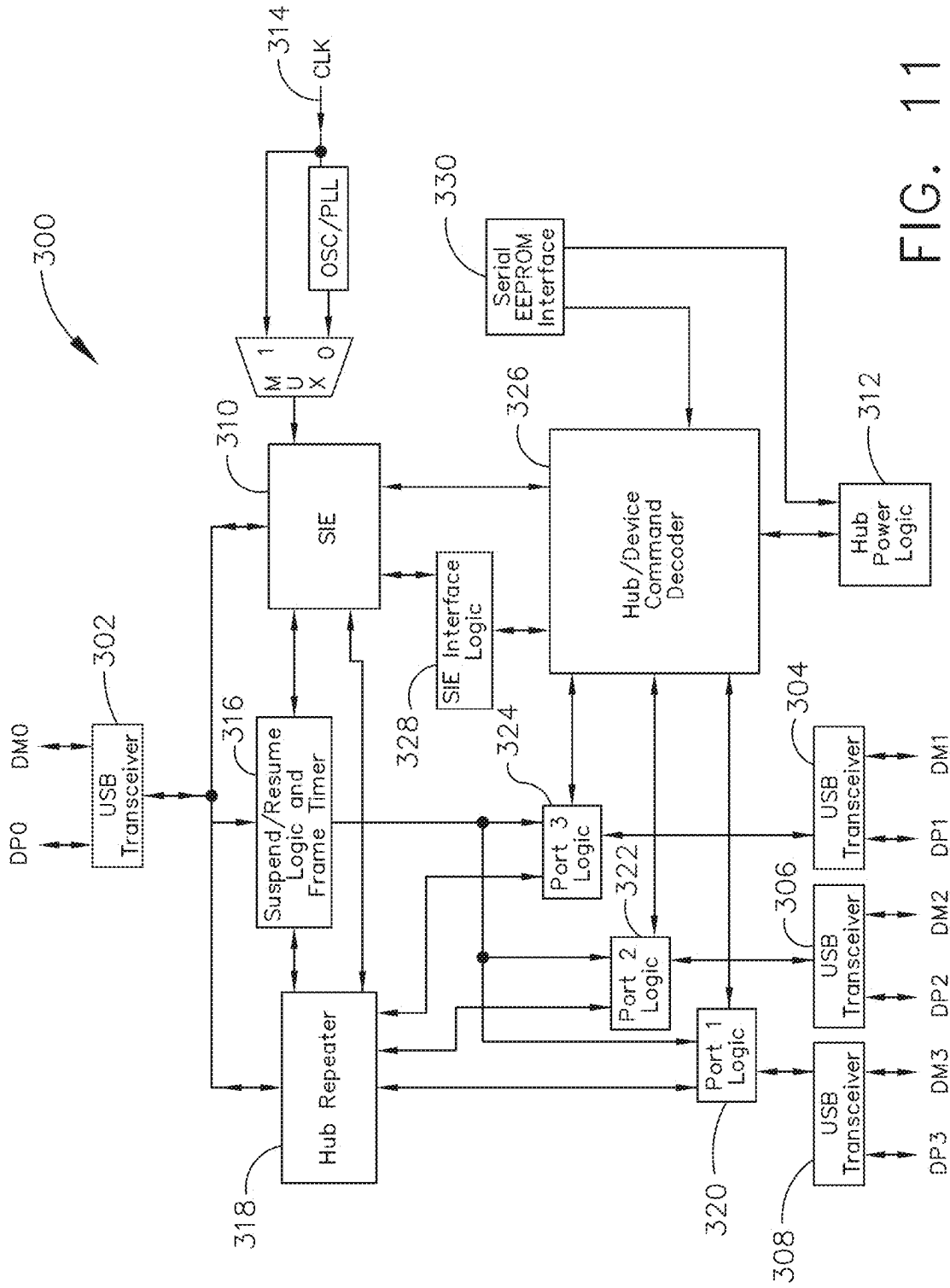


FIG. 11

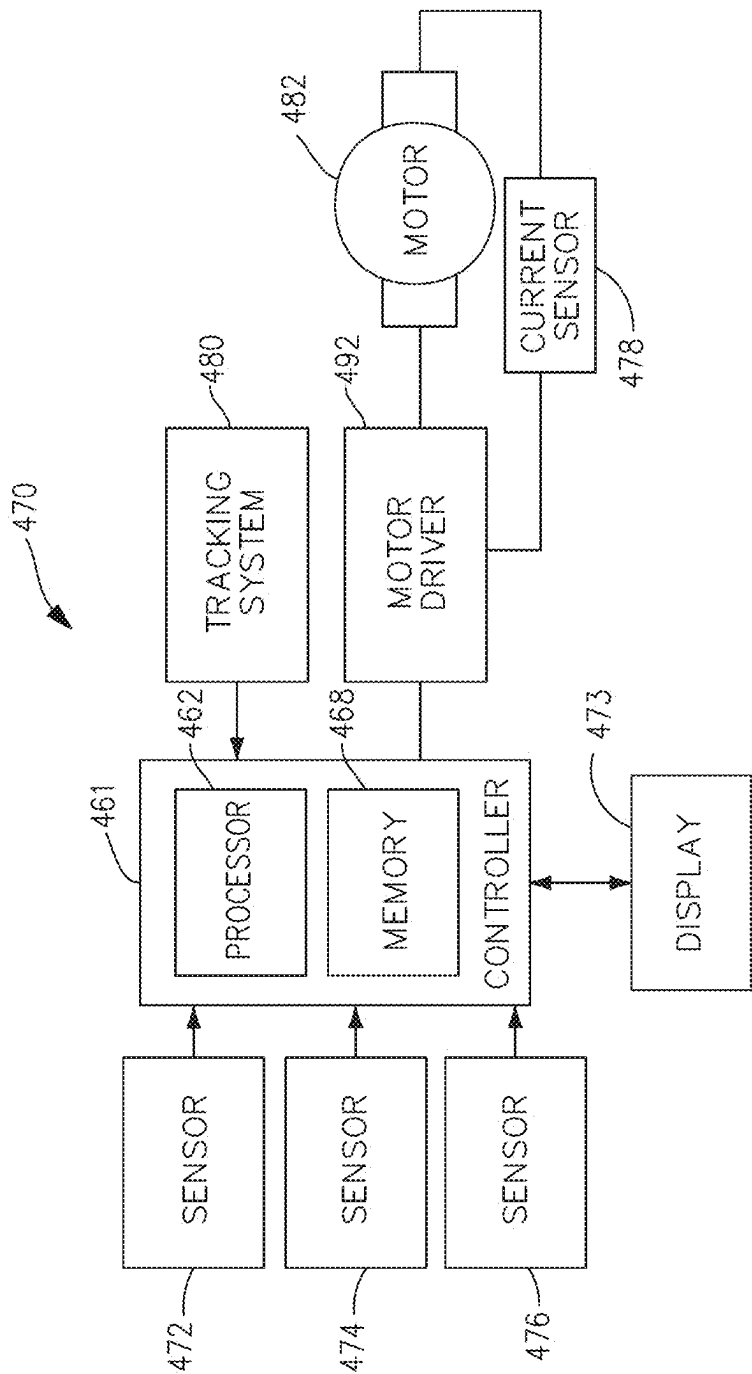


FIG. 12

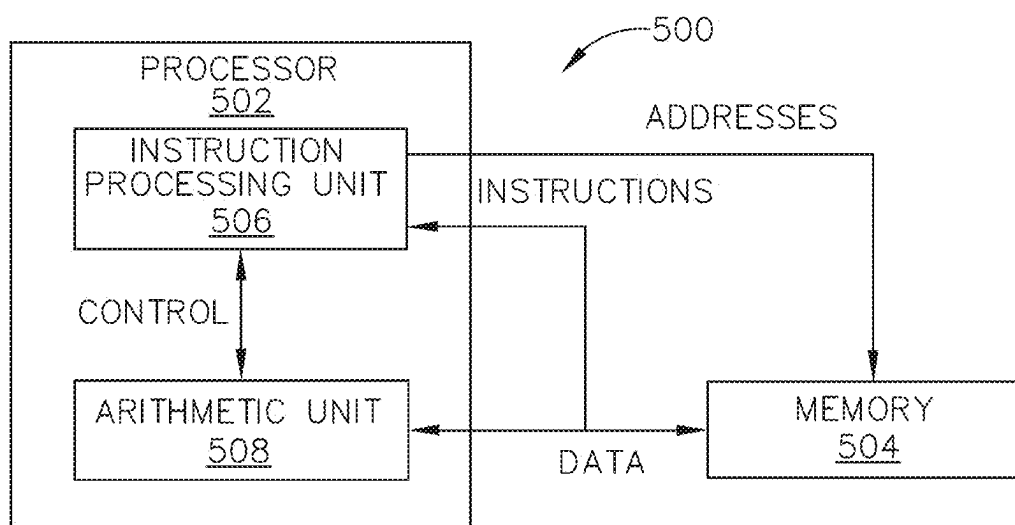


FIG. 13

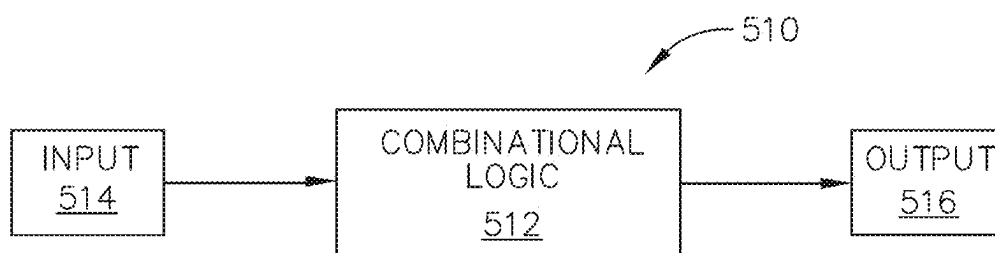


FIG. 14

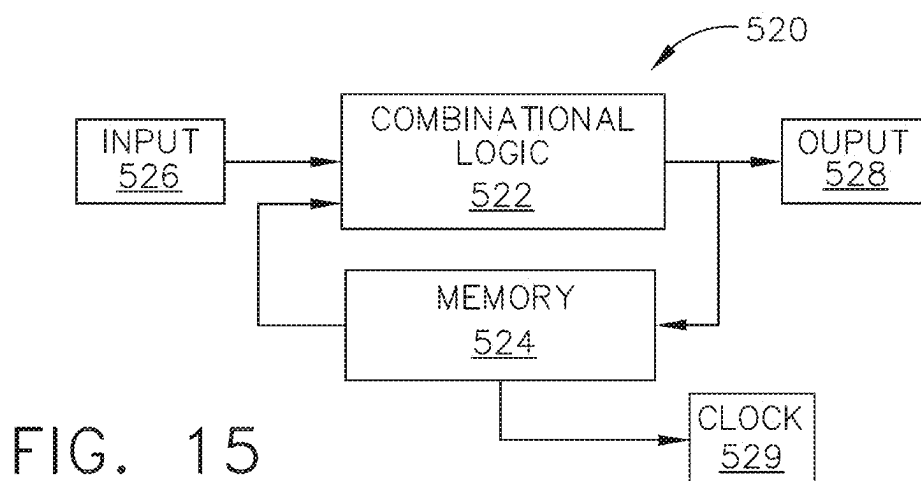


FIG. 15

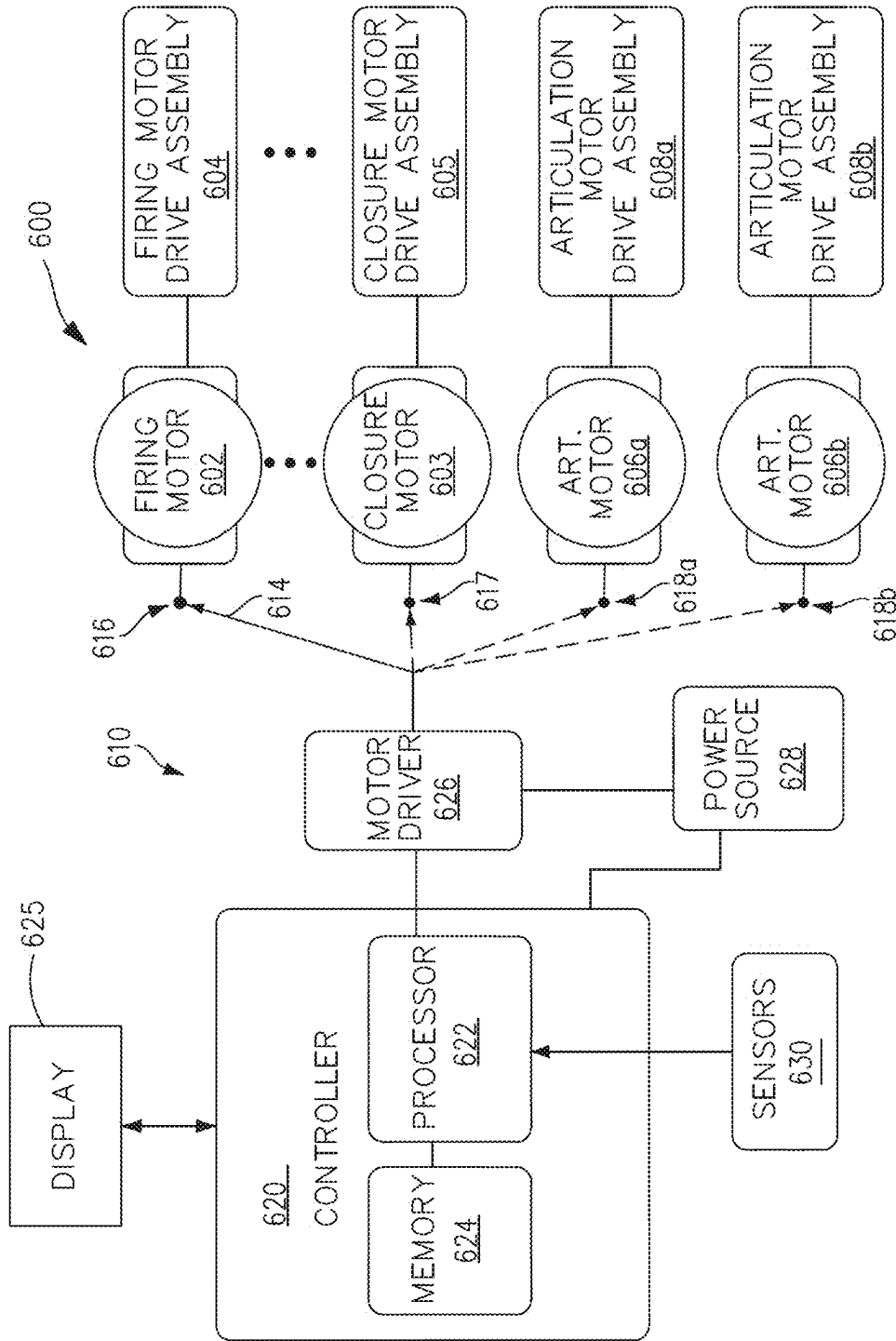


FIG. 16

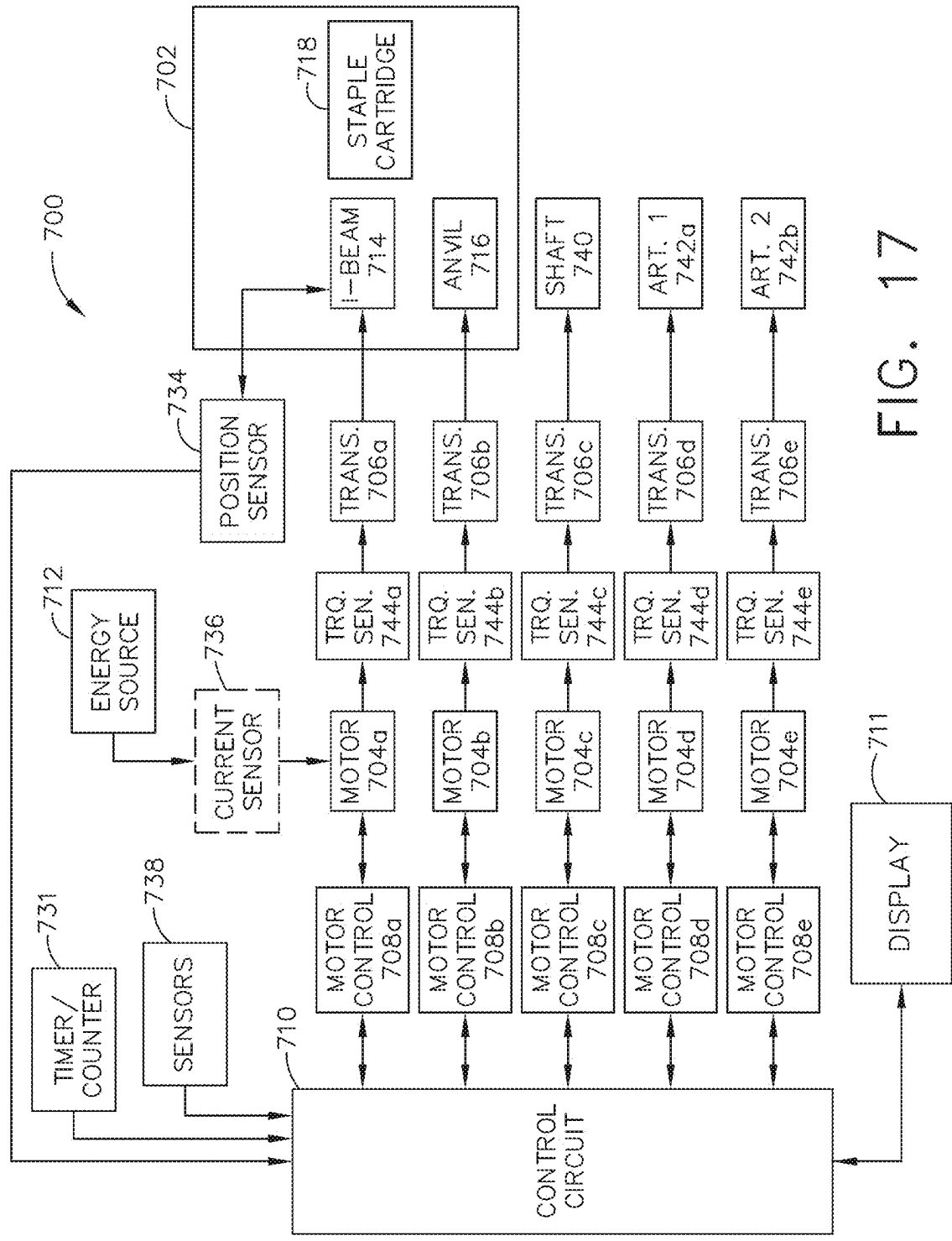


FIG. 17

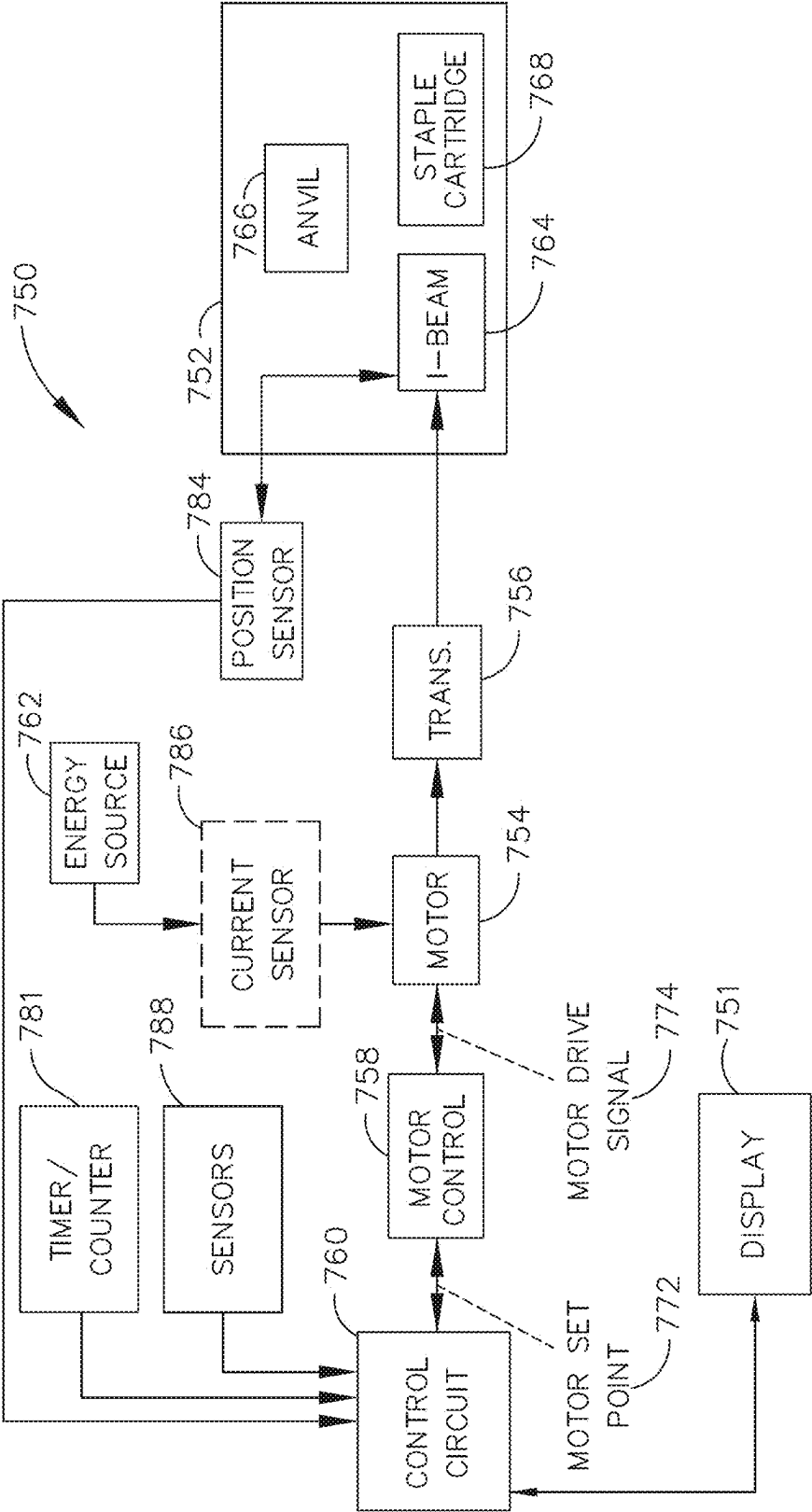


FIG. 18

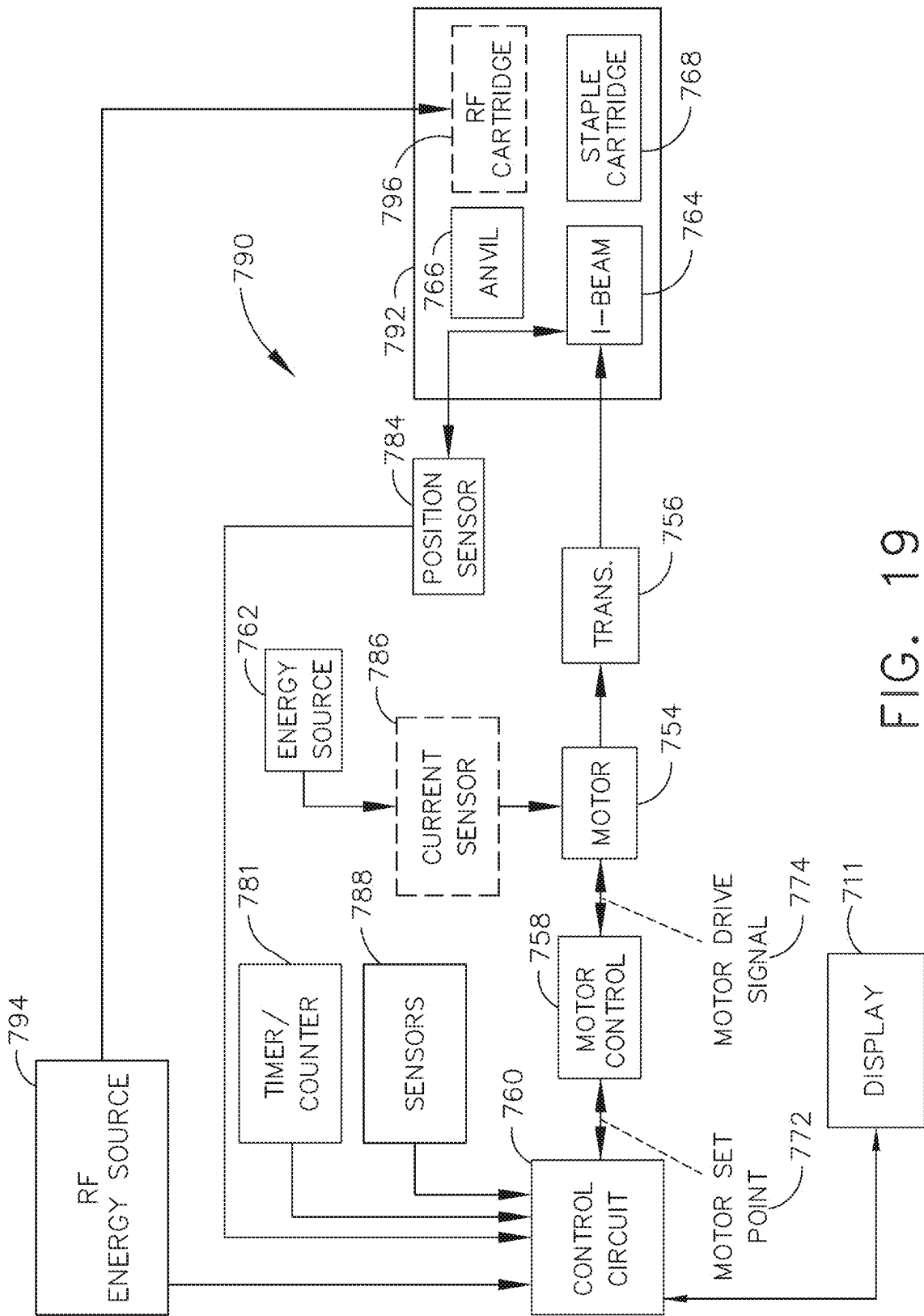


FIG. 19

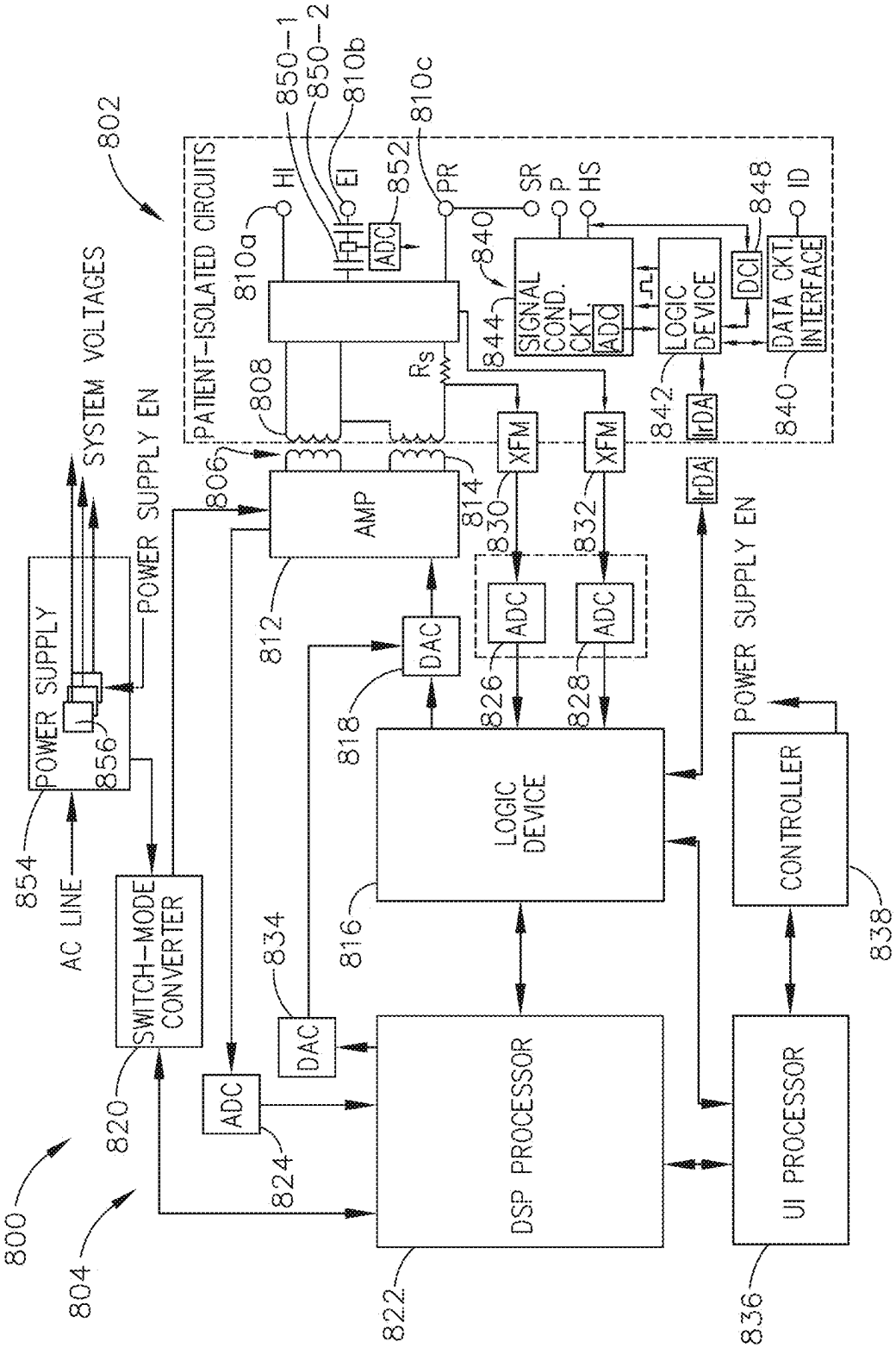


FIG. 20

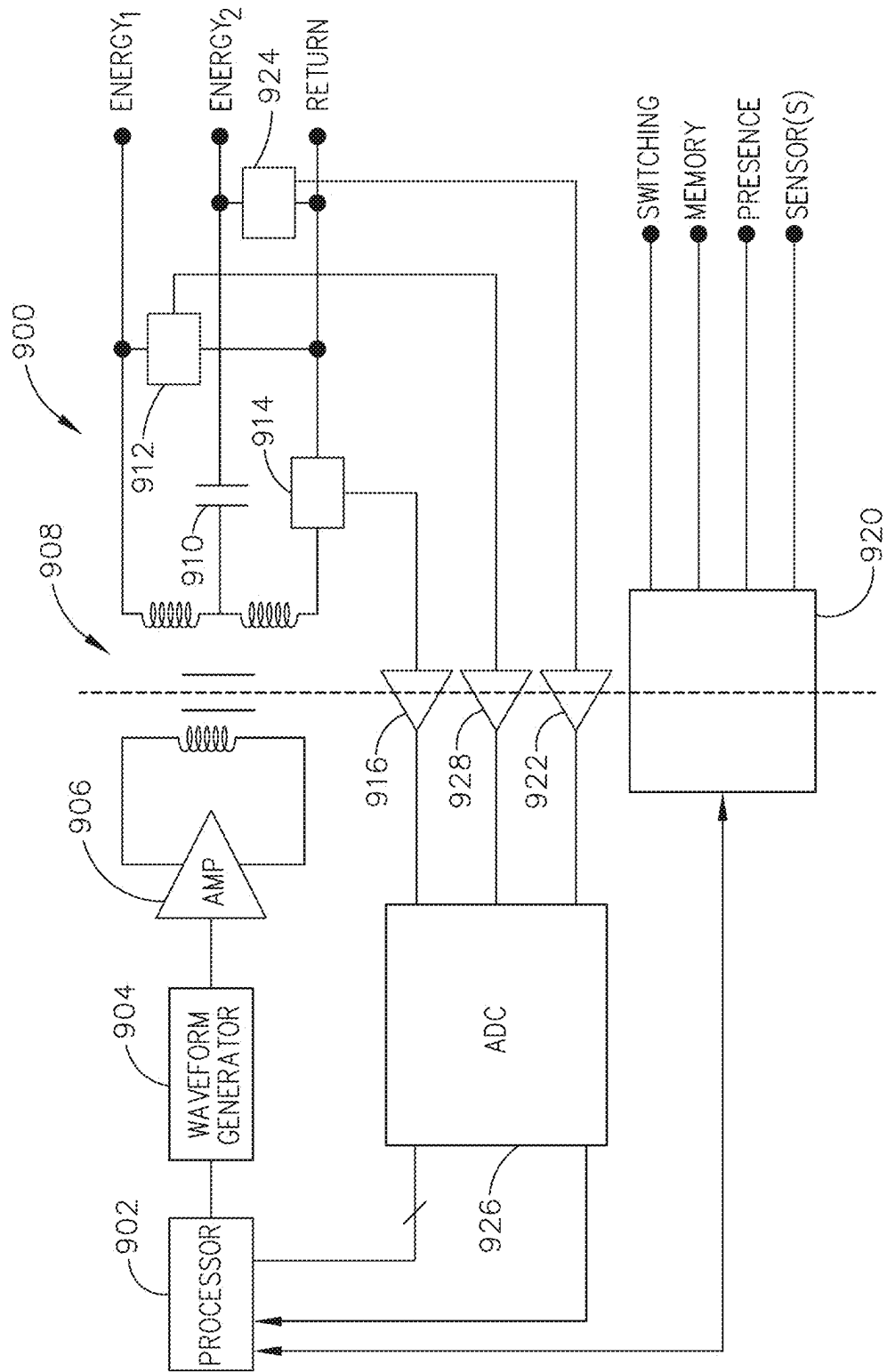


FIG. 21

COMPUTER IMPLEMENTED INTERACTIVE SURGICAL SYSTEMS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation patent application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 17/203,884, titled COMPUTER IMPLEMENTED INTERACTIVE SURGICAL SYSTEMS, now U.S. Patent Application Publication No. 2021/0205029, which claims priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 15/940,629, titled COMPUTER IMPLEMENTED INTERACTIVE SURGICAL SYSTEMS, filed Mar. 29, 2018, now U.S. Patent Application Publication No. 2019/0201112, which claims the benefit of priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application Ser. No. 62/649,310, titled COMPUTER IMPLEMENTED INTERACTIVE SURGICAL SYSTEMS, filed Mar. 28, 2018, U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, U.S. Provisional Patent Application Ser. No. 62/611,340, titled CLOUD-BASED MEDICAL ANALYTICS, filed Dec. 28, 2017, and U.S. Provisional Patent Application Ser. No. 62/611,339, titled ROBOT ASSISTED SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of each of which is herein incorporated by reference in its entirety.

BACKGROUND

[0002] The present disclosure relates to various surgical systems. Surgical procedures are typically performed in surgical operating theaters or rooms in a healthcare facility such as, for example, a hospital. A sterile field is typically created around the patient. The sterile field may include the scrubbed team members, who are properly attired, and all furniture and fixtures in the area. Various surgical devices and systems are utilized in performance of a surgical procedure.

SUMMARY

[0003] In one general aspect, a surgical hub is provided. The surgical hub is for use with a surgical instrument configured to deliver therapeutic energy to tissue at a surgical site of a surgical procedure. The surgical hub comprises: a hub enclosure, comprising a docking station including a docking port comprising data and power contacts; and a combo generator module removably retainable in the docking station. The combo generator module comprises: an ultrasonic energy generator component; a radio frequency (RF) energy generator component; a smoke evacuation component; and a connection port. At least one of the ultrasonic energy generator component and the radio frequency (RF) generator component are couplable to the surgical instrument through the connection port. The combo generator module further comprises at least one smoke evacuation component, configured to evacuate smoke generated by an application of therapeutic energy to the tissue by the surgical instrument.

[0004] In another general aspect, another surgical hub is provided which is modular. The modular surgical hub is for use with a surgical instrument configured to deliver therapeutic energy to tissue at a surgical site of a surgical procedure. The modular surgical enclosure comprises: a first

energy-generator module configured to generate a first therapeutic energy for application to the tissue; a first docking station comprising a first docking port that includes first data and power contacts; and a second energy-generator module configured to generate a second therapeutic energy, different than the first therapeutic energy, for application to the tissue. The first energy-generator module is slidably movable into an electrical engagement with the first data and power contacts, and also slidably movable out of the electrical engagement with the first data and power contacts. The second docking station comprises a second docking port that includes second data and power contacts. The second energy-generator module is slidably movable into an electrical engagement with the second data and power contacts, and is also slidably movable out of the electrical engagement with the second data and power contacts. The modular surgical enclosure further comprises a communication bus between the first docking port and the second docking port configured to facilitate communication between the first energy-generator module and the second energy-generator module.

[0005] In yet another aspect, another surgical hub is provided. The surgical hub is for use with a surgical instrument configured to deliver therapeutic energy to tissue at a surgical site of a surgical procedure. The surgical hub comprises: a hub enclosure, comprising docking stations including docking ports comprising data and power contacts; and a combo generator module slidably receivable in a first of the docking stations. The combo generator module comprises: an ultrasonic energy generator component; a radio frequency (RF) energy generator component; and a connection port. At least one of the ultrasonic energy generator component and the radio frequency (RF) generator component are couplable to the surgical instrument through the connection port. The surgical hub further comprises: a smoke evacuation module slidably receivable in a second one of the docking stations, where the smoke evacuation module is configured to evacuate smoke generated by an application of the therapeutic energy to the tissue by the surgical instrument; a processing module slidably receivable in a third one of the docking stations; a memory module slidably receivable in a fourth one of the docking stations; and an operating-room mapping module slidably receivable in a fifth one of the docking stations.

FIGURES

[0006] The features of various aspects are set forth with particularity in the appended claims. The various aspects, however, both as to organization and methods of operation, together with further objects and advantages thereof, may best be understood by reference to the following description, taken in conjunction with the accompanying drawings as follows.

[0007] FIG. 1 is a block diagram of a computer-implemented interactive surgical system, in accordance with at least one aspect of the present disclosure.

[0008] FIG. 2 is a surgical system being used to perform a surgical procedure in an operating room, in accordance with at least one aspect of the present disclosure.

[0009] FIG. 3 is a surgical hub paired with a visualization system, a robotic system, and an intelligent instrument, in accordance with at least one aspect of the present disclosure.

[0010] FIG. 4 is a partial perspective view of a surgical hub enclosure, and of a combo generator module slidably

receivable in a drawer of the surgical hub enclosure, in accordance with at least one aspect of the present disclosure.

[0011] FIG. 5 is a perspective view of a combo generator module with bipolar, ultrasonic, and monopolar contacts and a smoke evacuation component, in accordance with at least one aspect of the present disclosure.

[0012] FIG. 6 illustrates individual power bus attachments for a plurality of lateral docking ports of a lateral modular housing configured to receive a plurality of modules, in accordance with at least one aspect of the present disclosure.

[0013] FIG. 7 illustrates a vertical modular housing configured to receive a plurality of modules, in accordance with at least one aspect of the present disclosure.

[0014] FIG. 8 illustrates a surgical data network comprising a modular communication hub configured to connect modular devices located in one or more operating theaters of a healthcare facility, or any room in a healthcare facility specially equipped for surgical operations, to the cloud, in accordance with at least one aspect of the present disclosure.

[0015] FIG. 9 illustrates a computer-implemented interactive surgical system, in accordance with at least one aspect of the present disclosure.

[0016] FIG. 10 illustrates a surgical hub comprising a plurality of modules coupled to the modular control tower, in accordance with at least one aspect of the present disclosure.

[0017] FIG. 11 illustrates one aspect of a Universal Serial Bus (USB) network hub device, in accordance with at least one aspect of the present disclosure.

[0018] FIG. 12 illustrates a logic diagram of a control system of a surgical instrument or tool, in accordance with at least one aspect of the present disclosure.

[0019] FIG. 13 illustrates a control circuit configured to control aspects of the surgical instrument or tool, in accordance with at least one aspect of the present disclosure.

[0020] FIG. 14 illustrates a combinational logic circuit configured to control aspects of the surgical instrument or tool, in accordance with at least one aspect of the present disclosure.

[0021] FIG. 15 illustrates a sequential logic circuit configured to control aspects of the surgical instrument or tool, in accordance with at least one aspect of the present disclosure.

[0022] FIG. 16 illustrates a surgical instrument or tool comprising a plurality of motors which can be activated to perform various functions, in accordance with at least one aspect of the present disclosure.

[0023] FIG. 17 is a schematic diagram of a robotic surgical instrument configured to operate a surgical tool described herein, in accordance with at least one aspect of the present disclosure.

[0024] FIG. 18 illustrates a block diagram of a surgical instrument programmed to control the distal translation of a displacement member, in accordance with at least one aspect of the present disclosure.

[0025] FIG. 19 is a schematic diagram of a surgical instrument configured to control various functions, in accordance with at least one aspect of the present disclosure.

[0026] FIG. 20 is a simplified block diagram of a generator configured to provide inductorless tuning, among other benefits, in accordance with at least one aspect of the present disclosure.

[0027] FIG. 21 illustrates an example of a generator, which is one form of the generator of FIG. 20, in accordance with at least one aspect of the present disclosure.

DESCRIPTION

[0028] Applicant of the present application owns the following U.S. Provisional Patent Applications, filed on Mar. 28, 2018, each of which is herein incorporated by reference in its entirety:

[0029] U.S. Provisional Patent Application Ser. No. 62/649,302, titled INTERACTIVE SURGICAL SYSTEMS WITH ENCRYPTED COMMUNICATION CAPABILITIES;

[0030] U.S. Provisional Patent Application Ser. No. 62/649,294, titled DATA STRIPPING METHOD TO INTERROGATE PATIENT RECORDS AND CREATE ANONYMIZED RECORD;

[0031] U.S. Provisional Patent Application Ser. No. 62/649,300, titled SURGICAL HUB SITUATIONAL AWARENESS;

[0032] U.S. Provisional Patent Application Ser. No. 62/649,309, titled SURGICAL HUB SPATIAL AWARENESS TO DETERMINE DEVICES IN OPERATING THEATER;

[0033] U.S. Provisional Patent Application Ser. No. 62/649,310, titled COMPUTER IMPLEMENTED INTERACTIVE SURGICAL SYSTEMS;

[0034] U.S. Provisional Patent Application Ser. No. 62/649,291, titled USE OF LASER LIGHT AND RED-GREEN-BLUE COLORATION TO DETERMINE PROPERTIES OF BACK SCATTERED LIGHT;

[0035] U.S. Provisional Patent Application Ser. No. 62/649,296, titled ADAPTIVE CONTROL PROGRAM UPDATES FOR SURGICAL DEVICES;

[0036] U.S. Provisional Patent Application Ser. No. 62/649,333, titled CLOUD-BASED MEDICAL ANALYTICS FOR CUSTOMIZATION AND RECOMMENDATIONS TO A USER;

[0037] U.S. Provisional Patent Application Ser. No. 62/649,327, titled CLOUD-BASED MEDICAL ANALYTICS FOR SECURITY AND AUTHENTICATION TRENDS AND REACTIVE MEASURES;

[0038] U.S. Provisional Patent Application Ser. No. 62/649,315, titled DATA HANDLING AND PRIORITIZATION IN A CLOUD ANALYTICS NETWORK;

[0039] U.S. Provisional Patent Application Ser. No. 62/649,313, titled CLOUD INTERFACE FOR COUPLED SURGICAL DEVICES;

[0040] U.S. Provisional Patent Application Ser. No. 62/649,320, titled DRIVE ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS;

[0041] U.S. Provisional Patent Application Ser. No. 62/649,307, titled AUTOMATIC TOOL ADJUSTMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS; and

[0042] U.S. Provisional Patent Application Ser. No. 62/649,323, titled SENSING ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS.

[0043] Applicant of the present application owns the following U.S. Patent Applications, filed on Mar. 29, 2018, each of which is herein incorporated by reference in its entirety:

[0044] U.S. patent application Ser. No. 15/940,641, titled INTERACTIVE SURGICAL SYSTEMS WITH

- ENCRYPTED COMMUNICATION CAPABILITIES, now U.S. Pat. No. 10,944,728;
- [0045] U.S. patent application Ser. No. 15/940,648, titled INTERACTIVE SURGICAL SYSTEMS WITH CONDITION HANDLING OF DEVICES AND DATA CAPABILITIES, now U.S. Patent Application Publication No. 2019/0206004;
- [0046] U.S. patent application Ser. No. 15/940,656, titled SURGICAL HUB COORDINATION OF CONTROL AND COMMUNICATION OF OPERATING ROOM DEVICES, now U.S. Patent Application Publication No. 2019/0201141;
- [0047] U.S. patent application Ser. No. 15/940,666, titled SPATIAL AWARENESS OF SURGICAL HUBS IN OPERATING ROOMS, now U.S. Patent Application Publication No. 2019/0206551;
- [0048] U.S. patent application Ser. No. 15/940,670, titled COOPERATIVE UTILIZATION OF DATA DERIVED FROM SECONDARY SOURCES BY INTELLIGENT SURGICAL HUBS, now U.S. Patent Application Publication No. 2019/0201116;
- [0049] U.S. patent application Ser. No. 15/940,677, titled SURGICAL HUB CONTROL ARRANGEMENTS, now U.S. Patent Application Publication No. 2019/0201143;
- [0050] U.S. patent application Ser. No. 15/940,632, titled DATA STRIPPING METHOD TO INTERROGATE PATIENT RECORDS AND CREATE ANONYMIZED RECORD, now U.S. Patent Application Publication No. 2019/0205566;
- [0051] U.S. patent application Ser. No. 15/940,640, titled COMMUNICATION HUB AND STORAGE DEVICE FOR STORING PARAMETERS AND STATUS OF A SURGICAL DEVICE TO BE SHARED WITH CLOUD BASED ANALYTICS SYSTEMS, now U.S. Patent Application Publication No. 2019/0200863;
- [0052] U.S. patent application Ser. No. 15/940,645, titled SELF DESCRIBING DATA PACKETS GENERATED AT AN ISSUING INSTRUMENT, now U.S. Pat. No. 10,892,899;
- [0053] U.S. patent application Ser. No. 15/940,649, titled DATA PAIRING TO INTERCONNECT A DEVICE MEASURED PARAMETER WITH AN OUTCOME, now U.S. Patent Application Publication No. 2019/0205567;
- [0054] U.S. patent application Ser. No. 15/940,654, titled SURGICAL HUB SITUATIONAL AWARENESS, now U.S. Patent Application Publication No. 2019/0201140;
- [0055] U.S. patent application Ser. No. 15/940,663, titled SURGICAL SYSTEM DISTRIBUTED PROCESSING, now U.S. Patent Application Publication No. 2019/0201033;
- [0056] U.S. patent application Ser. No. 15/940,668, titled AGGREGATION AND REPORTING OF SURGICAL HUB DATA, now U.S. Patent Application Publication No. 2019/0201115;
- [0057] U.S. patent application Ser. No. 15/940,671, titled SURGICAL HUB SPATIAL AWARENESS TO DETERMINE DEVICES IN OPERATING THEATER, now U.S. Patent Application Publication No. 2019/0201104;
- [0058] U.S. patent application Ser. No. 15/940,686, titled DISPLAY OF ALIGNMENT OF STAPLE CARTRIDGE TO PRIOR LINEAR STAPLE LINE, now U.S. Patent Application Publication No. 2019/0201105;
- [0059] U.S. patent application Ser. No. 15/940,700, titled STERILE FIELD INTERACTIVE CONTROL DISPLAYS, now U.S. Patent Application Publication No. 2019/0205001;
- [0060] U.S. patent application Ser. No. 15/940,704, titled USE OF LASER LIGHT AND RED-GREEN-BLUE COLORATION TO DETERMINE PROPERTIES OF BACK SCATTERED LIGHT, now U.S. Patent Application Publication No. 2019/0206050;
- [0061] U.S. patent application Ser. No. 15/940,722, titled CHARACTERIZATION OF TISSUE IRREGULARITIES THROUGH THE USE OF MONO-CHROMATIC LIGHT REFRACTIVITY, now U.S. Patent Application Publication No. 2019/0200905; and
- [0062] U.S. patent application Ser. No. 15/940,742, titled DUAL CMOS ARRAY IMAGING, now U.S. Patent Application Publication No. 2019/0200906.
- [0063] Applicant of the present application owns the following U.S. patent applications, filed on Mar. 29, 2018, each of which is herein incorporated by reference in its entirety:
- [0064] U.S. patent application Ser. No. 15/940,627, titled DRIVE ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201111;
- [0065] U.S. patent application Ser. No. 15/940,637, titled COMMUNICATION ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201139;
- [0066] U.S. patent application Ser. No. 15/940,642, titled CONTROLS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201113;
- [0067] U.S. patent application Ser. No. 15/940,676, titled AUTOMATIC TOOL ADJUSTMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201142;
- [0068] U.S. patent application Ser. No. 15/940,680, titled CONTROLLERS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201135;
- [0069] U.S. patent application Ser. No. 15/940,683, titled COOPERATIVE SURGICAL ACTIONS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201145;
- [0070] U.S. patent application Ser. No. 15/940,690, titled DISPLAY ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201118; and
- [0071] U.S. patent application Ser. No. 15/940,711, titled SENSING ARRANGEMENTS FOR ROBOT-ASSISTED SURGICAL PLATFORMS, now U.S. Patent Application Publication No. 2019/0201120.
- [0072] Before explaining various aspects of surgical devices and generators in detail, it should be noted that the illustrative examples are not limited in application or use to the details of construction and arrangement of parts illustrated in the accompanying drawings and description. The illustrative examples may be implemented or incorporated in other aspects, variations and modifications, and may be practiced or carried out in various ways. Further, unless

otherwise indicated, the terms and expressions employed herein have been chosen for the purpose of describing the illustrative examples for the convenience of the reader and are not for the purpose of limitation thereof. Also, it will be appreciated that one or more of the following-described aspects, expressions of aspects, and/or examples, can be combined with any one or more of the other following-described aspects, expressions of aspects and/or examples.

[0073] Referring to FIG. 1, a computer-implemented interactive surgical system 100 includes one or more surgical systems 102 and a cloud-based system (e.g., the cloud 104 that may include a remote server 113 coupled to a storage device 105). Each surgical system 102 includes at least one surgical hub 106 in communication with the cloud 104 that may include a remote server 113. In one example, as illustrated in FIG. 1, the surgical system 102 includes a visualization system 108, a robotic system 110, and a handheld intelligent surgical instrument 112, which are configured to communicate with one another and/or the hub 106. In some aspects, a surgical system 102 may include an M number of hubs 106, an N number of visualization systems 108, an O number of robotic systems 110, and a P number of handheld intelligent surgical instruments 112, where M, N, O, and P are integers greater than or equal to one.

[0074] FIG. 3 depicts an example of a surgical system 102 being used to perform a surgical procedure on a patient who is lying down on an operating table 114 in a surgical operating room 116. A robotic system 110 is used in the surgical procedure as a part of the surgical system 102. The robotic system 110 includes a surgeon's console 118, a patient side cart 120 (surgical robot), and a surgical robotic hub 122. The patient side cart 120 can manipulate at least one removably coupled surgical tool 117 through a minimally invasive incision in the body of the patient while the surgeon views the surgical site through the surgeon's console 118. An image of the surgical site can be obtained by a medical imaging device 124, which can be manipulated by the patient side cart 120 to orient the imaging device 124. The robotic hub 122 can be used to process the images of the surgical site for subsequent display to the surgeon through the surgeon's console 118.

[0075] Other types of robotic systems can be readily adapted for use with the surgical system 102. Various examples of robotic systems and surgical tools that are suitable for use with the present disclosure are described in U.S. Provisional Patent Application Ser. No. 62/611,339, titled ROBOT ASSISTED SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety.

[0076] Various examples of cloud-based analytics that are performed by the cloud 104, and are suitable for use with the present disclosure, are described in U.S. Provisional Patent Application Ser. No. 62/611,340, titled CLOUD-BASED MEDICAL ANALYTICS, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety.

[0077] In various aspects, the imaging device 124 includes at least one image sensor and one or more optical components. Suitable image sensors include, but are not limited to, Charge-Coupled Device (CCD) sensors and Complementary Metal-Oxide Semiconductor (CMOS) sensors.

[0078] The optical components of the imaging device 124 may include one or more illumination sources and/or one or

more lenses. The one or more illumination sources may be directed to illuminate portions of the surgical field. The one or more image sensors may receive light reflected or refracted from the surgical field, including light reflected or refracted from tissue and/or surgical instruments.

[0079] The one or more illumination sources may be configured to radiate electromagnetic energy in the visible spectrum as well as the invisible spectrum. The visible spectrum, sometimes referred to as the optical spectrum or luminous spectrum, is that portion of the electromagnetic spectrum that is visible to (i.e., can be detected by) the human eye and may be referred to as visible light or simply light. A typical human eye will respond to wavelengths in air that are from about 380 nm to about 750 nm.

[0080] The invisible spectrum (i.e., the non-luminous spectrum) is that portion of the electromagnetic spectrum that lies below and above the visible spectrum (i.e., wavelengths below about 380 nm and above about 750 nm). The invisible spectrum is not detectable by the human eye. Wavelengths greater than about 750 nm are longer than the red visible spectrum, and they become invisible infrared (IR), microwave, and radio electromagnetic radiation. Wavelengths less than about 380 nm are shorter than the violet spectrum, and they become invisible ultraviolet, x-ray, and gamma ray electromagnetic radiation.

[0081] In various aspects, the imaging device 124 is configured for use in a minimally invasive procedure. Examples of imaging devices suitable for use with the present disclosure include, but not limited to, an arthroscope, angioscope, bronchoscope, choledochoscope, colonoscope, cytoscope, duodenoscope, enteroscope, esophago-gastro-duodenoscope (gastroscope), endoscope, laryngoscope, nasopharyngo-neproscope, sigmoidoscope, thoracoscope, and ureterscope.

[0082] In one aspect, the imaging device employs multi-spectrum monitoring to discriminate topography and underlying structures. A multi-spectral image is one that captures image data within specific wavelength ranges across the electromagnetic spectrum. The wavelengths may be separated by filters or by the use of instruments that are sensitive to particular wavelengths, including light from frequencies beyond the visible light range, e.g., IR and ultraviolet. Spectral imaging can allow extraction of additional information the human eye fails to capture with its receptors for red, green, and blue. The use of multi-spectral imaging is described in greater detail under the heading "Advanced Imaging Acquisition Module" in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety. Multi-spectrum monitoring can be a useful tool in relocating a surgical field after a surgical task is completed to perform one or more of the previously described tests on the treated tissue.

[0083] It is axiomatic that strict sterilization of the operating room and surgical equipment is required during any surgery. The strict hygiene and sterilization conditions required in a "surgical theater," i.e., an operating or treatment room, necessitate the highest possible sterility of all medical devices and equipment. Part of that sterilization process is the need to sterilize anything that comes in contact with the patient or penetrates the sterile field, including the imaging device 124 and its attachments and components. It will be appreciated that the sterile field may be considered

a specified area, such as within a tray or on a sterile towel, that is considered free of microorganisms, or the sterile field may be considered an area, immediately around a patient, who has been prepared for a surgical procedure. The sterile field may include the scrubbed team members, who are properly attired, and all furniture and fixtures in the area.

[0084] In various aspects, the visualization system **108** includes one or more imaging sensors, one or more image processing units, one or more storage arrays, and one or more displays that are strategically arranged with respect to the sterile field, as illustrated in FIG. 2. In one aspect, the visualization system **108** includes an interface for HL7, PACS, and EMR. Various components of the visualization system **108** are described under the heading “Advanced Imaging Acquisition Module” in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety.

[0085] As illustrated in FIG. 2, a primary display **119** is positioned in the sterile field to be visible to an operator at the operating table **114**. In addition, a visualization tower **111** is positioned outside the sterile field. The visualization tower **111** includes a first non-sterile display **107** and a second non-sterile display **109**, which face away from each other. The visualization system **108**, guided by the hub **106**, is configured to utilize the displays **107**, **109**, and **119** to coordinate information flow to operators inside and outside the sterile field. For example, the hub **106** may cause the visualization system **108** to display a snap-shot of a surgical site, as recorded by an imaging device **124**, on a non-sterile display **107** or **109**, while maintaining a live feed of the surgical site on the primary display **119**. The snap-shot on the non-sterile display **107** or **109** can permit a non-sterile operator to perform a diagnostic step relevant to the surgical procedure, for example.

[0086] In one aspect, the hub **106** is also configured to route a diagnostic input or feedback entered by a non-sterile operator at the visualization tower **111** to the primary display **119** within the sterile field, where it can be viewed by a sterile operator at the operating table. In one example, the input can be in the form of a modification to the snap-shot displayed on the non-sterile display **107** or **109**, which can be routed to the primary display **119** by the hub **106**.

[0087] Referring to FIG. 2, a surgical instrument **112** is being used in the surgical procedure as part of the surgical system **102**. The hub **106** is also configured to coordinate information flow to a display of the surgical instrument **112**. For example, in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety. A diagnostic input or feedback entered by a non-sterile operator at the visualization tower **111** can be routed by the hub **106** to the surgical instrument display **115** within the sterile field, where it can be viewed by the operator of the surgical instrument **112**. Example surgical instruments that are suitable for use with the surgical system **102** are described under the heading “Surgical Instrument Hardware” and in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, the disclosure of which is herein incorporated by reference in its entirety, for example.

[0088] Referring now to FIG. 3, a hub **106** is depicted in communication with a visualization system **108**, a robotic system **110**, and a handheld intelligent surgical instrument **112**. The hub **106** includes a hub display **135**, an imaging module **138**, a generator module **140**, a communication module **130**, a processor module **132**, and a storage array **134**. In certain aspects, as illustrated in FIG. 3, the hub **106** further includes a smoke evacuation module **126** and/or a suction/irrigation module **128**.

[0089] During a surgical procedure, energy application to tissue, for sealing and/or cutting, is generally associated with smoke evacuation, suction of excess fluid, and/or irrigation of the tissue. Fluid, power, and/or data lines from different sources are often entangled during the surgical procedure. Valuable time can be lost addressing this issue during a surgical procedure. Detangling the lines may necessitate disconnecting the lines from their respective modules, which may require resetting the modules. The hub modular enclosure **136** offers a unified environment for managing the power, data, and fluid lines, which reduces the frequency of entanglement between such lines.

[0090] Aspects of the present disclosure present a surgical hub for use in a surgical procedure that involves energy application to tissue at a surgical site. The surgical hub includes a hub enclosure and a combo generator module slidably receivable in a docking station of the hub enclosure. The docking station includes data and power contacts. The combo generator module includes two or more of an ultrasonic energy generator component, a bipolar RF energy generator component, and a monopolar RF energy generator component that are housed in a single unit. In one aspect, the combo generator module also includes a smoke evacuation component, at least one energy delivery cable for connecting the combo generator module to a surgical instrument, at least one smoke evacuation component configured to evacuate smoke, fluid, and/or particulates generated by the application of therapeutic energy to the tissue, and a fluid line extending from the remote surgical site to the smoke evacuation component.

[0091] In one aspect, the fluid line is a first fluid line and a second fluid line extends from the remote surgical site to a suction and irrigation module slidably received in the hub enclosure. In one aspect, the hub enclosure comprises a fluid interface.

[0092] Certain surgical procedures may require the application of more than one energy type to the tissue. One energy type may be more beneficial for cutting the tissue, while another different energy type may be more beneficial for sealing the tissue. For example, a bipolar generator can be used to seal the tissue while an ultrasonic generator can be used to cut the sealed tissue. Aspects of the present disclosure present a solution where a hub modular enclosure **136** is configured to accommodate different generators, and facilitate an interactive communication therebetween. One of the advantages of the hub modular enclosure **136** is enabling the quick removal and/or replacement of various modules.

[0093] Aspects of the present disclosure present a modular surgical enclosure for use in a surgical procedure that involves energy application to tissue. The modular surgical enclosure includes a first energy-generator module, configured to generate a first energy for application to the tissue, and a first docking station comprising a first docking port that includes first data and power contacts, wherein the first

energy-generator module is slidably movable into an electrical engagement with the power and data contacts and wherein the first energy-generator module is slidably movable out of the electrical engagement with the first power and data contacts,

[0094] Further to the above, the modular surgical enclosure also includes a second energy-generator module configured to generate a second energy, different than the first energy, for application to the tissue, and a second docking station comprising a second docking port that includes second data and power contacts, wherein the second energy-generator module is slidably movable into an electrical engagement with the power and data contacts, and wherein the second energy-generator module is slidably movable out of the electrical engagement with the second power and data contacts.

[0095] In addition, the modular surgical enclosure also includes a communication bus between the first docking port and the second docking port, configured to facilitate communication between the first energy-generator module and the second energy-generator module.

[0096] Referring to FIGS. 3-7, aspects of the present disclosure are presented for a hub modular enclosure 136 that allows the modular integration of a generator module 140, a smoke evacuation module 126, and a suction/irrigation module 128. The hub modular enclosure 136 further facilitates interactive communication between the modules 140, 126, 128. As illustrated in FIG. 5, the generator module 140 can be a generator module with integrated monopolar, bipolar, and ultrasonic components supported in a single housing unit 139 slidably insertable into the hub modular enclosure 136. As illustrated in FIG. 5, the generator module 140 can be configured to connect to a monopolar device 146, a bipolar device 147, and an ultrasonic device 148. Alternatively, the generator module 140 may comprise a series of monopolar, bipolar, and/or ultrasonic generator modules that interact through the hub modular enclosure 136. The hub modular enclosure 136 can be configured to facilitate the insertion of multiple generators and interactive communication between the generators docked into the hub modular enclosure 136 so that the generators would act as a single generator.

[0097] In one aspect, the hub modular enclosure 136 comprises a modular power and communication backplane 149 with external and wireless communication headers to enable the removable attachment of the modules 140, 126, 128 and interactive communication therebetween.

[0098] In one aspect, the hub modular enclosure 136 includes docking stations, or drawers, 151, herein also referred to as drawers, which are configured to slidably receive the modules 140, 126, 128. FIG. 4 illustrates a partial perspective view of a surgical hub enclosure 136, and a combo generator module 145 slidably receivable in a docking station 151 of the surgical hub enclosure 136. A docking port 152 with power and data contacts on a rear side of the combo generator module 145 is configured to engage a corresponding docking port 150 with power and data contacts of a corresponding docking station 151 of the hub modular enclosure 136 as the combo generator module 145 is slid into position within the corresponding docking station 151 of the hub module enclosure 136. In one aspect, the combo generator module 145 includes a bipolar, ultrasonic,

and monopolar module and a smoke evacuation module integrated together into a single housing unit 139, as illustrated in FIG. 5.

[0099] In various aspects, the smoke evacuation module 126 includes a fluid line 154 that conveys captured/collected smoke and/or fluid away from a surgical site and to, for example, the smoke evacuation module 126. Vacuum suction originating from the smoke evacuation module 126 can draw the smoke into an opening of a utility conduit at the surgical site. The utility conduit, coupled to the fluid line, can be in the form of a flexible tube terminating at the smoke evacuation module 126. The utility conduit and the fluid line define a fluid path extending toward the smoke evacuation module 126 that is received in the hub enclosure 136.

[0100] In various aspects, the suction/irrigation module 128 is coupled to a surgical tool comprising an aspiration fluid line and a suction fluid line. In one example, the aspiration and suction fluid lines are in the form of flexible tubes extending from the surgical site toward the suction/irrigation module 128. One or more drive systems can be configured to cause irrigation and aspiration of fluids to and from the surgical site.

[0101] In one aspect, the surgical tool includes a shaft having an end effector at a distal end thereof and at least one energy treatment associated with the end effector, an aspiration tube, and an irrigation tube. The aspiration tube can have an inlet port at a distal end thereof and the aspiration tube extends through the shaft. Similarly, an irrigation tube can extend through the shaft and can have an inlet port in proximity to the energy deliver implement. The energy deliver implement is configured to deliver ultrasonic and/or RF energy to the surgical site and is coupled to the generator module 140 by a cable extending initially through the shaft.

[0102] The irrigation tube can be in fluid communication with a fluid source, and the aspiration tube can be in fluid communication with a vacuum source. The fluid source and/or the vacuum source can be housed in the suction/irrigation module 128. In one example, the fluid source and/or the vacuum source can be housed in the hub enclosure 136 separately from the suction/irrigation module 128. In such example, a fluid interface can be configured to connect the suction/irrigation module 128 to the fluid source and/or the vacuum source.

[0103] In one aspect, the modules 140, 126, 128 and/or their corresponding docking stations on the hub modular enclosure 136 may include alignment features that are configured to align the docking ports of the modules into engagement with their counterparts in the docking stations of the hub modular enclosure 136. For example, as illustrated in FIG. 4, the combo generator module 145 includes side brackets 155 that are configured to slidably engage with corresponding brackets 156 of the corresponding docking station 151 of the hub modular enclosure 136. The brackets cooperate to guide the docking port contacts of the combo generator module 145 into an electrical engagement with the docking port contacts of the hub modular enclosure 136.

[0104] In some aspects, the drawers 151 of the hub modular enclosure 136 are the same, or substantially the same size, and the modules are adjusted in size to be received in the drawers 151. For example, the side brackets 155 and/or 156 can be larger or smaller depending on the size of the module. In other aspects, the drawers 151 are different in size and are each designed to accommodate a particular module.

[0105] Furthermore, the contacts of a particular module can be keyed for engagement with the contacts of a particular drawer to avoid inserting a module into a drawer with mismatching contacts.

[0106] As illustrated in FIG. 4, the docking port 150 of one drawer 151 can be coupled to the docking port 150 of another drawer 151 through a communications link 157 to facilitate an interactive communication between the modules housed in the hub modular enclosure 136. The docking ports 150 of the hub modular enclosure 136 may alternatively, or additionally, facilitate a wireless interactive communication between the modules housed in the hub modular enclosure 136. Any suitable wireless communication can be employed, such as for example Air Titan-Bluetooth.

[0107] FIG. 6 illustrates individual power bus attachments for a plurality of lateral docking ports of a lateral modular housing 160 configured to receive a plurality of modules of a surgical hub 206. The lateral modular housing 160 is configured to laterally receive and interconnect the modules 161. The modules 161 are slidably inserted into docking stations 162 of lateral modular housing 160, which includes a backplane for interconnecting the modules 161. As illustrated in FIG. 6, the modules 161 are arranged laterally in the lateral modular housing 160. Alternatively, the modules 161 may be arranged vertically in a lateral modular housing.

[0108] FIG. 7 illustrates a vertical modular housing 164 configured to receive a plurality of modules 165 of the surgical hub 106. The modules 165 are slidably inserted into docking stations, or drawers, 167 of vertical modular housing 164, which includes a backplane for interconnecting the modules 165. Although the drawers 167 of the vertical modular housing 164 are arranged vertically, in certain instances, a vertical modular housing 164 may include drawers that are arranged laterally. Furthermore, the modules 165 may interact with one another through the docking ports of the vertical modular housing 164. In the example of FIG. 7, a display 177 is provided for displaying data relevant to the operation of the modules 165. In addition, the vertical modular housing 164 includes a master module 178 housing a plurality of sub-modules that are slidably received in the master module 178.

[0109] In various aspects, the imaging module 138 comprises an integrated video processor and a modular light source and is adapted for use with various imaging devices. In one aspect, the imaging device is comprised of a modular housing that can be assembled with a light source module and a camera module. The housing can be a disposable housing. In at least one example, the disposable housing is removably coupled to a reusable controller, a light source module, and a camera module. The light source module and/or the camera module can be selectively chosen depending on the type of surgical procedure. In one aspect, the camera module comprises a CCD sensor. In another aspect, the camera module comprises a CMOS sensor. In another aspect, the camera module is configured for scanned beam imaging. Likewise, the light source module can be configured to deliver a white light or a different light, depending on the surgical procedure.

[0110] During a surgical procedure, removing a surgical device from the surgical field and replacing it with another surgical device that includes a different camera or a different light source can be inefficient. Temporarily losing sight of the surgical field may lead to undesirable consequences. The module imaging device of the present disclosure is config-

ured to permit the replacement of a light source module or a camera module midstream during a surgical procedure, without having to remove the imaging device from the surgical field.

[0111] In one aspect, the imaging device comprises a tubular housing that includes a plurality of channels. A first channel is configured to slidably receive the camera module, which can be configured for a snap-fit engagement with the first channel. A second channel is configured to slidably receive the light source module, which can be configured for a snap-fit engagement with the second channel. In another example, the camera module and/or the light source module can be rotated into a final position within their respective channels. A threaded engagement can be employed in lieu of the snap-fit engagement.

[0112] In various examples, multiple imaging devices are placed at different positions in the surgical field to provide multiple views. The imaging module 138 can be configured to switch between the imaging devices to provide an optimal view. In various aspects, the imaging module 138 can be configured to integrate the images from the different imaging device.

[0113] Various image processors and imaging devices suitable for use with the present disclosure are described in U.S. Pat. No. 7,995,045, titled COMBINED SBI AND CONVENTIONAL IMAGE PROCESSOR, which issued on Aug. 9, 2011, which is herein incorporated by reference in its entirety. In addition, U.S. Pat. No. 7,982,776, titled SBI MOTION ARTIFACT REMOVAL APPARATUS AND METHOD, which issued on Jul. 19, 2011, which is herein incorporated by reference in its entirety, describes various systems for removing motion artifacts from image data. Such systems can be integrated with the imaging module 138. Furthermore, U.S. Patent Application Publication No. 2011/0306840, titled CONTROLLABLE MAGNETIC SOURCE TO FIXTURE INTRACORPOREAL APPARATUS, which published on Dec. 15, 2011, and U.S. Patent Application Publication No. 2014/0243597, titled SYSTEM FOR PERFORMING A MINIMALLY INVASIVE SURGICAL PROCEDURE, which published on Aug. 28, 2014, now U.S. Pat. No. 10,098,527, each of which is herein incorporated by reference in its entirety.

[0114] FIG. 8 illustrates a surgical data network 201 comprising a modular communication hub 203 configured to connect modular devices located in one or more operating theaters of a healthcare facility, or any room in a healthcare facility specially equipped for surgical operations, to a cloud-based system (e.g., the cloud 204 that may include a remote server 213 coupled to a storage device 205). In one aspect, the modular communication hub 203 comprises a network hub 207 and/or a network switch 209 in communication with a network router. The modular communication hub 203 also can be coupled to a local computer system 210 to provide local computer processing and data manipulation. The surgical data network 201 may be configured as passive, intelligent, or switching. A passive surgical data network serves as a conduit for the data, enabling it to go from one device (or segment) to another and to the cloud computing resources. An intelligent surgical data network includes additional features to enable the traffic passing through the surgical data network to be monitored and to configure each port in the network hub 207 or network switch 209. An intelligent surgical data network may be referred to as a

manageable hub or switch. A switching hub reads the destination address of each packet and then forwards the packet to the correct port.

[0115] Modular devices $1a-1n$ located in the operating theater may be coupled to the modular communication hub 203. The network hub 207 and/or the network switch 209 may be coupled to a network router 211 to connect the devices $1a-1n$ to the cloud 204 or the local computer system 210. Data associated with the devices $1a-1n$ may be transferred to cloud-based computers via the router for remote data processing and manipulation. Data associated with the devices $1a-1n$ may also be transferred to the local computer system 210 for local data processing and manipulation. Modular devices $2a-2m$ located in the same operating theater also may be coupled to a network switch 209. The network switch 209 may be coupled to the network hub 207 and/or the network router 211 to connect to the devices $2a-2m$ to the cloud 204. Data associated with the devices $2a-2m$ may be transferred to the cloud 204 via the network router 211 for data processing and manipulation. Data associated with the devices $2a-2m$ may also be transferred to the local computer system 210 for local data processing and manipulation.

[0116] It will be appreciated that the surgical data network 201 may be expanded by interconnecting multiple network hubs 207 and/or multiple network switches 209 with multiple network routers 211. The modular communication hub 203 may be contained in a modular control tower configured to receive multiple devices $1a-1n/2a-2m$. The local computer system 210 also may be contained in a modular control tower. The modular communication hub 203 is connected to a display 212 to display images obtained by some of the devices $1a-1n/2a-2m$, for example during surgical procedures. In various aspects, the devices $1a-1n/2a-2m$ may include, for example, various modules such as an imaging module 138 coupled to an endoscope, a generator module 140 coupled to an energy-based surgical device, a smoke evacuation module 126, a suction/irrigation module 128, a communication module 130, a processor module 132, a storage array 134, a surgical device coupled to a display, and/or a non-contact sensor module, among other modular devices that may be connected to the modular communication hub 203 of the surgical data network 201.

[0117] In one aspect, the surgical data network 201 may comprise a combination of network hub(s), network switch(es), and network router(s) connecting the devices $1a-1n/2a-2m$ to the cloud. Any one of or all of the devices $1a-1n/2a-2m$ coupled to the network hub or network switch may collect data in real time and transfer the data to cloud computers for data processing and manipulation. It will be appreciated that cloud computing relies on sharing computing resources rather than having local servers or personal devices to handle software applications. The word “cloud” may be used as a metaphor for “the Internet,” although the term is not limited as such. Accordingly, the term “cloud computing” may be used herein to refer to “a type of Internet-based computing,” where different services—such as servers, storage, and applications—are delivered to the modular communication hub 203 and/or computer system 210 located in the surgical theater (e.g., a fixed, mobile, temporary, or field operating room or space) and to devices connected to the modular communication hub 203 and/or computer system 210 through the Internet. The cloud infrastructure may be maintained by a cloud service provider. In

this context, the cloud service provider may be the entity that coordinates the usage and control of the devices $1a-1n/2a-2m$ located in one or more operating theaters. The cloud computing services can perform a large number of calculations based on the data gathered by smart surgical instruments, robots, and other computerized devices located in the operating theater. The hub hardware enables multiple devices or connections to be connected to a computer that communicates with the cloud computing resources and storage.

[0118] Applying cloud computer data processing techniques on the data collected by the devices $1a-1n/2a-2m$, the surgical data network provides improved surgical outcomes, reduced costs, and improved patient satisfaction. At least some of the devices $1a-1n/2a-2m$ may be employed to view tissue states to assess leaks or perfusion of sealed tissue after a tissue sealing and cutting procedure. At least some of the devices $1a-1n/2a-2m$ may be employed to identify pathology, such as the effects of diseases, using the cloud-based computing to examine data including images of samples of body tissue for diagnostic purposes. This includes localization and margin confirmation of tissue and phenotypes. At least some of the devices $1a-1n/2a-2m$ may be employed to identify anatomical structures of the body using a variety of sensors integrated with imaging devices and techniques such as overlaying images captured by multiple imaging devices. The data gathered by the devices $1a-1n/2a-2m$, including image data, may be transferred to the cloud 204 or the local computer system 210 or both for data processing and manipulation including image processing and manipulation. The data may be analyzed to improve surgical procedure outcomes by determining if further treatment, such as the application of endoscopic intervention, emerging technologies, a targeted radiation, targeted intervention, and precise robotics to tissue-specific sites and conditions, may be pursued. Such data analysis may further employ outcome analytics processing, and using standardized approaches may provide beneficial feedback to either confirm surgical treatments and the behavior of the surgeon or suggest modifications to surgical treatments and the behavior of the surgeon.

[0119] In one implementation, the operating theater devices $1a-1n$ may be connected to the modular communication hub 203 over a wired channel or a wireless channel depending on the configuration of the devices $1a-1n$ to a network hub. The network hub 207 may be implemented, in one aspect, as a local network broadcast device that works on the physical layer of the Open System Interconnection (OSI) model. The network hub provides connectivity to the devices $1a-1n$ located in the same operating theater network. The network hub 207 collects data in the form of packets and sends them to the router in half duplex mode. The network hub 207 does not store any media access control/internet protocol (MAC/IP) to transfer the device data. Only one of the devices $1a-1n$ can send data at a time through the network hub 207. The network hub 207 has no routing tables or intelligence regarding where to send information and broadcasts all network data across each connection and to a remote server 213 (FIG. 9) over the cloud 204. The network hub 207 can detect basic network errors such as collisions, but having all information broadcast to multiple ports can be a security risk and cause bottlenecks.

[0120] In another implementation, the operating theater devices $2a-2m$ may be connected to a network switch 209

over a wired channel or a wireless channel. The network switch **209** works in the data link layer of the OSI model. The network switch **209** is a multicast device for connecting the devices **2a-2m** located in the same operating theater to the network. The network switch **209** sends data in the form of frames to the network router **211** and works in full duplex mode. Multiple devices **2a-2m** can send data at the same time through the network switch **209**. The network switch **209** stores and uses MAC addresses of the devices **2a-2m** to transfer data.

[0121] The network hub **207** and/or the network switch **209** are coupled to the network router **211** for connection to the cloud **204**. The network router **211** works in the network layer of the OSI model. The network router **211** creates a route for transmitting data packets received from the network hub **207** and/or network switch **211** to cloud-based computer resources for further processing and manipulation of the data collected by any one of or all the devices **1a-1n/2a-2m**. The network router **211** may be employed to connect two or more different networks located in different locations, such as, for example, different operating theaters of the same healthcare facility or different networks located in different operating theaters of different healthcare facilities. The network router **211** sends data in the form of packets to the cloud **204** and works in full duplex mode. Multiple devices can send data at the same time. The network router **211** uses IP addresses to transfer data.

[0122] In one example, the network hub **207** may be implemented as a USB hub, which allows multiple USB devices to be connected to a host computer. The USB hub may expand a single USB port into several tiers so that there are more ports available to connect devices to the host system computer. The network hub **207** may include wired or wireless capabilities to receive information over a wired channel or a wireless channel. In one aspect, a wireless USB short-range, high-bandwidth wireless radio communication protocol may be employed for communication between the devices **1a-1n** and devices **2a-2m** located in the operating theater.

[0123] In other examples, the operating theater devices **1a-1n/2a-2m** may communicate to the modular communication hub **203** via Bluetooth wireless technology standard for exchanging data over short distances (using short-wavelength UHF radio waves in the ISM band from 2.4 to 2.485 GHz) from fixed and mobile devices and building personal area networks (PANs). In other aspects, the operating theater devices **1a-1n/2a-2m** may communicate to the modular communication hub **203** via a number of wireless or wired communication standards or protocols, including but not limited to Wi-Fi (IEEE 802.11 family), WiMAX (IEEE 802.16 family), IEEE 802.20, long-term evolution (LTE), and Ev-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPRS, CDMA, TDMA, DECT, and Ethernet derivatives thereof, as well as any other wireless and wired protocols that are designated as 3G, 4G, 5G, and beyond. The computing module may include a plurality of communication modules. For instance, a first communication module may be dedicated to shorter-range wireless communications such as Wi-Fi and Bluetooth, and a second communication module may be dedicated to longer-range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, Ev-DO, and others.

[0124] The modular communication hub **203** may serve as a central connection for one or all of the operating theater

devices **1a-1n/2a-2m** and handles a data type known as frames. Frames carry the data generated by the devices **1a-1n/2a-2m**. When a frame is received by the modular communication hub **203**, it is amplified and transmitted to the network router **211**, which transfers the data to the cloud computing resources by using a number of wireless or wired communication standards or protocols, as described herein.

[0125] The modular communication hub **203** can be used as a standalone device or be connected to compatible network hubs and network switches to form a larger network. The modular communication hub **203** is generally easy to install, configure, and maintain, making it a good option for networking the operating theater devices **1a-1n/2a-2m**.

[0126] FIG. 9 illustrates a computer-implemented interactive surgical system **200**. The computer-implemented interactive surgical system **200** is similar in many respects to the computer-implemented interactive surgical system **100**. For example, the computer-implemented interactive surgical system **200** includes one or more surgical systems **202**, which are similar in many respects to the surgical systems **102**. Each surgical system **202** includes at least one surgical hub **206** in communication with a cloud **204** that may include a remote server **213**. In one aspect, the computer-implemented interactive surgical system **200** comprises a modular control tower **236** connected to multiple operating theater devices such as, for example, intelligent surgical instruments, robots, and other computerized devices located in the operating theater. As shown in FIG. 10, the modular control tower **236** comprises a modular communication hub **203** coupled to a computer system **210**. As illustrated in the example of FIG. 9, the modular control tower **236** is coupled to an imaging module **238** that is coupled to an endoscope **239**, a generator module **240** that is coupled to an energy device **241**, a smoke evacuator module **226**, a suction/irrigation module **228**, a communication module **230**, a processor module **232**, a storage array **234**, a smart device/instrument **235** optionally coupled to a display **237**, and a non-contact sensor module **242**. The operating theater devices are coupled to cloud computing resources and data storage via the modular control tower **236**. A robot hub **222** also may be connected to the modular control tower **236** and to the cloud computing resources. The devices/instruments **235**, visualization systems **208**, among others, may be coupled to the modular control tower **236** via wired or wireless communication standards or protocols, as described herein. The modular control tower **236** may be coupled to a hub display **215** (e.g., monitor, screen) to display and overlay images received from the imaging module, device/instrument display, and/or other visualization systems **208**. The hub display also may display data received from devices connected to the modular control tower in conjunction with images and overlaid images.

[0127] FIG. 10 illustrates a surgical hub **206** comprising a plurality of modules coupled to the modular control tower **236**. The modular control tower **236** comprises a modular communication hub **203**, e.g., a network connectivity device, and a computer system **210** to provide local processing, visualization, and imaging, for example. As shown in FIG. 10, the modular communication hub **203** may be connected in a tiered configuration to expand the number of modules (e.g., devices) that may be connected to the modular communication hub **203** and transfer data associated with the modules to the computer system **210**, cloud computing

resources, or both. As shown in FIG. 10, each of the network hubs/switches in the modular communication hub 203 includes three downstream ports and one upstream port. The upstream network hub/switch is connected to a processor to provide a communication connection to the cloud computing resources and a local display 217. Communication to the cloud 204 may be made either through a wired or a wireless communication channel.

[0128] The surgical hub 206 employs a non-contact sensor module 242 to measure the dimensions of the operating theater and generate a map of the surgical theater using either ultrasonic or laser-type non-contact measurement devices. An ultrasound-based non-contact sensor module scans the operating theater by transmitting a burst of ultrasound and receiving the echo when it bounces off the perimeter walls of an operating theater as described under the heading “Surgical Hub Spatial Awareness Within an Operating Room” in U.S. Provisional Patent Application Ser. No. 62/611,341, titled INTERACTIVE SURGICAL PLATFORM, filed Dec. 28, 2017, which is herein incorporated by reference in its entirety, in which the sensor module is configured to determine the size of the operating theater and to adjust Bluetooth-pairing distance limits. A laser-based non-contact sensor module scans the operating theater by transmitting laser light pulses, receiving laser light pulses that bounce off the perimeter walls of the operating theater, and comparing the phase of the transmitted pulse to the received pulse to determine the size of the operating theater and to adjust Bluetooth pairing distance limits, for example.

[0129] The computer system 210 comprises a processor 244 and a network interface 245. The processor 244 is coupled to a communication module 247, storage 248, memory 249, non-volatile memory 250, and input/output interface 251 via a system bus. The system bus can be any of several types of bus structure(s) including the memory bus or memory controller, a peripheral bus or external bus, and/or a local bus using any variety of available bus architectures including, but not limited to, 9-bit bus, Industrial Standard Architecture (ISA), Micro-Charmel Architecture (MSA), Extended ISA (EISA), Intelligent Drive Electronics (IDE), VESA Local Bus (VLB), Peripheral Component Interconnect (PCI), USB, Advanced Graphics Port (AGP), Personal Computer Memory Card International Association bus (PCMCIA), Small Computer Systems Interface (SCSI), or any other proprietary bus.

[0130] The processor 244 may be any single-core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In one aspect, the processor may be an LM4F230H5QR ARM Cortex-M4F Processor Core, available from Texas Instruments, for example, comprising an on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle serial random access memory (SRAM), an internal read-only memory (ROM) loaded with StellarisWare® software, a 2 KB electrically erasable programmable read-only memory (EEPROM), and/or one or more pulse width modulation (PWM) modules, one or more quadrature encoder inputs (QEI) analogs, one or more 12-bit analog-to-digital converters (ADCs) with 12 analog input channels, details of which are available for the product datasheet.

[0131] In one aspect, the processor 244 may comprise a safety controller comprising two controller-based families

such as TMS570 and RM4x, known under the trade name Hercules ARM Cortex R4, also by Texas Instruments. The safety controller may be configured specifically for IEC 61508 and ISO 26262 safety critical applications, among others, to provide advanced integrated safety features while delivering scalable performance, connectivity, and memory options.

[0132] The system memory includes volatile memory and non-volatile memory. The basic input/output system (BIOS), containing the basic routines to transfer information between elements within the computer system, such as during start-up, is stored in non-volatile memory. For example, the non-volatile memory can include ROM, programmable ROM (PROM), electrically programmable ROM (EPROM), EEPROM, or flash memory. Volatile memory includes random-access memory (RAM), which acts as external cache memory. Moreover, RAM is available in many forms such as SRAM, dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SL-DRAM), and direct Rambus RAM (DRRAM).

[0133] The computer system 210 also includes removable/non-removable, volatile/non-volatile computer storage media, such as for example disk storage. The disk storage includes, but is not limited to, devices like a magnetic disk drive, floppy disk drive, tape drive, Jaz drive, Zip drive, LS-60 drive, flash memory card, or memory stick. In addition, the disk storage can include storage media separately or in combination with other storage media including, but not limited to, an optical disc drive such as a compact disc ROM device (CD-ROM), compact disc recordable drive (CD-R Drive), compact disc rewritable drive (CD-RW Drive), or a digital versatile disc ROM drive (DVD-ROM). To facilitate the connection of the disk storage devices to the system bus, a removable or non-removable interface may be employed.

[0134] It is to be appreciated that the computer system 210 includes software that acts as an intermediary between users and the basic computer resources described in a suitable operating environment. Such software includes an operating system. The operating system, which can be stored on the disk storage, acts to control and allocate resources of the computer system. System applications take advantage of the management of resources by the operating system through program modules and program data stored either in the system memory or on the disk storage. It is to be appreciated that various components described herein can be implemented with various operating systems or combinations of operating systems.

[0135] A user enters commands or information into the computer system 210 through input device(s) coupled to the I/O interface 251. The input devices include, but are not limited to, a pointing device such as a mouse, trackball, stylus, touch pad, keyboard, microphone, joystick, game pad, satellite dish, scanner, TV tuner card, digital camera, digital video camera, web camera, and the like. These and other input devices connect to the processor through the system bus via interface port(s). The interface port(s) include, for example, a serial port, a parallel port, a game port, and a USB. The output device(s) use some of the same types of ports as input device(s). Thus, for example, a USB port may be used to provide input to the computer system and to output information from the computer system to an output device. An output adapter is provided to illustrate that there are some output devices like monitors, displays, speak-

ers, and printers, among other output devices that require special adapters. The output adapters include, by way of illustration and not limitation, video and sound cards that provide a means of connection between the output device and the system bus. It should be noted that other devices and/or systems of devices, such as remote computer(s), provide both input and output capabilities.

[0136] The computer system 210 can operate in a networked environment using logical connections to one or more remote computers, such as cloud computer(s), or local computers. The remote cloud computer(s) can be a personal computer, server, router, network PC, workstation, micro-processor-based appliance, peer device, or other common network node, and the like, and typically includes many or all of the elements described relative to the computer system. For purposes of brevity, only a memory storage device is illustrated with the remote computer(s). The remote computer(s) is logically connected to the computer system through a network interface and then physically connected via a communication connection. The network interface encompasses communication networks such as local area networks (LANs) and wide area networks (WANs). LAN technologies include Fiber Distributed Data Interface (FDDI), Copper Distributed Data Interface (CDDI), Ethernet/IEEE 802.3, Token Ring/IEEE 802.5 and the like. WAN technologies include, but are not limited to, point-to-point links, circuit-switching networks like Integrated Services Digital Networks (ISDN) and variations thereon, packet-switching networks, and Digital Subscriber Lines (DSL).

[0137] In various aspects, the computer system 210 of FIG. 10, the imaging module 238 and/or visualization system 208, and/or the processor module 232 of FIGS. 9-10, may comprise an image processor, image processing engine, media processor, or any specialized digital signal processor (DSP) used for the processing of digital images. The image processor may employ parallel computing with single instruction, multiple data (SIMD) or multiple instruction, multiple data (MIMD) technologies to increase speed and efficiency. The digital image processing engine can perform a range of tasks. The image processor may be a system on a chip with multicore processor architecture.

[0138] The communication connection(s) refers to the hardware/software employed to connect the network interface to the bus. While the communication connection is shown for illustrative clarity inside the computer system, it can also be external to the computer system 210. The hardware/software necessary for connection to the network interface includes, for illustrative purposes only, internal and external technologies such as modems, including regular telephone-grade modems, cable modems, and DSL modems, ISDN adapters, and Ethernet cards.

[0139] FIG. 11 illustrates a functional block diagram of one aspect of a USB network hub 300 device, according to one aspect of the present disclosure. In the illustrated aspect, the USB network hub device 300 employs a TUSB2036 integrated circuit hub by Texas Instruments. The USB network hub 300 is a CMOS device that provides an upstream USB transceiver port 302 and up to three downstream USB transceiver ports 304, 306, 308 in compliance with the USB 2.0 specification. The upstream USB transceiver port 302 is a differential root data port comprising a differential data minus (DM0) input paired with a differential data plus (DP0) input. The three downstream USB transceiver ports 304,

306, 308 are differential data ports where each port includes differential data plus (DP1-DP3) outputs paired with differential data minus (DM1-DM3) outputs.

[0140] The USB network hub 300 device is implemented with a digital state machine instead of a microcontroller, and no firmware programming is required. Fully compliant USB transceivers are integrated into the circuit for the upstream USB transceiver port 302 and all downstream USB transceiver ports 304, 306, 308. The downstream USB transceiver ports 304, 306, 308 support both full-speed and low-speed devices by automatically setting the slew rate according to the speed of the device attached to the ports. The USB network hub 300 device may be configured either in bus-powered or self-powered mode and includes a hub power logic 312 to manage power.

[0141] The USB network hub 300 device includes a serial interface engine 310 (SIE). The SIE 310 is the front end of the USB network hub 300 hardware and handles most of the protocol described in chapter 8 of the USB specification. The SIE 310 typically comprehends signaling up to the transaction level. The functions that it handles could include: packet recognition, transaction sequencing, SOP, EOP, RESET, and RESUME signal detection/generation, clock/data separation, non-return-to-zero invert (NRZI) data encoding/decoding and bit-stuffing, CRC generation and checking (token and data), packet ID (PID) generation and checking/decoding, and/or serial-parallel/parallel-serial conversion. The 310 receives a clock input 314 and is coupled to a suspend/resume logic and frame timer 316 circuit and a hub repeater circuit 318 to control communication between the upstream USB transceiver port 302 and the downstream USB transceiver ports 304, 306, 308 through port logic circuits 320, 322, 324. The SIE 310 is coupled to a command decoder 326 via interface logic to control commands from a serial EEPROM via a serial EEPROM interface 330.

[0142] In various aspects, the USB network hub 300 can connect 127 functions configured in up to six logical layers (tiers) to a single computer. Further, the USB network hub 300 can connect to all peripherals using a standardized four-wire cable that provides both communication and power distribution. The power configurations are bus-powered and self-powered modes. The USB network hub 300 may be configured to support four modes of power management: a bus-powered hub, with either individual-port power management or ganged-port power management, and the self-powered hub, with either individual-port power management or ganged-port power management. In one aspect, using a USB cable, the USB network hub 300, the upstream USB transceiver port 302 is plugged into a USB host controller, and the downstream USB transceiver ports 304, 306, 308 are exposed for connecting USB compatible devices, and so forth.

Surgical Instrument Hardware

[0143] FIG. 12 illustrates a logic diagram of a control system 470 of a surgical instrument or tool in accordance with one or more aspects of the present disclosure. The system 470 comprises a control circuit. The control circuit includes a microcontroller 461 comprising a processor 462 and a memory 468. One or more of sensors 472, 474, 476, for example, provide real-time feedback to the processor 462. A motor 482, driven by a motor driver 492, operably couples a longitudinally movable displacement member to drive the I-beam knife element. A tracking system 480 is

configured to determine the position of the longitudinally movable displacement member. The position information is provided to the processor **462**, which can be programmed or configured to determine the position of the longitudinally movable drive member as well as the position of a firing member, firing bar, and I-beam knife element. Additional motors may be provided at the tool driver interface to control I-beam firing, closure tube travel, shaft rotation, and articulation. A display **473** displays a variety of operating conditions of the instruments and may include touch screen functionality for data input. Information displayed on the display **473** may be overlaid with images acquired via endoscopic imaging modules.

[0144] In one aspect, the microcontroller **461** may be any single-core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In one aspect, the main microcontroller **461** may be an LM4F230H5QR ARM Cortex-M4F Processor Core, available from Texas Instruments, for example, comprising an on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle SRAM, and internal ROM loaded with StellarisWare® software, a 2 KB EEPROM, one or more PWM modules, one or more QEI analogs, and/or one or more 12-bit ADCs with 12 analog input channels, details of which are available for the product datasheet.

[0145] In one aspect, the microcontroller **461** may comprise a safety controller comprising two controller-based families such as TMS570 and RM4x, known under the trade name Hercules ARM Cortex R4, also by Texas Instruments. The safety controller may be configured specifically for IEC 61508 and ISO 26262 safety critical applications, among others, to provide advanced integrated safety features while delivering scalable performance, connectivity, and memory options.

[0146] The microcontroller **461** may be programmed to perform various functions such as precise control over the speed and position of the knife and articulation systems. In one aspect, the microcontroller **461** includes a processor **462** and a memory **468**. The electric motor **482** may be a brushed direct current (DC) motor with a gearbox and mechanical links to an articulation or knife system. In one aspect, a motor driver **492** may be an A3941 available from Allegro Microsystems, Inc. Other motor drivers may be readily substituted for use in the tracking system **480** comprising an absolute positioning system. A detailed description of an absolute positioning system is described in U.S. Patent Application Publication No. 2017/0296213, titled SYSTEMS AND METHODS FOR CONTROLLING A SURGICAL STAPLING AND CUTTING INSTRUMENT, which published on Oct. 19, 2017, now U.S. Pat. No. 9,958,764, which is herein incorporated by reference in its entirety.

[0147] The microcontroller **461** may be programmed to provide precise control over the speed and position of displacement members and articulation systems. The microcontroller **461** may be configured to compute a response in the software of the microcontroller **461**. The computed response is compared to a measured response of the actual system to obtain an “observed” response, which is used for actual feedback decisions. The observed response is a favorable, tuned value that balances the smooth, continuous

nature of the simulated response with the measured response, which can detect outside influences on the system.

[0148] In one aspect, the motor **482** may be controlled by the motor driver **492** and can be employed by the firing system of the surgical instrument or tool. In various forms, the motor **482** may be a brushed DC driving motor having a maximum rotational speed of approximately 25,000 RPM. In other arrangements, the motor **482** may include a brushless motor, a cordless motor, a synchronous motor, a stepper motor, or any other suitable electric motor. The motor driver **492** may comprise an H-bridge driver comprising field-effect transistors (FETs), for example. The motor **482** can be powered by a power assembly releasably mounted to the handle assembly or tool housing for supplying control power to the surgical instrument or tool. The power assembly may comprise a battery which may include a number of battery cells connected in series that can be used as the power source to power the surgical instrument or tool. In certain circumstances, the battery cells of the power assembly may be replaceable and/or rechargeable. In at least one example, the battery cells can be lithium-ion batteries which can be coupleable to and separable from the power assembly.

[0149] The motor driver **492** may be an A3941 available from Allegro Microsystems, Inc. The A3941 **492** is a full-bridge controller for use with external N-channel power metal-oxide semiconductor field-effect transistors (MOSFETs) specifically designed for inductive loads, such as brush DC motors. The driver **492** comprises a unique charge pump regulator that provides full (>10 V) gate drive for battery voltages down to 7 V and allows the A3941 to operate with a reduced gate drive, down to 5.5 V. A bootstrap capacitor may be employed to provide the above battery supply voltage required for N-channel MOSFETs. An internal charge pump for the high-side drive allows DC (100% duty cycle) operation. The full bridge can be driven in fast or slow decay modes using diode or synchronous rectification. In the slow decay mode, current recirculation can be through the high-side or the lowside FETs. The power FETs are protected from shoot-through by resistor-adjustable dead time. Integrated diagnostics provide indications of under-voltage, overtemperature, and power bridge faults and can be configured to protect the power MOSFETs under most short circuit conditions. Other motor drivers may be readily substituted for use in the tracking system **480** comprising an absolute positioning system.

[0150] The tracking system **480** comprises a controlled motor drive circuit arrangement comprising a position sensor **472** according to one aspect of this disclosure. The position sensor **472** for an absolute positioning system provides a unique position signal corresponding to the location of a displacement member. In one aspect, the displacement member represents a longitudinally movable drive member comprising a rack of drive teeth for meshing engagement with a corresponding drive gear of a gear reducer assembly. In other aspects, the displacement member represents the firing member, which could be adapted and configured to include a rack of drive teeth. In yet another aspect, the displacement member represents a firing bar or the I-beam, each of which can be adapted and configured to include a rack of drive teeth. Accordingly, as used herein, the term displacement member is used generically to refer to any movable member of the surgical instrument or tool such as the drive member, the firing member, the firing bar, the I-beam, or any element that can be displaced. In one aspect,

the longitudinally movable drive member is coupled to the firing member, the firing bar, and the I-beam. Accordingly, the absolute positioning system can, in effect, track the linear displacement of the I-beam by tracking the linear displacement of the longitudinally movable drive member. In various other aspects, the displacement member may be coupled to any position sensor 472 suitable for measuring linear displacement. Thus, the longitudinally movable drive member, the firing member, the firing bar, or the I-beam, or combinations thereof, may be coupled to any suitable linear displacement sensor. Linear displacement sensors may include contact or non-contact displacement sensors. Linear displacement sensors may comprise linear variable differential transformers (LVDT), differential variable reluctance transducers (DVRT), a slide potentiometer, a magnetic sensing system comprising a movable magnet and a series of linearly arranged Hall effect sensors, a magnetic sensing system comprising a fixed magnet and a series of movable, linearly arranged Hall effect sensors, an optical sensing system comprising a movable light source and a series of linearly arranged photo diodes or photo detectors, an optical sensing system comprising a fixed light source and a series of movable linearly arranged photo diodes or photo detectors, or any combination thereof.

[0151] The electric motor 482 can include a rotatable shaft that operably interfaces with a gear assembly that is mounted in meshing engagement with a set, or rack, of drive teeth on the displacement member. A sensor element may be operably coupled to a gear assembly such that a single revolution of the position sensor 472 element corresponds to some linear longitudinal translation of the displacement member. An arrangement of gearing and sensors can be connected to the linear actuator, via a rack and pinion arrangement, or a rotary actuator, via a spur gear or other connection. A power source supplies power to the absolute positioning system and an output indicator may display the output of the absolute positioning system. The displacement member represents the longitudinally movable drive member comprising a rack of drive teeth formed thereon for meshing engagement with a corresponding drive gear of the gear reducer assembly. The displacement member represents the longitudinally movable firing member, firing bar, I-beam, or combinations thereof.

[0152] A single revolution of the sensor element associated with the position sensor 472 is equivalent to a longitudinal linear displacement $d1$ of the of the displacement member, where $d1$ is the longitudinal linear distance that the displacement member moves from point "a" to point "b" after a single revolution of the sensor element coupled to the displacement member. The sensor arrangement may be connected via a gear reduction that results in the position sensor 472 completing one or more revolutions for the full stroke of the displacement member. The position sensor 472 may complete multiple revolutions for the full stroke of the displacement member.

[0153] A series of switches, where n is an integer greater than one, may be employed alone or in combination with a gear reduction to provide a unique position signal for more than one revolution of the position sensor 472. The state of the switches are fed back to the microcontroller 461 that applies logic to determine a unique position signal corresponding to the longitudinal linear displacement $d1+d2+...+dn$ of the displacement member. The output of the position sensor 472 is provided to the microcontroller 461. The

position sensor 472 of the sensor arrangement may comprise a magnetic sensor, an analog rotary sensor like a potentiometer, or an array of analog Hall-effect elements, which output a unique combination of position signals or values.

[0154] The position sensor 472 may comprise any number of magnetic sensing elements, such as, for example, magnetic sensors classified according to whether they measure the total magnetic field or the vector components of the magnetic field. The techniques used to produce both types of magnetic sensors encompass many aspects of physics and electronics. The technologies used for magnetic field sensing include search coil, fluxgate, optically pumped, nuclear precession, SQUID, Hall-effect, anisotropic magnetoresistance, giant magnetoresistance, magnetic tunnel junctions, giant magnetoimpedance, magnetostrictive/piezoelectric composites, magnetodiode, magnetotransistor, fiber-optic, magneto-optic, and microelectromechanical systems-based magnetic sensors, among others.

[0155] In one aspect, the position sensor 472 for the tracking system 480 comprising an absolute positioning system comprises a magnetic rotary absolute positioning system. The position sensor 472 may be implemented as an AS5055EQFT single-chip magnetic rotary position sensor available from Austria Microsystems, AG. The position sensor 472 is interfaced with the microcontroller 461 to provide an absolute positioning system. The position sensor 472 is a low-voltage and low-power component and includes four Hall-effect elements in an area of the position sensor 472 that is located above a magnet. A high-resolution ADC and a smart power management controller are also provided on the chip. A coordinate rotation digital computer (CORDIC) processor, also known as the digit-by-digit method and Volder's algorithm, is provided to implement a simple and efficient algorithm to calculate hyperbolic and trigonometric functions that require only addition, subtraction, bitshift, and table lookup operations. The angle position, alarm bits, and magnetic field information are transmitted over a standard serial communication interface, such as a serial peripheral interface (SPI) interface, to the microcontroller 461. The position sensor 472 provides 12 or 14 bits of resolution. The position sensor 472 may be an AS5055 chip provided in a small QFN 16-pin $4\times4\times0.85$ mm package.

[0156] The tracking system 480 comprising an absolute positioning system may comprise and/or be programmed to implement a feedback controller, such as a PID, state feedback, and adaptive controller. A power source converts the signal from the feedback controller into a physical input to the system: in this case the voltage. Other examples include a PWM of the voltage, current, and force. Other sensor(s) may be provided to measure physical parameters of the physical system in addition to the position measured by the position sensor 472. In some aspects, the other sensor(s) can include sensor arrangements such as those described in U.S. Pat. No. 9,345,481, titled STAPLE CARTRIDGE TISSUE THICKNESS SENSOR SYSTEM, which issued on May 24, 2016, which is herein incorporated by reference in its entirety; U.S. Patent Application Publication No. 2014/0263552, titled STAPLE CARTRIDGE TISSUE THICKNESS SENSOR SYSTEM, which published on Sep. 18, 2014, which is herein incorporated by reference in its entirety; and U.S. patent application Ser. No. 15/628,175, titled TECHNIQUES FOR ADAPTIVE CONTROL OF MOTOR VELOCITY OF A SURGICAL STAPLING AND

CUTTING INSTRUMENT, filed Jun. 20, 2017, now U.S. Pat. No. 10,881,399, which is herein incorporated by reference in its entirety. In a digital signal processing system, an absolute positioning system is coupled to a digital data acquisition system where the output of the absolute positioning system will have a finite resolution and sampling frequency. The absolute positioning system may comprise a compare-and-combine circuit to combine a computed response with a measured response using algorithms, such as a weighted average and a theoretical control loop, that drive the computed response towards the measured response. The computed response of the physical system takes into account properties like mass, inertial, viscous friction, inductance resistance, etc., to predict what the states and outputs of the physical system will be by knowing the input.

[0157] The absolute positioning system provides an absolute position of the displacement member upon power-up of the instrument, without retracting or advancing the displacement member to a reset (zero or home) position as may be required with conventional rotary encoders that merely count the number of steps forwards or backwards that the motor 482 has taken to infer the position of a device actuator, drive bar, knife, or the like.

[0158] A sensor 474, such as, for example, a strain gauge or a micro-strain gauge, is configured to measure one or more parameters of the end effector, such as, for example, the amplitude of the strain exerted on the anvil during a clamping operation, which can be indicative of the closure forces applied to the anvil. The measured strain is converted to a digital signal and provided to the processor 462. Alternatively, or in addition to the sensor 474, a sensor 476, such as, for example, a load sensor, can measure the closure force applied by the closure drive system to the anvil. The sensor 476, such as, for example, a load sensor, can measure the firing force applied to an I-beam in a firing stroke of the surgical instrument or tool. The I-beam is configured to engage a wedge sled, which is configured to upwardly cam staple drivers to force out staples into deforming contact with an anvil. The I-beam also includes a sharpened cutting edge that can be used to sever tissue as the I-beam is advanced distally by the firing bar. Alternatively, a current sensor 478 can be employed to measure the current drawn by the motor 482. The force required to advance the firing member can correspond to the current drawn by the motor 482, for example. The measured force is converted to a digital signal and provided to the processor 462.

[0159] In one form, the strain gauge sensor 474 can be used to measure the force applied to the tissue by the end effector. A strain gauge can be coupled to the end effector to measure the force on the tissue being treated by the end effector. A system for measuring forces applied to the tissue grasped by the end effector comprises a strain gauge sensor 474, such as, for example, a micro-strain gauge, that is configured to measure one or more parameters of the end effector, for example. In one aspect, the strain gauge sensor 474 can measure the amplitude or magnitude of the strain exerted on a jaw member of an end effector during a clamping operation, which can be indicative of the tissue compression. The measured strain is converted to a digital signal and provided to a processor 462 of the microcontroller 461. A load sensor 476 can measure the force used to operate the knife element, for example, to cut the tissue captured between the anvil and the staple cartridge. A

magnetic field sensor can be employed to measure the thickness of the captured tissue. The measurement of the magnetic field sensor also may be converted to a digital signal and provided to the processor 462.

[0160] The measurements of the tissue compression, the tissue thickness, and/or the force required to close the end effector on the tissue, as respectively measured by the sensors 474, 476, can be used by the microcontroller 461 to characterize the selected position of the firing member and/or the corresponding value of the speed of the firing member. In one instance, a memory 468 may store a technique, an equation, and/or a lookup table which can be employed by the microcontroller 461 in the assessment.

[0161] The control system 470 of the surgical instrument or tool also may comprise wired or wireless communication circuits to communicate with the modular communication hub as shown in FIGS. 8-11.

[0162] FIG. 13 illustrates a control circuit 500 configured to control aspects of the surgical instrument or tool according to one aspect of this disclosure. The control circuit 500 can be configured to implement various processes described herein. The control circuit 500 may comprise a microcontroller comprising one or more processors 502 (e.g., microprocessor, microcontroller) coupled to at least one memory circuit 504. The memory circuit 504 stores machine-executable instructions that, when executed by the processor 502, cause the processor 502 to execute machine instructions to implement various processes described herein. The processor 502 may be any one of a number of single-core or multicore processors known in the art. The memory circuit 504 may comprise volatile and non-volatile storage media. The processor 502 may include an instruction processing unit 506 and an arithmetic unit 508. The instruction processing unit may be configured to receive instructions from the memory circuit 504 of this disclosure.

[0163] FIG. 14 illustrates a combinational logic circuit 510 configured to control aspects of the surgical instrument or tool according to one aspect of this disclosure. The combinational logic circuit 510 can be configured to implement various processes described herein. The combinational logic circuit 510 may comprise a finite state machine comprising a combinational logic 512 configured to receive data associated with the surgical instrument or tool at an input 514, process the data by the combinational logic 512, and provide an output 516.

[0164] FIG. 15 illustrates a sequential logic circuit 520 configured to control aspects of the surgical instrument or tool according to one aspect of this disclosure. The sequential logic circuit 520 or the combinational logic 522 can be configured to implement various processes described herein. The sequential logic circuit 520 may comprise a finite state machine. The sequential logic circuit 520 may comprise a combinational logic 522, at least one memory circuit 524, and a clock 529, for example. The at least one memory circuit 524 can store a current state of the finite state machine. In certain instances, the sequential logic circuit 520 may be synchronous or asynchronous. The combinational logic 522 is configured to receive data associated with the surgical instrument or tool from an input 526, process the data by the combinational logic 522, and provide an output 528. In other aspects, the circuit may comprise a combination of a processor (e.g., processor 502, FIG. 13) and a finite state machine to implement various processes herein. In other aspects, the finite state machine may comprise a

combination of a combinational logic circuit (e.g., combinational logic circuit **510**, FIG. **14**) and the sequential logic circuit **520**.

[0165] FIG. **16** illustrates a surgical instrument or tool comprising a plurality of motors which can be activated to perform various functions. In certain instances, a first motor can be activated to perform a first function, a second motor can be activated to perform a second function, a third motor can be activated to perform a third function, a fourth motor can be activated to perform a fourth function, and so on. In certain instances, the plurality of motors of robotic surgical instrument **600** can be individually activated to cause firing, closure, and/or articulation motions in the end effector. The firing, closure, and/or articulation motions can be transmitted to the end effector through a shaft assembly, for example.

[0166] In certain instances, the surgical instrument system or tool may include a firing motor **602**. The firing motor **602** may be operably coupled to a firing motor drive assembly **604** which can be configured to transmit firing motions, generated by the motor **602** to the end effector, in particular to displace the I-beam element. In certain instances, the firing motions generated by the motor **602** may cause the staples to be deployed from the staple cartridge into tissue captured by the end effector and/or the cutting edge of the I-beam element to be advanced to cut the captured tissue, for example. The I-beam element may be retracted by reversing the direction of the motor **602**.

[0167] In certain instances, the surgical instrument or tool may include a closure motor **603**. The closure motor **603** may be operably coupled to a closure motor drive assembly **605** which can be configured to transmit closure motions, generated by the motor **603** to the end effector, in particular to displace a closure tube to close the anvil and compress tissue between the anvil and the staple cartridge. The closure motions may cause the end effector to transition from an open configuration to an approximated configuration to capture tissue, for example. The end effector may be transitioned to an open position by reversing the direction of the motor **603**.

[0168] In certain instances, the surgical instrument or tool may include one or more articulation motors **606a**, **606b**, for example. The motors **606a**, **606b** may be operably coupled to respective articulation motor drive assemblies **608a**, **608b**, which can be configured to transmit articulation motions generated by the motors **606a**, **606b** to the end effector. In certain instances, the articulation motions may cause the end effector to articulate relative to the shaft, for example.

[0169] As described above, the surgical instrument or tool may include a plurality of motors which may be configured to perform various independent functions. In certain instances, the plurality of motors of the surgical instrument or tool can be individually or separately activated to perform one or more functions while the other motors remain inactive. For example, the articulation motors **606a**, **606b** can be activated to cause the end effector to be articulated while the firing motor **602** remains inactive. Alternatively, the firing motor **602** can be activated to fire the plurality of staples, and/or to advance the cutting edge, while the articulation motor **606** remains inactive. Furthermore the closure motor **603** may be activated simultaneously with the firing motor **602** to cause the closure tube and the I-beam element to advance distally as described in more detail hereinbelow.

[0170] In certain instances, the surgical instrument or tool may include a common control module **610** which can be employed with a plurality of motors of the surgical instrument or tool. In certain instances, the common control module **610** may accommodate one of the plurality of motors at a time. For example, the common control module **610** can be couplable to and separable from the plurality of motors of the robotic surgical instrument individually. In certain instances, a plurality of the motors of the surgical instrument or tool may share one or more common control modules such as the common control module **610**. In certain instances, a plurality of motors of the surgical instrument or tool can be individually and selectively engaged with the common control module **610**. In certain instances, the common control module **610** can be selectively switched from interfacing with one of a plurality of motors of the surgical instrument or tool to interfacing with another one of the plurality of motors of the surgical instrument or tool.

[0171] In at least one example, the common control module **610** can be selectively switched between operable engagement with the articulation motors **606a**, **606b** and operable engagement with either the firing motor **602** or the closure motor **603**. In at least one example, as illustrated in FIG. **16**, a switch **614** can be moved or transitioned between a plurality of positions and/or states. In a first position **616**, the switch **614** may electrically couple the common control module **610** to the firing motor **602**; in a second position **617**, the switch **614** may electrically couple the common control module **610** to the closure motor **603**; in a third position **618a**, the switch **614** may electrically couple the common control module **610** to the first articulation motor **606a**; and in a fourth position **618b**, the switch **614** may electrically couple the common control module **610** to the second articulation motor **606b**, for example. In certain instances, separate common control modules **610** can be electrically coupled to the firing motor **602**, the closure motor **603**, and the articulations motor **606a**, **606b** at the same time. In certain instances, the switch **614** may be a mechanical switch, an electromechanical switch, a solid-state switch, or any suitable switching mechanism.

[0172] Each of the motors **602**, **603**, **606a**, **606b** may comprise a torque sensor to measure the output torque on the shaft of the motor. The force on an end effector may be sensed in any conventional manner, such as by force sensors on the outer sides of the jaws or by a torque sensor for the motor actuating the jaws.

[0173] In various instances, as illustrated in FIG. **16**, the common control module **610** may comprise a motor driver **626** which may comprise one or more H-Bridge FETs. The motor driver **626** may modulate the power transmitted from a power source **628** to a motor coupled to the common control module **610** based on input from a microcontroller **620** (the “controller”), for example. In certain instances, the microcontroller **620** can be employed to determine the current drawn by the motor, for example, while the motor is coupled to the common control module **610**, as described above.

[0174] In certain instances, the microcontroller **620** may include a microprocessor **622** (the “processor”) and one or more non-transitory computer-readable mediums or memory units **624** (the “memory”). In certain instances, the memory **624** may store various program instructions, which when executed may cause the processor **622** to perform a plurality of functions and/or calculations described herein.

In certain instances, one or more of the memory units **624** may be coupled to the processor **622**, for example.

[0175] In certain instances, the power source **628** can be employed to supply power to the microcontroller **620**, for example. In certain instances, the power source **628** may comprise a battery (or “battery pack” or “power pack”), such as a lithium-ion battery, for example. In certain instances, the battery pack may be configured to be releasably mounted to a handle for supplying power to the surgical instrument **600**. A number of battery cells connected in series may be used as the power source **628**. In certain instances, the power source **628** may be replaceable and/or rechargeable, for example.

[0176] In various instances, the processor **622** may control the motor driver **626** to control the position, direction of rotation, and/or velocity of a motor that is coupled to the common control module **610**. In certain instances, the processor **622** can signal the motor driver **626** to stop and/or disable a motor that is coupled to the common control module **610**. It should be understood that the term “processor” as used herein includes any suitable microprocessor, microcontroller, or other basic computing device that incorporates the functions of a computer’s central processing unit (CPU) on an integrated circuit or, at most, a few integrated circuits. The processor is a multipurpose, programmable device that accepts digital data as input, processes it according to instructions stored in its memory, and provides results as output. It is an example of sequential digital logic, as it has internal memory. Processors operate on numbers and symbols represented in the binary numeral system.

[0177] In one instance, the processor **622** may be any single-core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In certain instances, the microcontroller **620** may be an LM4F230H5QR, available from Texas Instruments, for example. In at least one example, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising an on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle SRAM, an internal ROM loaded with StellarisWare® software, a 2 KB EEPROM, one or more PWM modules, one or more QEI analogs, one or more 12-bit ADCs with 12 analog input channels, among other features that are readily available for the product datasheet. Other microcontrollers may be readily substituted for use with the module **4410**. Accordingly, the present disclosure should not be limited in this context.

[0178] In certain instances, the memory **624** may include program instructions for controlling each of the motors of the surgical instrument **600** that are couplable to the common control module **610**. For example, the memory **624** may include program instructions for controlling the firing motor **602**, the closure motor **603**, and the articulation motors **606a**, **606b**. Such program instructions may cause the processor **622** to control the firing, closure, and articulation functions in accordance with inputs from algorithms or control programs of the surgical instrument or tool.

[0179] In certain instances, one or more mechanisms and/or sensors such as, for example, sensors **630** can be employed to alert the processor **622** to the program instructions that should be used in a particular setting. For example, the sensors **630** may alert the processor **622** to use the program instructions associated with firing, closing, and

articulating the end effector. In certain instances, the sensors **630** may comprise position sensors which can be employed to sense the position of the switch **614**, for example. Accordingly, the processor **622** may use the program instructions associated with firing the I-beam of the end effector upon detecting, through the sensors **630** for example, that the switch **614** is in the first position **616**; the processor **622** may use the program instructions associated with closing the anvil upon detecting, through the sensors **630** for example, that the switch **614** is in the second position **617**; and the processor **622** may use the program instructions associated with articulating the end effector upon detecting, through the sensors **630** for example, that the switch **614** is in the third or fourth position **618a**, **618b**.

[0180] FIG. 17 is a schematic diagram of a robotic surgical instrument **700** configured to operate a surgical tool described herein according to one aspect of this disclosure. The robotic surgical instrument **700** may be programmed or configured to control distal/proximal translation of a displacement member, distal/proximal displacement of a closure tube, shaft rotation, and articulation, either with single or multiple articulation drive links. In one aspect, the surgical instrument **700** may be programmed or configured to individually control a firing member, a closure member, a shaft member, and/or one or more articulation members. The surgical instrument **700** comprises a control circuit **710** configured to control motor-driven firing members, closure members, shaft members, and/or one or more articulation members.

[0181] In one aspect, the robotic surgical instrument **700** comprises a control circuit **710** configured to control an anvil **716** and an I-beam **714** (including a sharp cutting edge) portion of an end effector **702**, a removable staple cartridge **718**, a shaft **740**, and one or more articulation members **742a**, **742b** via a plurality of motors **704a-704e**. A position sensor **734** may be configured to provide position feedback of the I-beam **714** to the control circuit **710**. Other sensors **738** may be configured to provide feedback to the control circuit **710**. A timer/counter **731** provides timing and counting information to the control circuit **710**. An energy source **712** may be provided to operate the motors **704a-704e**, and a current sensor **736** provides motor current feedback to the control circuit **710**. The motors **704a-704e** can be operated individually by the control circuit **710** in an open-loop or closed-loop feedback control.

[0182] In one aspect, the control circuit **710** may comprise one or more microcontrollers, microprocessors, or other suitable processors for executing instructions that cause the processor or processors to perform one or more tasks. In one aspect, a timer/counter **731** provides an output signal, such as the elapsed time or a digital count, to the control circuit **710** to correlate the position of the I-beam **714** as determined by the position sensor **734** with the output of the timer/counter **731** such that the control circuit **710** can determine the position of the I-beam **714** at a specific time (t) relative to a starting position or the time (t) when the I-beam **714** is at a specific position relative to a starting position. The timer/counter **731** may be configured to measure elapsed time, count external events, or time external events.

[0183] In one aspect, the control circuit **710** may be programmed to control functions of the end effector **702** based on one or more tissue conditions. The control circuit **710** may be programmed to sense tissue conditions, such as thickness, either directly or indirectly, as described herein.

The control circuit **710** may be programmed to select a firing control program or closure control program based on tissue conditions. A firing control program may describe the distal motion of the displacement member. Different firing control programs may be selected to better treat different tissue conditions. For example, when thicker tissue is present, the control circuit **710** may be programmed to translate the displacement member at a lower velocity and/or with lower power. When thinner tissue is present, the control circuit **710** may be programmed to translate the displacement member at a higher velocity and/or with higher power. A closure control program may control the closure force applied to the tissue by the anvil **716**. Other control programs control the rotation of the shaft **740** and the articulation members **742a**, **742b**.

[0184] In one aspect, the control circuit **710** may generate motor set point signals. The motor set point signals may be provided to various motor controllers **708a-708e**. The motor controllers **708a-708e** may comprise one or more circuits configured to provide motor drive signals to the motors **704a-704e** to drive the motors **704a-704e** as described herein. In some examples, the motors **704a-704e** may be brushed DC electric motors. For example, the velocity of the motors **704a-704e** may be proportional to the respective motor drive signals. In some examples, the motors **704a-704e** may be brushless DC electric motors, and the respective motor drive signals may comprise a PWM signal provided to one or more stator windings of the motors **704a-704e**. Also, in some examples, the motor controllers **708a-708e** may be omitted and the control circuit **710** may generate the motor drive signals directly.

[0185] In one aspect, the control circuit **710** may initially operate each of the motors **704a-704e** in an open-loop configuration for a first open-loop portion of a stroke of the displacement member. Based on the response of the robotic surgical instrument **700** during the open-loop portion of the stroke, the control circuit **710** may select a firing control program in a closed-loop configuration. The response of the instrument may include a translation distance of the displacement member during the open-loop portion, a time elapsed during the open-loop portion, the energy provided to one of the motors **704a-704e** during the open-loop portion, a sum of pulse widths of a motor drive signal, etc. After the open-loop portion, the control circuit **710** may implement the selected firing control program for a second portion of the displacement member stroke. For example, during a closed-loop portion of the stroke, the control circuit **710** may modulate one of the motors **704a-704e** based on translation data describing a position of the displacement member in a closed-loop manner to translate the displacement member at a constant velocity.

[0186] In one aspect, the motors **704a-704e** may receive power from an energy source **712**. The energy source **712** may be a DC power supply driven by a main alternating current power source, a battery, a super capacitor, or any other suitable energy source. The motors **704a-704e** may be mechanically coupled to individual movable mechanical elements such as the I-beam **714**, anvil **716**, shaft **740**, articulation **742a**, and articulation **742b** via respective transmissions **706a-706e**. The transmissions **706a-706e** may include one or more gears or other linkage components to couple the motors **704a-704e** to movable mechanical elements. A position sensor **734** may sense a position of the I-beam **714**. The position sensor **734** may be or include any

type of sensor that is capable of generating position data that indicate a position of the I-beam **714**. In some examples, the position sensor **734** may include an encoder configured to provide a series of pulses to the control circuit **710** as the I-beam **714** translates distally and proximally. The control circuit **710** may track the pulses to determine the position of the I-beam **714**. Other suitable position sensors may be used, including, for example, a proximity sensor. Other types of position sensors may provide other signals indicating motion of the I-beam **714**. Also, in some examples, the position sensor **734** may be omitted. Where any of the motors **704a-704e** is a stepper motor, the control circuit **710** may track the position of the I-beam **714** by aggregating the number and direction of steps that the motor **704** has been instructed to execute. The position sensor **734** may be located in the end effector **702** or at any other portion of the instrument. The outputs of each of the motors **704a-704e** include a torque sensor **744a-744e** to sense force and have an encoder to sense rotation of the drive shaft.

[0187] In one aspect, the control circuit **710** is configured to drive a firing member such as the I-beam **714** portion of the end effector **702**. The control circuit **710** provides a motor set point to a motor control **708a**, which provides a drive signal to the motor **704a**. The output shaft of the motor **704a** is coupled to a torque sensor **744a**. The torque sensor **744a** is coupled to a transmission **706a** which is coupled to the I-beam **714**. The transmission **706a** comprises movable mechanical elements such as rotating elements and a firing member to control the movement of the I-beam **714** distally and proximally along a longitudinal axis of the end effector **702**. In one aspect, the motor **704a** may be coupled to the knife gear assembly, which includes a knife gear reduction set that includes a first knife drive gear and a second knife drive gear. A torque sensor **744a** provides a firing force feedback signal to the control circuit **710**. The firing force signal represents the force required to fire or displace the I-beam **714**. A position sensor **734** may be configured to provide the position of the I-beam **714** along the firing stroke or the position of the firing member as a feedback signal to the control circuit **710**. The end effector **702** may include additional sensors **738** configured to provide feedback signals to the control circuit **710**. When ready to use, the control circuit **710** may provide a firing signal to the motor control **708a**. In response to the firing signal, the motor **704a** may drive the firing member distally along the longitudinal axis of the end effector **702** from a proximal stroke start position to a stroke end position distal to the stroke start position. As the firing member translates distally, an I-beam **714**, with a cutting element positioned at a distal end, advances distally to cut tissue located between the staple cartridge **718** and the anvil **716**.

[0188] In one aspect, the control circuit **710** is configured to drive a closure member such as the anvil **716** portion of the end effector **702**. The control circuit **710** provides a motor set point to a motor control **708b**, which provides a drive signal to the motor **704b**. The output shaft of the motor **704b** is coupled to a torque sensor **744b**. The torque sensor **744b** is coupled to a transmission **706b** which is coupled to the anvil **716**. The transmission **706b** comprises movable mechanical elements such as rotating elements and a closure member to control the movement of the anvil **716** from the open and closed positions. In one aspect, the motor **704b** is coupled to a closure gear assembly, which includes a closure reduction gear set that is supported in meshing engagement

with the closure spur gear. The torque sensor **744b** provides a closure force feedback signal to the control circuit **710**. The closure force feedback signal represents the closure force applied to the anvil **716**. The position sensor **734** may be configured to provide the position of the closure member as a feedback signal to the control circuit **710**. Additional sensors **738** in the end effector **702** may provide the closure force feedback signal to the control circuit **710**. The pivotable anvil **716** is positioned opposite the staple cartridge **718**. When ready to use, the control circuit **710** may provide a closure signal to the motor control **708b**. In response to the closure signal, the motor **704b** advances a closure member to grasp tissue between the anvil **716** and the staple cartridge **718**.

[0189] In one aspect, the control circuit **710** is configured to rotate a shaft member such as the shaft **740** to rotate the end effector **702**. The control circuit **710** provides a motor set point to a motor control **708c**, which provides a drive signal to the motor **704c**. The output shaft of the motor **704c** is coupled to a torque sensor **744c**. The torque sensor **744c** is coupled to a transmission **706c** which is coupled to the shaft **740**. The transmission **706c** comprises movable mechanical elements such as rotating elements to control the rotation of the shaft **740** clockwise or counterclockwise up to and over 360°. In one aspect, the motor **704c** is coupled to the rotational transmission assembly, which includes a tube gear segment that is formed on (or attached to) the proximal end of the proximal closure tube for operable engagement by a rotational gear assembly that is operably supported on the tool mounting plate. The torque sensor **744c** provides a rotation force feedback signal to the control circuit **710**. The rotation force feedback signal represents the rotation force applied to the shaft **740**. The position sensor **734** may be configured to provide the position of the closure member as a feedback signal to the control circuit **710**. Additional sensors **738** such as a shaft encoder may provide the rotational position of the shaft **740** to the control circuit **710**.

[0190] In one aspect, the control circuit **710** is configured to articulate the end effector **702**. The control circuit **710** provides a motor set point to a motor control **708d**, which provides a drive signal to the motor **704d**. The output shaft of the motor **704d** is coupled to a torque sensor **744d**. The torque sensor **744d** is coupled to a transmission **706d** which is coupled to an articulation member **742a**. The transmission **706d** comprises movable mechanical elements such as articulation elements to control the articulation of the end effector **702** $\pm 65^\circ$. In one aspect, the motor **704d** is coupled to an articulation nut, which is rotatably journaled on the proximal end portion of the distal spine portion and is rotatably driven thereon by an articulation gear assembly. The torque sensor **744d** provides an articulation force feedback signal to the control circuit **710**. The articulation force feedback signal represents the articulation force applied to the end effector **702**. Sensors **738**, such as an articulation encoder, may provide the articulation position of the end effector **702** to the control circuit **710**.

[0191] In another aspect, the articulation function of the robotic surgical system **700** may comprise two articulation members, or links, **742a**, **742b**. These articulation members **742a**, **742b** are driven by separate disks on the robot interface (the rack) which are driven by the two motors **708d**, **708e**. When the separate firing motor **704a** is provided, each of articulation links **742a**, **742b** can be antago-

nistically driven with respect to the other link in order to provide a resistive holding motion and a load to the head when it is not moving and to provide an articulation motion as the head is articulated. The articulation members **742a**, **742b** attach to the head at a fixed radius as the head is rotated. Accordingly, the mechanical advantage of the push-and-pull link changes as the head is rotated. This change in the mechanical advantage may be more pronounced with other articulation link drive systems.

[0192] In one aspect, the one or more motors **704a-704e** may comprise a brushed DC motor with a gearbox and mechanical links to a firing member, closure member, or articulation member. Another example includes electric motors **704a-704e** that operate the movable mechanical elements such as the displacement member, articulation links, closure tube, and shaft. An outside influence is an unmeasured, unpredictable influence of things like tissue, surrounding bodies, and friction on the physical system. Such outside influence can be referred to as drag, which acts in opposition to one of electric motors **704a-704e**. The outside influence, such as drag, may cause the operation of the physical system to deviate from a desired operation of the physical system.

[0193] In one aspect, the position sensor **734** may be implemented as an absolute positioning system. In one aspect, the position sensor **734** may comprise a magnetic rotary absolute positioning system implemented as an AS5055EQFT single-chip magnetic rotary position sensor available from Austria Microsystems, AG. The position sensor **734** may interface with the control circuit **710** to provide an absolute positioning system. The position may include multiple Hall-effect elements located above a magnet and coupled to a CORDIC processor, also known as the digit-by-digit method and Volder's algorithm, that is provided to implement a simple and efficient algorithm to calculate hyperbolic and trigonometric functions that require only addition, subtraction, bitshift, and table lookup operations.

[0194] In one aspect, the control circuit **710** may be in communication with one or more sensors **738**. The sensors **738** may be positioned on the end effector **702** and adapted to operate with the robotic surgical instrument **700** to measure the various derived parameters such as the gap distance versus time, tissue compression versus time, and anvil strain versus time. The sensors **738** may comprise a magnetic sensor, a magnetic field sensor, a strain gauge, a load cell, a pressure sensor, a force sensor, a torque sensor, an inductive sensor such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor for measuring one or more parameters of the end effector **702**. The sensors **738** may include one or more sensors. The sensors **738** may be located on the staple cartridge **718** deck to determine tissue location using segmented electrodes. The torque sensors **744a-744e** may be configured to sense force such as firing force, closure force, and/or articulation force, among others. Accordingly, the control circuit **710** can sense (1) the closure load experienced by the distal closure tube and its position, (2) the firing member at the rack and its position, (3) what portion of the staple cartridge **718** has tissue on it, and (4) the load and position on both articulation rods.

[0195] In one aspect, the one or more sensors **738** may comprise a strain gauge, such as a micro-strain gauge, configured to measure the magnitude of the strain in the

anvil **716** during a clamped condition. The strain gauge provides an electrical signal whose amplitude varies with the magnitude of the strain. The sensors **738** may comprise a pressure sensor configured to detect a pressure generated by the presence of compressed tissue between the anvil **716** and the staple cartridge **718**. The sensors **738** may be configured to detect impedance of a tissue section located between the anvil **716** and the staple cartridge **718** that is indicative of the thickness and/or fullness of tissue located therebetween.

[0196] In one aspect, the sensors **738** may be implemented as one or more limit switches, electromechanical devices, solid-state switches, Hall-effect devices, magneto-resistive (MR) devices, giant magneto-resistive (GMR) devices, magnetometers, among others. In other implementations, the sensors **738** may be implemented as solid-state switches that operate under the influence of light, such as optical sensors, IR sensors, ultraviolet sensors, among others. Still, the switches may be solid-state devices such as transistors (e.g., FET, junction FET, MOSFET, bipolar, and the like). In other implementations, the sensors **738** may include electrical conductorless switches, ultrasonic switches, accelerometers, and inertial sensors, among others.

[0197] In one aspect, the sensors **738** may be configured to measure forces exerted on the anvil **716** by the closure drive system. For example, one or more sensors **738** can be at an interaction point between the closure tube and the anvil **716** to detect the closure forces applied by the closure tube to the anvil **716**. The forces exerted on the anvil **716** can be representative of the tissue compression experienced by the tissue section captured between the anvil **716** and the staple cartridge **718**. The one or more sensors **738** can be positioned at various interaction points along the closure drive system to detect the closure forces applied to the anvil **716** by the closure drive system. The one or more sensors **738** may be sampled in real time during a clamping operation by the processor of the control circuit **710**. The control circuit **710** receives real-time sample measurements to provide and analyze time-based information and assess, in real time, closure forces applied to the anvil **716**.

[0198] In one aspect, a current sensor **736** can be employed to measure the current drawn by each of the motors **704a-704e**. The force required to advance any of the movable mechanical elements such as the I-beam **714** corresponds to the current drawn by one of the motors **704a-704e**. The force is converted to a digital signal and provided to the control circuit **710**. The control circuit **710** can be configured to simulate the response of the actual system of the instrument in the software of the controller. A displacement member can be actuated to move an I-beam **714** in the end effector **702** at or near a target velocity. The robotic surgical instrument **700** can include a feedback controller, which can be one of any feedback controllers, including, but not limited to a PID, a state feedback, a linear-quadratic (LQR), and/or an adaptive controller, for example. The robotic surgical instrument **700** can include a power source to convert the signal from the feedback controller into a physical input such as case voltage, PWM voltage, frequency modulated voltage, current, torque, and/or force, for example. Additional details are disclosed in U.S. patent application Ser. No. 15/636,829, titled CLOSED LOOP VELOCITY CONTROL TECHNIQUES FOR ROBOTIC SURGICAL INSTRUMENT, filed Jun. 29, 2017, now U.S. Pat. No. 10,932,772, which is herein incorporated by reference in its entirety.

[0199] FIG. **18** illustrates a block diagram of a surgical instrument **750** programmed to control the distal translation of a displacement member according to one aspect of this disclosure. In one aspect, the surgical instrument **750** is programmed to control the distal translation of a displacement member such as the I-beam **764**. The surgical instrument **750** comprises an end effector **752** that may comprise an anvil **766**, an I-beam **764** (including a sharp cutting edge), and a removable staple cartridge **768**.

[0200] The position, movement, displacement, and/or translation of a linear displacement member, such as the I-beam **764**, can be measured by an absolute positioning system, sensor arrangement, and position sensor **784**. Because the I-beam **764** is coupled to a longitudinally movable drive member, the position of the I-beam **764** can be determined by measuring the position of the longitudinally movable drive member employing the position sensor **784**. Accordingly, in the following description, the position, displacement, and/or translation of the I-beam **764** can be achieved by the position sensor **784** as described herein. A control circuit **760** may be programmed to control the translation of the displacement member, such as the I-beam **764**. The control circuit **760**, in some examples, may comprise one or more microcontrollers, microprocessors, or other suitable processors for executing instructions that cause the processor or processors to control the displacement member, e.g., the I-beam **764**, in the manner described. In one aspect, a timer/counter **781** provides an output signal, such as the elapsed time or a digital count, to the control circuit **760** to correlate the position of the I-beam **764** as determined by the position sensor **784** with the output of the timer/counter **781** such that the control circuit **760** can determine the position of the I-beam **764** at a specific time (t) relative to a starting position. The timer/counter **781** may be configured to measure elapsed time, count external events, or time external events.

[0201] The control circuit **760** may generate a motor set point signal **772**. The motor set point signal **772** may be provided to a motor controller **758**. The motor controller **758** may comprise one or more circuits configured to provide a motor drive signal **774** to the motor **754** to drive the motor **754** as described herein. In some examples, the motor **754** may be a brushed DC electric motor. For example, the velocity of the motor **754** may be proportional to the motor drive signal **774**. In some examples, the motor **754** may be a brushless DC electric motor and the motor drive signal **774** may comprise a PWM signal provided to one or more stator windings of the motor **754**. Also, in some examples, the motor controller **758** may be omitted, and the control circuit **760** may generate the motor drive signal **774** directly.

[0202] The motor **754** may receive power from an energy source **762**. The energy source **762** may be or include a battery, a super capacitor, or any other suitable energy source. The motor **754** may be mechanically coupled to the I-beam **764** via a transmission **756**. The transmission **756** may include one or more gears or other linkage components to couple the motor **754** to the I-beam **764**. A position sensor **784** may sense a position of the I-beam **764**. The position sensor **784** may be or include any type of sensor that is capable of generating position data that indicate a position of the I-beam **764**. In some examples, the position sensor **784** may include an encoder configured to provide a series of pulses to the control circuit **760** as the I-beam **764** translates distally and proximally. The control circuit **760** may track

the pulses to determine the position of the I-beam **764**. Other suitable position sensors may be used, including, for example, a proximity sensor. Other types of position sensors may provide other signals indicating motion of the I-beam **764**. Also, in some examples, the position sensor **784** may be omitted. Where the motor **754** is a stepper motor, the control circuit **760** may track the position of the I-beam **764** by aggregating the number and direction of steps that the motor **754** has been instructed to execute. The position sensor **784** may be located in the end effector **752** or at any other portion of the instrument.

[0203] The control circuit **760** may be in communication with one or more sensors **788**. The sensors **788** may be positioned on the end effector **752** and adapted to operate with the surgical instrument **750** to measure the various derived parameters such as gap distance versus time, tissue compression versus time, and anvil strain versus time. The sensors **788** may comprise a magnetic sensor, a magnetic field sensor, a strain gauge, a pressure sensor, a force sensor, an inductive sensor such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor for measuring one or more parameters of the end effector **752**. The sensors **788** may include one or more sensors.

[0204] The one or more sensors **788** may comprise a strain gauge, such as a micro-strain gauge, configured to measure the magnitude of the strain in the anvil **766** during a clamped condition. The strain gauge provides an electrical signal whose amplitude varies with the magnitude of the strain. The sensors **788** may comprise a pressure sensor configured to detect a pressure generated by the presence of compressed tissue between the anvil **766** and the staple cartridge **768**. The sensors **788** may be configured to detect impedance of a tissue section located between the anvil **766** and the staple cartridge **768** that is indicative of the thickness and/or fullness of tissue located therebetween.

[0205] The sensors **788** may be configured to measure forces exerted on the anvil **766** by a closure drive system. For example, one or more sensors **788** can be at an interaction point between a closure tube and the anvil **766** to detect the closure forces applied by a closure tube to the anvil **766**. The forces exerted on the anvil **766** can be representative of the tissue compression experienced by the tissue section captured between the anvil **766** and the staple cartridge **768**. The one or more sensors **788** can be positioned at various interaction points along the closure drive system to detect the closure forces applied to the anvil **766** by the closure drive system. The one or more sensors **788** may be sampled in real time during a clamping operation by a processor of the control circuit **760**. The control circuit **760** receives real-time sample measurements to provide and analyze time-based information and assess, in real time, closure forces applied to the anvil **766**.

[0206] A current sensor **786** can be employed to measure the current drawn by the motor **754**. The force required to advance the I-beam **764** corresponds to the current drawn by the motor **754**. The force is converted to a digital signal and provided to the control circuit **760**.

[0207] The control circuit **760** can be configured to simulate the response of the actual system of the instrument in the software of the controller. A displacement member can be actuated to move an I-beam **764** in the end effector **752** at or near a target velocity. The surgical instrument **750** can include a feedback controller, which can be one of any

feedback controllers, including, but not limited to a PID, a state feedback, LQR, and/or an adaptive controller, for example. The surgical instrument **750** can include a power source to convert the signal from the feedback controller into a physical input such as case voltage, PWM voltage, frequency modulated voltage, current, torque, and/or force, for example.

[0208] The actual drive system of the surgical instrument **750** is configured to drive the displacement member, cutting member, or I-beam **764**, by a brushed DC motor with gearbox and mechanical links to an articulation and/or knife system. Another example is the electric motor **754** that operates the displacement member and the articulation driver, for example, of an interchangeable shaft assembly. An outside influence is an unmeasured, unpredictable influence of things like tissue, surrounding bodies and friction on the physical system. Such outside influence can be referred to as drag which acts in opposition to the electric motor **754**. The outside influence, such as drag, may cause the operation of the physical system to deviate from a desired operation of the physical system.

[0209] Various example aspects are directed to a surgical instrument **750** comprising an end effector **752** with motor-driven surgical stapling and cutting implements. For example, a motor **754** may drive a displacement member distally and proximally along a longitudinal axis of the end effector **752**. The end effector **752** may comprise a pivotable anvil **766** and, when configured for use, a staple cartridge **768** positioned opposite the anvil **766**. A clinician may grasp tissue between the anvil **766** and the staple cartridge **768**, as described herein. When ready to use the instrument **750**, the clinician may provide a firing signal, for example by depressing a trigger of the instrument **750**. In response to the firing signal, the motor **754** may drive the displacement member distally along the longitudinal axis of the end effector **752** from a proximal stroke begin position to a stroke end position distal of the stroke begin position. As the displacement member translates distally, an I-beam **764** with a cutting element positioned at a distal end, may cut the tissue between the staple cartridge **768** and the anvil **766**.

[0210] In various examples, the surgical instrument **750** may comprise a control circuit **760** programmed to control the distal translation of the displacement member, such as the I-beam **764**, for example, based on one or more tissue conditions. The control circuit **760** may be programmed to sense tissue conditions, such as thickness, either directly or indirectly, as described herein. The control circuit **760** may be programmed to select a firing control program based on tissue conditions. A firing control program may describe the distal motion of the displacement member. Different firing control programs may be selected to better treat different tissue conditions. For example, when thicker tissue is present, the control circuit **760** may be programmed to translate the displacement member at a lower velocity and/or with lower power. When thinner tissue is present, the control circuit **760** may be programmed to translate the displacement member at a higher velocity and/or with higher power.

[0211] In some examples, the control circuit **760** may initially operate the motor **754** in an open loop configuration for a first open loop portion of a stroke of the displacement member. Based on a response of the instrument **750** during the open loop portion of the stroke, the control circuit **760** may select a firing control program. The response of the instrument may include, a translation distance of the dis-

placement member during the open loop portion, a time elapsed during the open loop portion, energy provided to the motor 754 during the open loop portion, a sum of pulse widths of a motor drive signal, etc. After the open loop portion, the control circuit 760 may implement the selected firing control program for a second portion of the displacement member stroke. For example, during the closed loop portion of the stroke, the control circuit 760 may modulate the motor 754 based on translation data describing a position of the displacement member in a closed loop manner to translate the displacement member at a constant velocity. Additional details are disclosed in U.S. patent application Ser. No. 15/720,852, titled SYSTEM AND METHODS FOR CONTROLLING A DISPLAY OF A SURGICAL INSTRUMENT, filed Sep. 29, 2017, now U.S. Pat. No. 10,743,872, which is herein incorporated by reference in its entirety.

[0212] FIG. 19 is a schematic diagram of a surgical instrument 790 configured to control various functions according to one aspect of this disclosure. In one aspect, the surgical instrument 790 is programmed to control distal translation of a displacement member such as the I-beam 764. The surgical instrument 790 comprises an end effector 792 that may comprise an anvil 766, an I-beam 764, and a removable staple cartridge 768 which may be interchanged with an RF cartridge 796 (shown in dashed line).

[0213] In one aspect, sensors 788 may be implemented as a limit switch, electromechanical device, solid-state switches, Hall-effect devices, MR devices, GMR devices, magnetometers, among others. In other implementations, the sensors 638 may be solid-state switches that operate under the influence of light, such as optical sensors, IR sensors, ultraviolet sensors, among others. Still, the switches may be solid-state devices such as transistors (e.g., FET, junction FET, MOSFET, bipolar, and the like). In other implementations, the sensors 788 may include electrical conductorless switches, ultrasonic switches, accelerometers, and inertial sensors, among others.

[0214] In one aspect, the position sensor 784 may be implemented as an absolute positioning system comprising a magnetic rotary absolute positioning system implemented as an AS5055EQFT single-chip magnetic rotary position sensor available from Austria Microsystems, AG. The position sensor 784 may interface with the control circuit 760 to provide an absolute positioning system. The position may include multiple Hall-effect elements located above a magnet and coupled to a CORDIC processor, also known as the digit-by-digit method and Volder's algorithm, that is provided to implement a simple and efficient algorithm to calculate hyperbolic and trigonometric functions that require only addition, subtraction, bitshift, and table lookup operations.

[0215] In one aspect, the I-beam 764 may be implemented as a knife member comprising a knife body that operably supports a tissue cutting blade thereon and may further include anvil engagement tabs or features and channel engagement features or a foot. In one aspect, the staple cartridge 768 may be implemented as a standard (mechanical) surgical fastener cartridge. In one aspect, the RF cartridge 796 may be implemented as an RF cartridge. These and other sensors arrangements are described in commonly owned U.S. patent application Ser. No. 15/628,175, titled TECHNIQUES FOR ADAPTIVE CONTROL OF MOTOR VELOCITY OF A SURGICAL STAPLING AND CUT-

TING INSTRUMENT, filed Jun. 20, 2017, now U.S. Pat. No. 10,881,399, which is herein incorporated by reference in its entirety.

[0216] The position, movement, displacement, and/or translation of a linear displacement member, such as the I-beam 764, can be measured by an absolute positioning system, sensor arrangement, and position sensor represented as position sensor 784. Because the I-beam 764 is coupled to the longitudinally movable drive member, the position of the I-beam 764 can be determined by measuring the position of the longitudinally movable drive member employing the position sensor 784. Accordingly, in the following description, the position, displacement, and/or translation of the I-beam 764 can be achieved by the position sensor 784 as described herein. A control circuit 760 may be programmed to control the translation of the displacement member, such as the I-beam 764, as described herein. The control circuit 760, in some examples, may comprise one or more micro-controllers, microprocessors, or other suitable processors for executing instructions that cause the processor or processors to control the displacement member, e.g., the I-beam 764, in the manner described. In one aspect, a timer/counter 781 provides an output signal, such as the elapsed time or a digital count, to the control circuit 760 to correlate the position of the I-beam 764 as determined by the position sensor 784 with the output of the timer/counter 781 such that the control circuit 760 can determine the position of the I-beam 764 at a specific time (t) relative to a starting position. The timer/counter 781 may be configured to measure elapsed time, count external events, or time external events.

[0217] The control circuit 760 may generate a motor set point signal 772. The motor set point signal 772 may be provided to a motor controller 758. The motor controller 758 may comprise one or more circuits configured to provide a motor drive signal 774 to the motor 754 to drive the motor 754 as described herein. In some examples, the motor 754 may be a brushed DC electric motor. For example, the velocity of the motor 754 may be proportional to the motor drive signal 774. In some examples, the motor 754 may be a brushless DC electric motor and the motor drive signal 774 may comprise a PWM signal provided to one or more stator windings of the motor 754. Also, in some examples, the motor controller 758 may be omitted, and the control circuit 760 may generate the motor drive signal 774 directly.

[0218] The motor 754 may receive power from an energy source 762. The energy source 762 may be or include a battery, a super capacitor, or any other suitable energy source. The motor 754 may be mechanically coupled to the I-beam 764 via a transmission 756. The transmission 756 may include one or more gears or other linkage components to couple the motor 754 to the I-beam 764. A position sensor 784 may sense a position of the I-beam 764. The position sensor 784 may be or include any type of sensor that is capable of generating position data that indicate a position of the I-beam 764. In some examples, the position sensor 784 may include an encoder configured to provide a series of pulses to the control circuit 760 as the I-beam 764 translates distally and proximally. The control circuit 760 may track the pulses to determine the position of the I-beam 764. Other suitable position sensors may be used, including, for example, a proximity sensor. Other types of position sensors may provide other signals indicating motion of the I-beam 764. Also, in some examples, the position sensor 784 may be

omitted. Where the motor **754** is a stepper motor, the control circuit **760** may track the position of the I-beam **764** by aggregating the number and direction of steps that the motor has been instructed to execute. The position sensor **784** may be located in the end effector **792** or at any other portion of the instrument.

[0219] The control circuit **760** may be in communication with one or more sensors **788**. The sensors **788** may be positioned on the end effector **792** and adapted to operate with the surgical instrument **790** to measure the various derived parameters such as gap distance versus time, tissue compression versus time, and anvil strain versus time. The sensors **788** may comprise a magnetic sensor, a magnetic field sensor, a strain gauge, a pressure sensor, a force sensor, an inductive sensor such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor for measuring one or more parameters of the end effector **792**. The sensors **788** may include one or more sensors.

[0220] The one or more sensors **788** may comprise a strain gauge, such as a micro-strain gauge, configured to measure the magnitude of the strain in the anvil **766** during a clamped condition. The strain gauge provides an electrical signal whose amplitude varies with the magnitude of the strain. The sensors **788** may comprise a pressure sensor configured to detect a pressure generated by the presence of compressed tissue between the anvil **766** and the staple cartridge **768**. The sensors **788** may be configured to detect impedance of a tissue section located between the anvil **766** and the staple cartridge **768** that is indicative of the thickness and/or fullness of tissue located therebetween.

[0221] The sensors **788** may be configured to measure forces exerted on the anvil **766** by the closure drive system. For example, one or more sensors **788** can be at an interaction point between a closure tube and the anvil **766** to detect the closure forces applied by a closure tube to the anvil **766**. The forces exerted on the anvil **766** can be representative of the tissue compression experienced by the tissue section captured between the anvil **766** and the staple cartridge **768**. The one or more sensors **788** can be positioned at various interaction points along the closure drive system to detect the closure forces applied to the anvil **766** by the closure drive system. The one or more sensors **788** may be sampled in real time during a clamping operation by a processor portion of the control circuit **760**. The control circuit **760** receives real-time sample measurements to provide and analyze time-based information and assess, in real time, closure forces applied to the anvil **766**.

[0222] A current sensor **786** can be employed to measure the current drawn by the motor **754**. The force required to advance the I-beam **764** corresponds to the current drawn by the motor **754**. The force is converted to a digital signal and provided to the control circuit **760**.

[0223] An RF energy source **794** is coupled to the end effector **792** and is applied to the RF cartridge **796** when the RF cartridge **796** is loaded in the end effector **792** in place of the staple cartridge **768**. The control circuit **760** controls the delivery of the RF energy to the RF cartridge **796**.

[0224] Additional details are disclosed in U.S. patent application Ser. No. 15/636,096, titled SURGICAL SYSTEM COUPLABLE WITH STAPLE CARTRIDGE AND RADIO FREQUENCY CARTRIDGE, AND METHOD OF USING SAME, filed Jun. 28, 2017, now U.S. Patent Appli-

cation Publication No. 2019/0000478, which is herein incorporated by reference in its entirety.

Generator Hardware

[0225] FIG. 20 is a simplified block diagram of a generator **800** configured to provide inductorless tuning, among other benefits. Additional details of the generator **800** are described in U.S. Pat. No. 9,060,775, titled SURGICAL GENERATOR FOR ULTRASONIC AND ELECTROSURGICAL DEVICES, which issued on Jun. 23, 2015, which is herein incorporated by reference in its entirety. The generator **800** may comprise a patient isolated stage **802** in communication with a non-isolated stage **804** via a power transformer **806**. A secondary winding **808** of the power transformer **806** is contained in the isolated stage **802** and may comprise a tapped configuration (e.g., a center-tapped or a non-center-tapped configuration) to define drive signal outputs **810a**, **810b**, **810c** for delivering drive signals to different surgical instruments, such as, for example, an ultrasonic surgical instrument, an RF electrosurgical instrument, and a multifunction surgical instrument which includes ultrasonic and RF energy modes that can be delivered alone or simultaneously. In particular, drive signal outputs **810a**, **810c** may output an ultrasonic drive signal (e.g., a 420V root-mean-square (RMS) drive signal) to an ultrasonic surgical instrument, and drive signal outputs **810b**, **810c** may output an RF electrosurgical drive signal (e.g., a 100V RMS drive signal) to an RF electrosurgical instrument, with the drive signal output **810b** corresponding to the center tap of the power transformer **806**.

[0226] In certain forms, the ultrasonic and electrosurgical drive signals may be provided simultaneously to distinct surgical instruments and/or to a single surgical instrument, such as the multifunction surgical instrument, having the capability to deliver both ultrasonic and electrosurgical energy to tissue. It will be appreciated that the electrosurgical signal, provided either to a dedicated electrosurgical instrument and/or to a combined multifunction ultrasonic/electrosurgical instrument may be either a therapeutic or sub-therapeutic level signal where the sub-therapeutic signal can be used, for example, to monitor tissue or instrument conditions and provide feedback to the generator. For example, the ultrasonic and RF signals can be delivered separately or simultaneously from a generator with a single output port in order to provide the desired output signal to the surgical instrument, as will be discussed in more detail below. Accordingly, the generator can combine the ultrasonic and electrosurgical RF energies and deliver the combined energies to the multifunction ultrasonic/electrosurgical instrument. Bipolar electrodes can be placed on one or both jaws of the end effector. One jaw may be driven by ultrasonic energy in addition to electrosurgical RF energy, working simultaneously. The ultrasonic energy may be employed to dissect tissue, while the electrosurgical RF energy may be employed for vessel sealing.

[0227] The non-isolated stage **804** may comprise a power amplifier **812** having an output connected to a primary winding **814** of the power transformer **806**. In certain forms, the power amplifier **812** may comprise a push-pull amplifier. For example, the non-isolated stage **804** may further comprise a logic device **816** for supplying a digital output to a digital-to-analog converter (DAC) circuit **818**, which in turn supplies a corresponding analog signal to an input of the power amplifier **812**. In certain forms, the logic device **816**

may comprise a programmable gate array (PGA), a FPGA, programmable logic device (PLD), among other logic circuits, for example. The logic device **816**, by virtue of controlling the input of the power amplifier **812** via the DAC circuit **818**, may therefore control any of a number of parameters (e.g., frequency, waveform shape, waveform amplitude) of drive signals appearing at the drive signal outputs **810a**, **810b**, **810c**. In certain forms and as discussed below, the logic device **816**, in conjunction with a processor (e.g., a DSP discussed below), may implement a number of DSP-based and/or other control algorithms to control parameters of the drive signals output by the generator **800**.

[0228] Power may be supplied to a power rail of the power amplifier **812** by a switch-mode regulator **820**, e.g., a power converter. In certain forms, the switch-mode regulator **820** may comprise an adjustable buck regulator, for example. The non-isolated stage **804** may further comprise a first processor **822**, which in one form may comprise a DSP processor such as an Analog Devices ADSP-21469 SHARC DSP, available from Analog Devices, Norwood, MA, for example, although in various forms any suitable processor may be employed. In certain forms the DSP processor **822** may control the operation of the switch-mode regulator **820** responsive to voltage feedback data received from the power amplifier **812** by the DSP processor **822** via an ADC circuit **824**. In one form, for example, the DSP processor **822** may receive as input, via the ADC circuit **824**, the waveform envelope of a signal (e.g., an RF signal) being amplified by the power amplifier **812**. The DSP processor **822** may then control the switch-mode regulator **820** (e.g., via a PWM output) such that the rail voltage supplied to the power amplifier **812** tracks the waveform envelope of the amplified signal. By dynamically modulating the rail voltage of the power amplifier **812** based on the waveform envelope, the efficiency of the power amplifier **812** may be significantly improved relative to a fixed rail voltage amplifier schemes.

[0229] In certain forms, the logic device **816**, in conjunction with the DSP processor **822**, may implement a digital synthesis circuit such as a direct digital synthesizer control scheme to control the waveform shape, frequency, and/or amplitude of drive signals output by the generator **800**. In one form, for example, the logic device **816** may implement a DDS control algorithm by recalling waveform samples stored in a dynamically updated lookup table (LUT), such as a RAM LUT, which may be embedded in an FPGA. This control algorithm is particularly useful for ultrasonic applications in which an ultrasonic transducer, such as an ultrasonic transducer, may be driven by a clean sinusoidal current at its resonant frequency. Because other frequencies may excite parasitic resonances, minimizing or reducing the total distortion of the motional branch current may correspondingly minimize or reduce undesirable resonance effects. Because the waveform shape of a drive signal output by the generator **800** is impacted by various sources of distortion present in the output drive circuit (e.g., the power transformer **806**, the power amplifier **812**), voltage and current feedback data based on the drive signal may be input into an algorithm, such as an error control algorithm implemented by the DSP processor **822**, which compensates for distortion by suitably pre-distorting or modifying the waveform samples stored in the LUT on a dynamic, ongoing basis (e.g., in real time). In one form, the amount or degree of pre-distortion applied to the LUT samples may be based on the error between a computed motional branch current and

a desired current waveform shape, with the error being determined on a sample-by-sample basis. In this way, the pre-distorted LUT samples, when processed through the drive circuit, may result in a motional branch drive signal having the desired waveform shape (e.g., sinusoidal) for optimally driving the ultrasonic transducer. In such forms, the LUT waveform samples will therefore not represent the desired waveform shape of the drive signal, but rather the waveform shape that is required to ultimately produce the desired waveform shape of the motional branch drive signal when distortion effects are taken into account.

[0230] The non-isolated stage **804** may further comprise a first ADC circuit **826** and a second ADC circuit **828** coupled to the output of the power transformer **806** via respective isolation transformers **830**, **832** for respectively sampling the voltage and current of drive signals output by the generator **800**. In certain forms, the ADC circuits **826**, **828** may be configured to sample at high speeds (e.g., 80 mega samples per second (MSPS)) to enable oversampling of the drive signals. In one form, for example, the sampling speed of the ADC circuits **826**, **828** may enable approximately 200× (depending on frequency) oversampling of the drive signals. In certain forms, the sampling operations of the ADC circuit **826**, **828** may be performed by a single ADC circuit receiving input voltage and current signals via a two-way multiplexer. The use of high-speed sampling in forms of the generator **800** may enable, among other things, calculation of the complex current flowing through the motional branch (which may be used in certain forms to implement DDS-based waveform shape control described above), accurate digital filtering of the sampled signals, and calculation of real power consumption with a high degree of precision. Voltage and current feedback data output by the ADC circuits **826**, **828** may be received and processed (e.g., first-in-first-out (FIFO) buffer, multiplexer) by the logic device **816** and stored in data memory for subsequent retrieval by, for example, the DSP processor **822**. As noted above, voltage and current feedback data may be used as input to an algorithm for pre-distorting or modifying LUT waveform samples on a dynamic and ongoing basis. In certain forms, this may require each stored voltage and current feedback data pair to be indexed based on, or otherwise associated with, a corresponding LUT sample that was output by the logic device **816** when the voltage and current feedback data pair was acquired. Synchronization of the LUT samples and the voltage and current feedback data in this manner contributes to the correct timing and stability of the pre-distortion algorithm.

[0231] In certain forms, the voltage and current feedback data may be used to control the frequency and/or amplitude (e.g., current amplitude) of the drive signals. In one form, for example, voltage and current feedback data may be used to determine impedance phase. The frequency of the drive signal may then be controlled to minimize or reduce the difference between the determined impedance phase and an impedance phase setpoint (e.g.,) 0°, thereby minimizing or reducing the effects of harmonic distortion and correspondingly enhancing impedance phase measurement accuracy. The determination of phase impedance and a frequency control signal may be implemented in the DSP processor **822**, for example, with the frequency control signal being supplied as input to a DDS control algorithm implemented by the logic device **816**.

[0232] In another form, for example, the current feedback data may be monitored in order to maintain the current amplitude of the drive signal at a current amplitude setpoint. The current amplitude setpoint may be specified directly or determined indirectly based on specified voltage amplitude and power setpoints. In certain forms, control of the current amplitude may be implemented by control algorithm, such as, for example, a proportional-integral-derivative (PID) control algorithm, in the DSP processor 822. Variables controlled by the control algorithm to suitably control the current amplitude of the drive signal may include, for example, the scaling of the LUT waveform samples stored in the logic device 816 and/or the full-scale output voltage of the DAC circuit 818 (which supplies the input to the power amplifier 812) via a DAC circuit 834.

[0233] The non-isolated stage 804 may further comprise a second processor 836 for providing, among other things user interface (UI) functionality. In one form, the UI processor 836 may comprise an Atmel AT91SAM9263 processor having an ARM 926EJ-S core, available from Atmel Corporation, San Jose, California, for example. Examples of UI functionality supported by the UI processor 836 may include audible and visual user feedback, communication with peripheral devices (e.g., via a USB interface), communication with a foot switch, communication with an input device (e.g., a touch screen display) and communication with an output device (e.g., a speaker). The UI processor 836 may communicate with the DSP processor 822 and the logic device 816 (e.g., via SPI buses). Although the UI processor 836 may primarily support UI functionality, it may also coordinate with the DSP processor 822 to implement hazard mitigation in certain forms. For example, the UI processor 836 may be programmed to monitor various aspects of user input and/or other inputs (e.g., touch screen inputs, foot switch inputs, temperature sensor inputs) and may disable the drive output of the generator 800 when an erroneous condition is detected.

[0234] In certain forms, both the DSP processor 822 and the UI processor 836, for example, may determine and monitor the operating state of the generator 800. For the DSP processor 822, the operating state of the generator 800 may dictate, for example, which control and/or diagnostic processes are implemented by the DSP processor 822. For the UI processor 836, the operating state of the generator 800 may dictate, for example, which elements of a UI (e.g., display screens, sounds) are presented to a user. The respective DSP and UI processors 822, 836 may independently maintain the current operating state of the generator 800 and recognize and evaluate possible transitions out of the current operating state. The DSP processor 822 may function as the master in this relationship and determine when transitions between operating states are to occur. The UI processor 836 may be aware of valid transitions between operating states and may confirm if a particular transition is appropriate. For example, when the DSP processor 822 instructs the UI processor 836 to transition to a specific state, the UI processor 836 may verify that requested transition is valid. In the event that a requested transition between states is determined to be invalid by the UI processor 836, the UI processor 836 may cause the generator 800 to enter a failure mode.

[0235] The non-isolated stage 804 may further comprise a controller 838 for monitoring input devices (e.g., a capacitive touch sensor used for turning the generator 800 on and

off, a capacitive touch screen). In certain forms, the controller 838 may comprise at least one processor and/or other controller device in communication with the UI processor 836. In one form, for example, the controller 838 may comprise a processor (e.g., a Meg168 8-bit controller available from Atmel) configured to monitor user input provided via one or more capacitive touch sensors. In one form, the controller 838 may comprise a touch screen controller (e.g., a QT5480 touch screen controller available from Atmel) to control and manage the acquisition of touch data from a capacitive touch screen.

[0236] In certain forms, when the generator 800 is in a “power off” state, the controller 838 may continue to receive operating power (e.g., via a line from a power supply of the generator 800, such as the power supply 854 discussed below). In this way, the controller 838 may continue to monitor an input device (e.g., a capacitive touch sensor located on a front panel of the generator 800) for turning the generator 800 on and off. When the generator 800 is in the power off state, the controller 838 may wake the power supply (e.g., enable operation of one or more DC/DC voltage converters 856 of the power supply 854) if activation of the “on/off” input device by a user is detected. The controller 838 may therefore initiate a sequence for transitioning the generator 800 to a “power on” state. Conversely, the controller 838 may initiate a sequence for transitioning the generator 800 to the power off state if activation of the “on/off” input device is detected when the generator 800 is in the power on state. In certain forms, for example, the controller 838 may report activation of the “on/off” input device to the UI processor 836, which in turn implements the necessary process sequence for transitioning the generator 800 to the power off state. In such forms, the controller 838 may have no independent ability for causing the removal of power from the generator 800 after its power on state has been established.

[0237] In certain forms, the controller 838 may cause the generator 800 to provide audible or other sensory feedback for alerting the user that a power on or power off sequence has been initiated. Such an alert may be provided at the beginning of a power on or power off sequence and prior to the commencement of other processes associated with the sequence.

[0238] In certain forms, the isolated stage 802 may comprise an instrument interface circuit 840 to, for example, provide a communication interface between a control circuit of a surgical instrument (e.g., a control circuit comprising handpiece switches) and components of the non-isolated stage 804, such as, for example, the logic device 816, the DSP processor 822, and/or the UI processor 836. The instrument interface circuit 840 may exchange information with components of the non-isolated stage 804 via a communication link that maintains a suitable degree of electrical isolation between the isolated and non-isolated stages 802, 804, such as, for example, an IR-based communication link. Power may be supplied to the instrument interface circuit 840 using, for example, a low-dropout voltage regulator powered by an isolation transformer driven from the non-isolated stage 804.

[0239] In one form, the instrument interface circuit 840 may comprise a logic circuit 842 (e.g., logic circuit, programmable logic circuit, PGA, FPGA, PLD) in communication with a signal conditioning circuit 844. The signal conditioning circuit 844 may be configured to receive a

periodic signal from the logic circuit **842** (e.g., a 2 kHz square wave) to generate a bipolar interrogation signal having an identical frequency. The interrogation signal may be generated, for example, using a bipolar current source fed by a differential amplifier. The interrogation signal may be communicated to a surgical instrument control circuit (e.g., by using a conductive pair in a cable that connects the generator **800** to the surgical instrument) and monitored to determine a state or configuration of the control circuit. The control circuit may comprise a number of switches, resistors, and/or diodes to modify one or more characteristics (e.g., amplitude, rectification) of the interrogation signal such that a state or configuration of the control circuit is uniquely discernable based on the one or more characteristics. In one form, for example, the signal conditioning circuit **844** may comprise an ADC circuit for generating samples of a voltage signal appearing across inputs of the control circuit resulting from passage of interrogation signal therethrough. The logic circuit **842** (or a component of the non-isolated stage **804**) may then determine the state or configuration of the control circuit based on the ADC circuit samples.

[0240] In one form, the instrument interface circuit **840** may comprise a first data circuit interface **846** to enable information exchange between the logic circuit **842** (or other element of the instrument interface circuit **840**) and a first data circuit disposed in or otherwise associated with a surgical instrument. In certain forms, for example, a first data circuit may be disposed in a cable integrally attached to a surgical instrument handpiece or in an adaptor for interfacing a specific surgical instrument type or model with the generator **800**. The first data circuit may be implemented in any suitable manner and may communicate with the generator according to any suitable protocol, including, for example, as described herein with respect to the first data circuit. In certain forms, the first data circuit may comprise a non-volatile storage device, such as an EEPROM device. In certain forms, the first data circuit interface **846** may be implemented separately from the logic circuit **842** and comprise suitable circuitry (e.g., discrete logic devices, a processor) to enable communication between the logic circuit **842** and the first data circuit. In other forms, the first data circuit interface **846** may be integral with the logic circuit **842**.

[0241] In certain forms, the first data circuit may store information pertaining to the particular surgical instrument with which it is associated. Such information may include, for example, a model number, a serial number, a number of operations in which the surgical instrument has been used, and/or any other type of information. This information may be read by the instrument interface circuit **840** (e.g., by the logic circuit **842**), transferred to a component of the non-isolated stage **804** (e.g., to logic device **816**, DSP processor **822**, and/or UI processor **836**) for presentation to a user via an output device and/or for controlling a function or operation of the generator **800**. Additionally, any type of information may be communicated to the first data circuit for storage therein via the first data circuit interface **846** (e.g., using the logic circuit **842**). Such information may comprise, for example, an updated number of operations in which the surgical instrument has been used and/or dates and/or times of its usage.

[0242] As discussed previously, a surgical instrument may be detachable from a handpiece (e.g., the multifunction surgical instrument may be detachable from the handpiece)

to promote instrument interchangeability and/or disposability. In such cases, conventional generators may be limited in their ability to recognize particular instrument configurations being used and to optimize control and diagnostic processes accordingly. The addition of readable data circuits to surgical instruments to address this issue is problematic from a compatibility standpoint, however. For example, designing a surgical instrument to remain backwardly compatible with generators that lack the requisite data reading functionality may be impractical due to, for example, differing signal schemes, design complexity, and cost. Forms of instruments discussed herein address these concerns by using data circuits that may be implemented in existing surgical instruments economically and with minimal design changes to preserve compatibility of the surgical instruments with current generator platforms.

[0243] Additionally, forms of the generator **800** may enable communication with instrument-based data circuits. For example, the generator **800** may be configured to communicate with a second data circuit contained in an instrument (e.g., the multifunction surgical instrument). In some forms, the second data circuit may be implemented in a many similar to that of the first data circuit described herein. The instrument interface circuit **840** may comprise a second data circuit interface **848** to enable this communication. In one form, the second data circuit interface **848** may comprise a tri-state digital interface, although other interfaces may also be used. In certain forms, the second data circuit may generally be any circuit for transmitting and/or receiving data. In one form, for example, the second data circuit may store information pertaining to the particular surgical instrument with which it is associated. Such information may include, for example, a model number, a serial number, a number of operations in which the surgical instrument has been used, and/or any other type of information.

[0244] In some forms, the second data circuit may store information about the electrical and/or ultrasonic properties of an associated ultrasonic transducer, end effector, or ultrasonic drive system. For example, the first data circuit may indicate a burn-in frequency slope, as described herein. Additionally or alternatively, any type of information may be communicated to second data circuit for storage therein via the second data circuit interface **848** (e.g., using the logic circuit **842**). Such information may comprise, for example, an updated number of operations in which the instrument has been used and/or dates and/or times of its usage. In certain forms, the second data circuit may transmit data acquired by one or more sensors (e.g., an instrument-based temperature sensor). In certain forms, the second data circuit may receive data from the generator **800** and provide an indication to a user (e.g., a light emitting diode indication or other visible indication) based on the received data.

[0245] In certain forms, the second data circuit and the second data circuit interface **848** may be configured such that communication between the logic circuit **842** and the second data circuit can be effected without the need to provide additional conductors for this purpose (e.g., dedicated conductors of a cable connecting a handpiece to the generator **800**). In one form, for example, information may be communicated to and from the second data circuit using a one-wire bus communication scheme implemented on existing cabling, such as one of the conductors used transmit interrogation signals from the signal conditioning circuit **844**

to a control circuit in a handpiece. In this way, design changes or modifications to the surgical instrument that might otherwise be necessary are minimized or reduced. Moreover, because different types of communications implemented over a common physical channel can be frequency-band separated, the presence of a second data circuit may be “invisible” to generators that do not have the requisite data reading functionality, thus enabling backward compatibility of the surgical instrument.

[0246] In certain forms, the isolated stage **802** may comprise at least one blocking capacitor **850-1** connected to the drive signal output **810b** to prevent passage of DC current to a patient. A single blocking capacitor may be required to comply with medical regulations or standards, for example. While failure in single-capacitor designs is relatively uncommon, such failure may nonetheless have negative consequences. In one form, a second blocking capacitor **850-2** may be provided in series with the blocking capacitor **850-1**, with current leakage from a point between the blocking capacitors **850-1**, **850-2** being monitored by, for example, an ADC circuit **852** for sampling a voltage induced by leakage current. The samples may be received by the logic circuit **842**, for example. Based changes in the leakage current (as indicated by the voltage samples), the generator **800** may determine when at least one of the blocking capacitors **850-1**, **850-2** has failed, thus providing a benefit over single-capacitor designs having a single point of failure.

[0247] In certain forms, the non-isolated stage **804** may comprise a power supply **854** for delivering DC power at a suitable voltage and current. The power supply may comprise, for example, a 400 W power supply for delivering a 48 VDC system voltage. The power supply **854** may further comprise one or more DC/DC voltage converters **856** for receiving the output of the power supply to generate DC outputs at the voltages and currents required by the various components of the generator **800**. As discussed above in connection with the controller **838**, one or more of the DC/DC voltage converters **856** may receive an input from the controller **838** when activation of the “on/off” input device by a user is detected by the controller **838** to enable operation of, or wake, the DC/DC voltage converters **856**.

[0248] FIG. 21 illustrates an example of a generator **900**, which is one form of the generator **800** (FIG. 20). The generator **900** is configured to deliver multiple energy modalities to a surgical instrument. The generator **900** provides RF and ultrasonic signals for delivering energy to a surgical instrument either independently or simultaneously. The RF and ultrasonic signals may be provided alone or in combination and may be provided simultaneously. As noted above, at least one generator output can deliver multiple energy modalities (e.g., ultrasonic, bipolar or monopolar RF, irreversible and/or reversible electroporation, and/or microwave energy, among others) through a single port, and these signals can be delivered separately or simultaneously to the end effector to treat tissue. The generator **900** comprises a processor **902** coupled to a waveform generator **904**. The processor **902** and waveform generator **904** are configured to generate a variety of signal waveforms based on information stored in a memory coupled to the processor **902**, not shown for clarity of disclosure. The digital information associated with a waveform is provided to the waveform generator **904** which includes one or more DAC circuits to convert the digital input into an analog

output. The analog output is fed to an amplifier **1106** for signal conditioning and amplification. The conditioned and amplified output of the amplifier **906** is coupled to a power transformer **908**. The signals are coupled across the power transformer **908** to the secondary side, which is in the patient isolation side. A first signal of a first energy modality is provided to the surgical instrument between the terminals labeled ENERGY1 and RETURN. A second signal of a second energy modality is coupled across a capacitor **910** and is provided to the surgical instrument between the terminals labeled ENERGY2 and RETURN. It will be appreciated that more than two energy modalities may be output and thus the subscript “n” may be used to designate that up to n ENERGYn terminals may be provided, where n is a positive integer greater than 1. It also will be appreciated that up to “n” return paths RETURNn may be provided without departing from the scope of the present disclosure.

[0249] A first voltage sensing circuit **912** is coupled across the terminals labeled ENERGY1 and the RETURN path to measure the output voltage therebetween. A second voltage sensing circuit **924** is coupled across the terminals labeled ENERGY2 and the RETURN path to measure the output voltage therebetween. A current sensing circuit **914** is disposed in series with the RETURN leg of the secondary side of the power transformer **908** as shown to measure the output current for either energy modality. If different return paths are provided for each energy modality, then a separate current sensing circuit should be provided in each return leg. The outputs of the first and second voltage sensing circuits **912**, **924** are provided to respective isolation sensing transformers **916**, **922** and the output of the current sensing circuit **914** is provided to another isolation transformer **918**. The outputs of the isolation transformers **916**, **928**, **922** in the on the primary side of the power transformer **908** (non-patient isolated side) are provided to a one or more ADC circuit **926**. The digitized output of the ADC circuit **926** is provided to the processor **902** for further processing and computation. The output voltages and output current feedback information can be employed to adjust the output voltage and current provided to the surgical instrument and to compute output impedance, among other parameters. Input/output communications between the processor **902** and patient isolated circuits is provided through an interface circuit **920**. Sensors also may be in electrical communication with the processor **902** by way of the interface circuit **920**.

[0250] In one aspect, the impedance may be determined by the processor **902** by dividing the output of either the first voltage sensing circuit **912** coupled across the terminals labeled ENERGY1/RETURN or the second voltage sensing circuit **924** coupled across the terminals labeled ENERGY2/RETURN by the output of the current sensing circuit **914** disposed in series with the RETURN leg of the secondary side of the power transformer **908**. The outputs of the first and second voltage sensing circuits **912**, **924** are provided to separate isolation transformers **916**, **922** and the output of the current sensing circuit **914** is provided to another isolation transformer **916**. The digitized voltage and current sensing measurements from the ADC circuit **926** are provided to the processor **902** for computing impedance. As an example, the first energy modality ENERGY1 may be ultrasonic energy and the second energy modality ENERGY2 may be RF energy. Nevertheless, in addition to ultrasonic and bipolar or monopolar RF energy modalities, other energy modalities include irreversible and/or revers-

ible electroporation and/or microwave energy, among others. Also, although the example illustrated in FIG. 21 shows a single return path RETURN may be provided for two or more energy modalities, in other aspects, multiple return paths RETURNn may be provided for each energy modality ENERGYn. Thus, as described herein, the ultrasonic transducer impedance may be measured by dividing the output of the first voltage sensing circuit 912 by the current sensing circuit 914 and the tissue impedance may be measured by dividing the output of the second voltage sensing circuit 924 by the current sensing circuit 914.

[0251] As shown in FIG. 21, the generator 900 comprising at least one output port can include a power transformer 908 with a single output and with multiple taps to provide power in the form of one or more energy modalities, such as ultrasonic, bipolar or monopolar RF, irreversible and/or reversible electroporation, and/or microwave energy, among others, for example, to the end effector depending on the type of treatment of tissue being performed. For example, the generator 900 can deliver energy with higher voltage and lower current to drive an ultrasonic transducer, with lower voltage and higher current to drive RF electrodes for sealing tissue, or with a coagulation waveform for spot coagulation using either monopolar or bipolar RF electrosurgical electrodes. The output waveform from the generator 900 can be steered, switched, or filtered to provide the frequency to the end effector of the surgical instrument. The connection of an ultrasonic transducer to the generator 900 output would be preferably located between the output labeled ENERGY1 and RETURN as shown in FIG. 21. In one example, a connection of RF bipolar electrodes to the generator 900 output would be preferably located between the output labeled ENERGY2 and RETURN. In the case of monopolar output, the preferred connections would be active electrode (e.g., pencil or other probe) to the ENERGY2 output and a suitable return pad connected to the RETURN output.

[0252] Additional details are disclosed in U.S. Patent Application Publication No. 2017/0086914, titled TECHNIQUES FOR OPERATING GENERATOR FOR DIGITALLY GENERATING ELECTRICAL SIGNAL WAVEFORMS AND SURGICAL INSTRUMENTS, which published on Mar. 30, 2017, now U.S. Pat. No. 10,624,691, which is herein incorporated by reference in its entirety.

[0253] As used throughout this description, the term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a non-solid medium. The term does not imply that the associated devices do not contain any wires, although in some aspects they might not. The communication module may implement any of a number of wireless or wired communication standards or protocols, including but not limited to Wi-Fi (IEEE 802.11 family), WiMAX (IEEE 802.16 family), IEEE 802.20, long term evolution (LTE), Ev-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPRS, CDMA, TDMA, DECT, Bluetooth, Ethernet derivatives thereof, as well as any other wireless and wired protocols that are designated as 3G, 4G, 5G, and beyond. The computing module may include a plurality of communication modules. For instance, a first communication module may be dedicated to shorter range wireless communications such as Wi-Fi and Bluetooth and a second communication module

may be dedicated to longer range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, Ev-DO, and others.

[0254] As used herein a processor or processing unit is an electronic circuit which performs operations on some external data source, usually memory or some other data stream. The term is used herein to refer to the central processor (central processing unit) in a system or computer systems (especially systems on a chip (SoCs)) that combine a number of specialized “processors.”

[0255] As used herein, a system on a chip or system on chip (SoC or SOC) is an integrated circuit (also known as an “IC” or “chip”) that integrates all components of a computer or other electronic systems. It may contain digital, analog, mixed-signal, and often radio-frequency functions—all on a single substrate. A SoC integrates a microcontroller (or microprocessor) with advanced peripherals like graphics processing unit (GPU), Wi-Fi module, or coprocessor. A SoC may or may not contain built-in memory.

[0256] As used herein, a microcontroller or controller is a system that integrates a microprocessor with peripheral circuits and memory. A microcontroller (or MCU for microcontroller unit) may be implemented as a small computer on a single integrated circuit. It may be similar to a SoC; an SoC may include a microcontroller as one of its components. A microcontroller may contain one or more core processing units (CPUs) along with memory and programmable input/output peripherals. Program memory in the form of Ferroelectric RAM, NOR flash or OTP ROM is also often included on chip, as well as a small amount of RAM. Microcontrollers may be employed for embedded applications, in contrast to the microprocessors used in personal computers or other general purpose applications consisting of various discrete chips.

[0257] As used herein, the term controller or microcontroller may be a stand-alone IC or chip device that interfaces with a peripheral device. This may be a link between two parts of a computer or a controller on an external device that manages the operation of (and connection with) that device.

[0258] Any of the processors or microcontrollers described herein, may be implemented by any single core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In one aspect, the processor may be an LM4F230H5QR ARM Cortex-M4F Processor Core, available from Texas Instruments, for example, comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle serial random access memory (SRAM), internal read-only memory (ROM) loaded with StellarisWare® software, 2 KB electrically erasable programmable read-only memory (EEPROM), one or more pulse width modulation (PWM) modules, one or more quadrature encoder inputs (QEI) analog, one or more 12-bit Analog-to-Digital Converters (ADC) with 12 analog input channels, details of which are available for the product datasheet.

[0259] In one aspect, the processor may comprise a safety controller comprising two controller-based families such as TMS570 and RM4x known under the trade name Hercules ARM Cortex R4, also by Texas Instruments. The safety controller may be configured specifically for IEC 61508 and ISO 26262 safety critical applications, among others, to

provide advanced integrated safety features while delivering scalable performance, connectivity, and memory options.

[0260] Modular devices include the modules (as described in connection with FIGS. 3 and 9, for example) that are receivable within a surgical hub and the surgical devices or instruments that can be connected to the various modules in order to connect or pair with the corresponding surgical hub. The modular devices include, for example, intelligent surgical instruments, medical imaging devices, suction/irrigation devices, smoke evacuators, energy generators, ventilators, insufflators, and displays. The modular devices described herein can be controlled by control algorithms. The control algorithms can be executed on the modular device itself, on the surgical hub to which the particular modular device is paired, or on both the modular device and the surgical hub (e.g., via a distributed computing architecture). In some exemplifications, the modular devices' control algorithms control the devices based on data sensed by the modular device itself (i.e., by sensors in, on, or connected to the modular device). This data can be related to the patient being operated on (e.g., tissue properties or insufflation pressure) or the modular device itself (e.g., the rate at which a knife is being advanced, motor current, or energy levels). For example, a control algorithm for a surgical stapling and cutting instrument can control the rate at which the instrument's motor drives its knife through tissue according to resistance encountered by the knife as it advances.

[0261] Various aspects of the subject matter described herein are set out in the following numbered examples.

[0262] Example 1. A surgical hub for use with a surgical instrument configured to deliver therapeutic energy to tissue at a surgical site of a surgical procedure, wherein the surgical hub comprises: a hub enclosure, comprising a docking station including a docking port comprising data and power contacts; and a combo generator module removably retainable in the docking station, wherein the combo generator module comprises: an ultrasonic energy generator component; a radio frequency (RF) energy generator component; a smoke evacuation component; a connection port, wherein at least one of the ultrasonic energy generator component and the radio frequency (RF) generator component is coupleable to the surgical instrument through the connection port; and at least one smoke evacuation component configured to evacuate smoke generated by an application of therapeutic energy to the tissue by the surgical instrument.

[0263] Example 2. The surgical hub of Example 1, wherein the docking station is a first docking station, wherein the docking port is a first docking port, and wherein the hub enclosure comprises a second docking station comprising a second docking port that has data and power contacts.

[0264] Example 3. The surgical hub of Example 2, further comprising a suction and irrigation module removably retainable in the second docking station.

[0265] Example 4. The surgical hub of Example 3, wherein the combo generator module comprises a third docking port connectable to the first docking port of the first docking station.

[0266] Example 5. The surgical hub of Example 4, wherein the suction and irrigation module comprises a fourth docking port connectable to the second docking port of the second docking station.

[0267] Example 6. The surgical hub of Example 5, wherein the hub enclosure comprises a communication link between the second docking port and the first docking port.

[0268] Example 7. The surgical hub of any of Examples 1-6, wherein the combo generator module comprises a fluid line extendable to the remote surgical site for passing the smoke evacuated from the remote surgical site to the combo generator module.

[0269] Example 8. The surgical hub of any one of Examples 1-7, wherein the docking station comprises brackets configured to slidably receive and guide the combo generator module into a working connection with the power and data contacts of the docking port.

[0270] Example 9. The surgical hub of any one of Examples 1-8, wherein the combo generator module comprises side brackets configured to movably engage the brackets of the docking station.

[0271] Example 10. A modular surgical hub for use with a surgical instrument configured to deliver therapeutic energy to tissue at a surgical site of a surgical procedure, wherein the modular surgical enclosure comprises: a first energy-generator module configured to generate a first therapeutic energy for application to the tissue; a first docking station comprising a first docking port that includes first data and power contacts, wherein the first energy-generator module is slidably movable into an electrical engagement with the first data and power contacts, and wherein the first energy-generator module is slidably movable out of the electrical engagement with the first data and power contacts; a second energy-generator module configured to generate a second therapeutic energy, different than the first therapeutic energy, for application to the tissue; a second docking station comprising a second docking port that includes second data and power contacts, wherein the second energy-generator module is slidably movable into an electrical engagement with the second data and power contacts, and wherein the second energy-generator module is slidably movable out of the electrical engagement with the second data and power contacts; and a communication bus between the first docking port and the second docking port configured to facilitate communication between the first energy-generator module and the second energy-generator module.

[0272] Example 11. The modular surgical hub of Example 10, wherein the first docking station comprises brackets configured to slidably receive and guide the first energy-generator module into the electrical engagement with the first data and power contacts.

[0273] Example 12. The modular surgical hub of Example 11, wherein the second docking station comprises brackets configured to slidably receive and guide the second energy-generator module into the electrical engagement with the second data and power contacts.

[0274] Example 13. The modular surgical hub of any one of Examples 10-12, wherein the first therapeutic energy is an ultrasonic energy.

[0275] Example 14. The modular surgical hub of any one of Examples 10-12, wherein the second therapeutic energy is a radio frequency (RF) energy.

[0276] Example 15. The modular surgical hub of any one of Examples 10-14, further comprising a smoke evacuation module configured to evacuate smoke generated at the remote surgical site by application of the first therapeutic energy to the tissue.

[0277] Example 16. The modular surgical hub of Example 15, further comprising a third docking station comprising a third docking port that includes third data and power contacts.

[0278] Example 17. The modular surgical hub of Example 16, further comprising a suction and irrigation module slidably movable into an electrical engagement with the third data and power contacts, and wherein the suction and irrigation module is slidably movable out of the electrical engagement with the third data and power contacts.

[0279] Example 18. A surgical hub for use with a surgical instrument configured to deliver therapeutic energy to tissue at a surgical site of a surgical procedure, wherein the surgical hub comprises: a hub enclosure, comprising docking stations including docking ports comprising data and power contacts; a combo generator module slidably receivable in a first of the docking stations, wherein the combo generator module comprises: an ultrasonic energy generator component; a radio frequency (RF) energy generator component; and a connection port, wherein at least one of the ultrasonic energy generator component and the radio frequency (RF) generator component is couplable to the surgical instrument through the connection port; a smoke evacuation module slidably receivable in a second one of the docking stations, wherein the smoke evacuation module is configured to evacuate smoke generated by an application of the therapeutic energy to the tissue by the surgical instrument; a processing module slidably receivable in a third one of the docking stations; a memory module slidably receivable in a fourth one of the docking stations; and an operating-room mapping module slidably receivable in a fifth one of the docking stations.

[0280] Example 19. The surgical hub of Example 18, wherein the docking stations comprise brackets configured to slidably guide the modules into electrical engagements with the power and data contacts of the docking ports.

[0281] Example 20. The surgical hub of any one of Examples 18-19, comprising a display.

[0282] While several forms have been illustrated and described, it is not the intention of the applicant to restrict or limit the scope of the appended claims to such detail. Numerous modifications, variations, changes, substitutions, combinations, and equivalents to those forms may be implemented and will occur to those skilled in the art without departing from the scope of the present disclosure. Moreover, the structure of each element associated with the described forms can be alternatively described as a means for providing the function performed by the element. Also, where materials are disclosed for certain components, other materials may be used. It is therefore to be understood that the foregoing description and the appended claims are intended to cover all such modifications, combinations, and variations as falling within the scope of the disclosed forms. The appended claims are intended to cover all such modifications, variations, changes, substitutions, modifications, and equivalents.

[0283] The foregoing detailed description has set forth various forms of the devices and/or processes via the use of block diagrams, flowcharts, and/or examples. Insofar as such block diagrams, flowcharts, and/or examples contain one or more functions and/or operations, it will be understood by those within the art that each function and/or operation within such block diagrams, flowcharts, and/or examples can be implemented, individually and/or collec-

tively, by a wide range of hardware, software, firmware, or virtually any combination thereof. Those skilled in the art will recognize that some aspects of the forms disclosed herein, in whole or in part, can be equivalently implemented in integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more processors (e.g., as one or more programs running on one or more micro-processors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software and or firmware would be well within the skill of one of skill in the art in light of this disclosure. In addition, those skilled in the art will appreciate that the mechanisms of the subject matter described herein are capable of being distributed as one or more program products in a variety of forms, and that an illustrative form of the subject matter described herein applies regardless of the particular type of signal bearing medium used to actually carry out the distribution.

[0284] Instructions used to program logic to perform various disclosed aspects can be stored within a memory in the system, such as dynamic random access memory (DRAM), cache, flash memory, or other storage. Furthermore, the instructions can be distributed via a network or by way of other computer readable media. Thus a machine-readable medium may include any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computer), but is not limited to, floppy diskettes, optical disks, compact disc, read-only memory (CD-ROMs), and magneto-optical disks, read-only memory (ROMs), random access memory (RAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), magnetic or optical cards, flash memory, or a tangible, machine-readable storage used in the transmission of information over the Internet via electrical, optical, acoustical or other forms of propagated signals (e.g., carrier waves, infrared signals, digital signals, etc.). Accordingly, the non-transitory computer-readable medium includes any type of tangible machine-readable medium suitable for storing or transmitting electronic instructions or information in a form readable by a machine (e.g., a computer).

[0285] As used in any aspect herein, the term “control circuit” may refer to, for example, hardwired circuitry, programmable circuitry (e.g., a computer processor comprising one or more individual instruction processing cores, processing unit, processor, microcontroller, microcontroller unit, controller, digital signal processor (DSP), programmable logic device (PLD), programmable logic array (PLA), or field programmable gate array (FPGA)), state machine circuitry, firmware that stores instructions executed by programmable circuitry, and any combination thereof. The control circuit may, collectively or individually, be embodied as circuitry that forms part of a larger system, for example, an integrated circuit (IC), an application-specific integrated circuit (ASIC), a system on-chip (SoC), desktop computers, laptop computers, tablet computers, servers, smart phones, etc. Accordingly, as used herein “control circuit” includes, but is not limited to, electrical circuitry having at least one discrete electrical circuit, electrical circuitry having at least one integrated circuit, electrical circuitry having at least one application specific integrated circuit, electrical circuitry forming a general purpose com-

puting device configured by a computer program (e.g., a general purpose computer configured by a computer program which at least partially carries out processes and/or devices described herein, or a microprocessor configured by a computer program which at least partially carries out processes and/or devices described herein), electrical circuitry forming a memory device (e.g., forms of random access memory), and/or electrical circuitry forming a communications device (e.g., a modem, communications switch, or optical-electrical equipment). Those having skill in the art will recognize that the subject matter described herein may be implemented in an analog or digital fashion or some combination thereof.

[0286] As used in any aspect herein, the term “logic” may refer to an app, software, firmware and/or circuitry configured to perform any of the aforementioned operations. Software may be embodied as a software package, code, instructions, instruction sets and/or data recorded on non-transitory computer readable storage medium. Firmware may be embodied as code, instructions or instruction sets and/or data that are hard-coded (e.g., nonvolatile) in memory devices.

[0287] As used in any aspect herein, the terms “component,” “system,” “module” and the like can refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution.

[0288] As used in any aspect herein, an “algorithm” refers to a self-consistent sequence of steps leading to a desired result, where a “step” refers to a manipulation of physical quantities and/or logic states which may, though need not necessarily, take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It is common usage to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like. These and similar terms may be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities and/or states.

[0289] A network may include a packet switched network. The communication devices may be capable of communicating with each other using a selected packet switched network communications protocol. One example communications protocol may include an Ethernet communications protocol which may be capable permitting communication using a Transmission Control Protocol/Internet Protocol (TCP/IP). The Ethernet protocol may comply or be compatible with the Ethernet standard published by the Institute of Electrical and Electronics Engineers (IEEE) titled “IEEE 802.3 Standard”, published in December 2008 and/or later versions of this standard. Alternatively or additionally, the communication devices may be capable of communicating with each other using an X.25 communications protocol. The X.25 communications protocol may comply or be compatible with a standard promulgated by the International Telecommunication Union-Telecommunication Standardization Sector (ITU-T). Alternatively or additionally, the communication devices may be capable of communicating with each other using a frame relay communications protocol. The frame relay communications protocol may comply or be compatible with a standard promulgated by Consultative Committee for International Telegraph and Telephone (CCITT) and/or the American National Standards Institute (ANSI). Alternatively or additionally, the transceivers may be capable of communicating with each other using an

Asynchronous Transfer Mode (ATM) communications protocol. The ATM communications protocol may comply or be compatible with an ATM standard published by the ATM Forum titled “ATM-MPLS Network Interworking 2.0” published August 2001, and/or later versions of this standard. Of course, different and/or after-developed connection-oriented network communication protocols are equally contemplated herein.

[0290] Unless specifically stated otherwise as apparent from the foregoing disclosure, it is appreciated that, throughout the foregoing disclosure, discussions using terms such as “processing,” “computing,” “calculating,” “determining,” “displaying,” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system’s registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

[0291] One or more components may be referred to herein as “configured to,” “configurable to,” “operable/operative to,” “adapted/adaptable,” “able to,” “conformable/conformed to,” etc. Those skilled in the art will recognize that “configured to” can generally encompass active-state components and/or inactive-state components and/or standby-state components, unless context requires otherwise.

[0292] The terms “proximal” and “distal” are used herein with reference to a clinician manipulating the handle portion of the surgical instrument. The term “proximal” refers to the portion closest to the clinician and the term “distal” refers to the portion located away from the clinician. It will be further appreciated that, for convenience and clarity, spatial terms such as “vertical,” “horizontal,” “up,” and “down” may be used herein with respect to the drawings. However, surgical instruments are used in many orientations and positions, and these terms are not intended to be limiting and/or absolute.

[0293] Those skilled in the art will recognize that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to claims containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations.

[0294] In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in

the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that typically a disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms unless context dictates otherwise. For example, the phrase “A or B” will be typically understood to include the possibilities of “A” or “B” or “A and B.”

[0295] With respect to the appended claims, those skilled in the art will appreciate that recited operations therein may generally be performed in any order. Also, although various operational flow diagrams are presented in a sequence(s), it should be understood that the various operations may be performed in other orders than those which are illustrated, or may be performed concurrently. Examples of such alternate orderings may include overlapping, interleaved, interrupted, reordered, incremental, preparatory, supplemental, simultaneous, reverse, or other variant orderings, unless context dictates otherwise. Furthermore, terms like “responsive to,” “related to,” or other past-tense adjectives are generally not intended to exclude such variants, unless context dictates otherwise.

[0296] It is worthy to note that any reference to “one aspect,” “an aspect,” “an exemplification,” “one exemplification,” and the like means that a particular feature, structure, or characteristic described in connection with the aspect is included in at least one aspect. Thus, appearances of the phrases “in one aspect,” “in an aspect,” “in an exemplification,” and “in one exemplification” in various places throughout the specification are not necessarily all referring to the same aspect. Furthermore, the particular features, structures or characteristics may be combined in any suitable manner in one or more aspects.

[0297] Any patent application, patent, non-patent publication, or other disclosure material referred to in this specification and/or listed in any Application Data Sheet is incorporated by reference herein, to the extent that the incorporated materials is not inconsistent herewith. As such, and to the extent necessary, the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by reference. Any material, or portion thereof, that is said to be incorporated by reference herein, but which conflicts with existing definitions, statements, or other disclosure material set forth herein will only be incorporated to

the extent that no conflict arises between that incorporated material and the existing disclosure material.

[0298] In summary, numerous benefits have been described which result from employing the concepts described herein. The foregoing description of the one or more forms has been presented for purposes of illustration and description. It is not intended to be exhaustive or limiting to the precise form disclosed. Modifications or variations are possible in light of the above teachings. The one or more forms were chosen and described in order to illustrate principles and practical application to thereby enable one of ordinary skill in the art to utilize the various forms and with various modifications as are suited to the particular use contemplated. It is intended that the claims submitted herewith define the overall scope.

What is claimed is:

1. A modular surgical system, comprising:
 - a cart including a plurality of shelves;
 - a backplane;
 - a display operably positionable on the cart, wherein the display comprises an electrical connector configured to electrically couple to the backplane based on the display being operably positioned on the cart;
 - a generator module operably positionable on the cart, wherein the generator module comprises:
 - an electrical connector configured to electrically couple to the backplane based on the generator module being operably positioned on the cart; and
 - a plurality of energy ports, wherein the generator module is configured to provide electrosurgical energy to a surgical instrument via the plurality of energy ports; and
 - a master module operably positionable on the cart, wherein the master module comprises an electrical connector configured to electrically couple to the backplane based on the master module being operably positioned on the cart, and wherein the master module is configured to control the display and the generator module via the backplane.
2. The modular surgical system of claim 1, wherein the master module is configured to display data associated with the generator module on the display.
3. The modular surgical system of claim 1, wherein the plurality of energy ports comprises:
 - a monopolar port, wherein the generator module is configured to provide monopolar energy through the monopolar port;
 - a bipolar port, wherein the generator module is configured to provide bipolar energy through the bipolar port; and
 - an ultrasonic port, wherein the generator module is configured to provide ultrasonic energy through the ultrasonic port.
4. The modular surgical system of claim 1, wherein the generator module is configured to receive power signals via the backplane.
5. The modular surgical system of claim 1, wherein the generator module is configured to receive data signals via the backplane.
6. The modular surgical system of claim 1, further comprising a surgical module operably positionable on the cart, wherein the surgical module comprises an electrical connector configured to electrically couple to the backplane when the surgical module is operably positioned on the cart,

and wherein the master module is configured to control the surgical module via the backplane.

7. The modular surgical system of claim 6, wherein the surgical module comprises a smoke evacuator.

8. The modular surgical system of claim 6, wherein the surgical module comprises an insufflator.

9. The modular surgical system of claim 1, wherein the cart comprises wheels.

10. A modular cart configured to support a plurality of surgical modules thereon, wherein the modular cart comprises:

- a frame;
- a backplane; and
- a plurality of modular sections defined by the frame, wherein the plurality of modular sections comprises:
 - a first section configured to operably support a display, wherein the display comprises an electrical connector configured to electrically couple to the backplane based on the display being operably supported by the first section;
 - a second section configured to operably support a generator module, wherein the generator module is configured to provide electrosurgical energy to a surgical instrument operably coupled thereto, and wherein the generator module comprises an electrical connector configured to electrically couple to the backplane based on the generator module being operably supported by the second section; and
 - a third section configured to operably support a master module, wherein the master module comprises an electrical connector configured to electrically couple to the backplane based on the master module being operably supported by the third section, and wherein the master module is configured to control the display and the generator module via the backplane.

11. The modular cart of claim 10, further comprising wheels extending from the frame.

12. The modular cart of claim 10, wherein the generator module is configured to receive power signals via the backplane.

13. The modular cart of claim 10, wherein the generator module is configured to receive data signals via the backplane.

14. The modular cart of claim 10, wherein the cart is further configured to operably support a surgical module, and wherein the surgical module comprises an electrical connector configured to electrically couple to the backplane based on the surgical module being operably supported by the cart.

15. A modular surgical system, comprising:

- a cart including a plurality of compartments;
- a backplane;
- a display operably supportable by the cart, wherein the display comprises an electrical connector configured to electrically couple to the backplane based on the display being operably supported by the cart, and wherein the display is configured to receive power and data signals via the backplane;
- a generator module operably supportable by the cart, wherein the generator module comprises:
 - an electrical connector configured to electrically couple to the backplane based on the generator module being operably supported by the cart, wherein the generator module is configured to receive power and data signals via the backplane; and
- a port assembly, wherein the generator module is configured to provide electrosurgical energy to a surgical instrument operably coupled to the port assembly; and
- a master module operably supportable by the cart, wherein the master module comprises an electrical connector configured to electrically couple to the backplane based on the master module being operably supported by the cart, and wherein the master module is configured to control the display and the generator module via the backplane.

16. The modular surgical system of claim 15, wherein the master module is configured to display data associated with the generator module on the display.

17. The modular surgical system of claim 15, wherein the port assembly comprises:

- a monopolar port, wherein the generator module is configured to provide monopolar energy through the monopolar port;
- a bipolar port, wherein the generator module is configured to provide bipolar energy through the bipolar port; and
- an ultrasonic port, wherein the generator module is configured to provide ultrasonic energy through the ultrasonic port.

18. The modular surgical system of claim 1, further comprising a surgical module operably supportable by the cart, wherein the surgical module comprises an electrical connector configured to electrically couple to the backplane based on the surgical module being operably supported by the cart, and wherein the master module is configured to control the surgical module via the backplane.

19. The modular surgical system of claim 18, wherein the surgical module comprises a smoke evacuator.

20. The modular surgical system of claim 18, wherein the surgical module comprises an insufflator.

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